

The Synapse Cookbook

Community (CC0). Ararat Synapse by Lukas Gebauer.

Contents

Introduction	7
About Synapse	7
Getting Started	8
A 60-second mental model	8
What you need	8
Hello, network — fetch a web page	8
Hello, socket — the raw version	9
Compiling it	9
Going encrypted (one line)	10
Where to go next	10
The Architecture of Synapse	11
The Blocking Model & Threads	12
The era it came from	12
What “blocking with timeouts” means	12
Why this makes concurrency simple	12
Practical notes	13
Cross-Platform: the synsock Seam	14
The idea	14
Why a seam beats scattered IFDEFs	14
Companion seams	14
The SSL Plugin Seam	16
The problem it solves	16
How it works	16
TLS “midway” — upgrading a live socket	17
Why it is beautiful	17
Rolling Its Own Crypto & Encoding	18
What’s in the box	18
Why roll your own here	18
The lesson (and the caveat)	19
The Socket Class Family	20
The hierarchy	20
What unifies them	20
Choosing one	21
TBlockSocket — the base class	22

The lifecycle	22
The LastError model	22
Sending — low level and high level	23
Receiving — low level and high level	23
Readiness and waiting	24
Timeouts and other knobs	24
A worked example — a tiny POP3-style dialogue	24
TTCPSocket — TCP	26
The client pattern	26
The listen / accept pattern	27
The threaded server	27
SSL/TLS upgrade	28
Other TCP-specific methods	29
Honest caveats	29
TUDPSocket — UDP	30
Do I need a thread? No.	30
The two addressing styles	30
Worked example — a client that sends and waits for a reply	30
Worked example — a UDP server	31
Broadcast	32
Multicast	32
Honest caveats	33
TSocksSocket — the proxy layer	34
The idea: proxying is configuration, not a code path	34
Using it — the whole change is three lines	34
SOCKS4, SOCKS4a, SOCKS5 — and who resolves the name	35
It works for UDP too	35
Status and internals	35
Not to be confused with the HTTP tunnel	36
Why it is nicely designed	36
Synapse Protocol Classes	37
One shape, repeated	37
The request/response rhythm	37
TLS drops in from the side	38
The clients	38
TPOP3Send	40
The class at a glance	40
Authentication: where the hand-rolled crypto earns its keep	41
TLS: implicit tunnel or STLS upgrade	41
Worked example: fetch and print the first message	41
Peeking without downloading	42
Deletion is deferred	43
Streaming large messages	43
TLDPASend — LDAP directory queries	44
The class at a glance	44
Login, then bind — they are two steps	45
TLS	45
The result model	45
Worked example: bind and search	46

Writing to the directory	47
TSMTPSend — sending mail	48
The class at a glance	48
What Login does for you	49
STARTTLS via the plugin	49
The send rhythm	49
Worked example: send a plain message	49
Composing the body with mimemess	50
The shortcut functions	51
THTTPSend — talking HTTP	52
The class at a glance	52
HTTPS is automatic	52
The convenience functions	53
Worked example: a quick GET	53
Worked example: a form POST	54
Object-level usage: full control	54
When to use which	55
TIMAPSend — the mailbox client	57
The class at a glance	57
What Login does for you	58
FullSSL vs. STARTTLS	58
Sequence numbers vs. UIDs	59
Selecting a folder is a mode switch	59
Deleting is two steps	59
Worked example: fetch a message from a folder	59
Searching	61
Escaping and the raw command	61
When to use which verb	61
TFTPSend — file transfer over two connections	62
The class at a glance	62
What Login does for you	63
The data-connection model	63
DataStream vs. DirectFile	64
FTPS: explicit and implicit	64
Worked example: download a file	64
Worked example: upload a file	65
Listing a directory	66
The convenience functions	67
A note on ftptsend — a different protocol	67
TDNSSend	68
The class at a glance	68
How answers are shaped	69
Reverse lookups are automatic	69
UDP vs TCP — and the truncation caveat	69
Worked example: resolve a host (A record)	70
Worked example: look up mail exchangers (MX)	70
No TLS here	71
The rest of the protocol clients	72
NNTP — Usenet news (nntpsend, TNNTPSend, port 119)	72

SNMP — device monitoring (snmpsend, TSNMPSend, port 161)	73
SNTP — network time (sntpsend, TSNTPSend, port 123)	74
Ping — ICMP echo (pingsend, TPINGSend, no TCP port)	74
Telnet — terminal sessions (tlntsend, TTelnetSend, port 23)	75
Syslog — remote logging (slogsend, TSyslogSend, port 514, UDP)	75
Where to go next	76
MIME & Mail Messages	77
A message is a tree of parts	77
The message: TMimeMess	78
The headers: TMessHeader	79
Encodings: transfer encoding vs. inline header encoding	79
Where to go next	80
Building a Message	81
The shape you are building	81
Worked example	81
What the calls actually did	83
Variations	83
Caveats	83
Parsing a Message	84
The two-step decode	84
Walking the tree	84
Worked example	84
Reading the results	86
Using the WalkPart hook instead	86
Caveats	87
TBlockSerial — the sockets model, applied to RS-232	88
The lifecycle	88
Naming the device — ‘COM1’ vs ‘/dev/ttyS0’	89
A privilege caveat (Unix)	89
Config — baud, bits, parity, stop bits, flow	89
Sending — identical to the socket API	90
Receiving — the same three you already reach for	90
Readiness and buffers	91
The LastError model — reports, not exceptions	91
Modem control lines — RTS, DTR, and the status inputs	92
A worked example — 9600 8N1, command and terminated reply	93
Encoding & Crypto	95
The three units	95
You usually don’t call these directly	95
The recipes	95
One caveat up front	96
Hashing & HMAC	97
The output is raw bytes, not text	97
HMAC — keyed message authentication	98
Real protocol usage — you often don’t call these yourself	98
Long-hash variants	98
The honest caveat	98
Transfer Encodings	100

Base64	100
Quoted-printable	100
URL encoding	101
No security caveat here — but a correctness one	102
Charsets	103
Charsets are a TMimeChar enum, not numeric IDs	103
Name ↔ enum: GetCPFFromID / GetIDFromCP	103
Converting: CharSetConversion	104
The richer variants	104
Detection helpers	104
iconv: the reach beyond the built-in tables	105
No security dimension — just correctness	105
The Utility Units	106
How they fit together	106
A note on strings	106
The synutil Toolbox	108
Splitting strings	108
GetBetween — respects nesting	108
The Fetch family — consume tokens left to right	109
Trimming — spaces only	109
Dates and times	109
Header lists	110
Email address extraction	110
Hex, binary, and byte packing	111
Line-terminator and quoting helpers	111
URL parsing	112
Streams and temp files	112
IP validation lives next door	112
When to use what	112
ASN.1 BER with asnlutil	113
The TLV idea	113
The type constants	113
Encoding	114
Decoding	114
OIDs and MIBs	115
Debugging	115
Summary	115
IP Addresses and Host Info	117
synaip — parse and format addresses	117
Validation	117
IPv4 conversion	117
IPv6 conversion	118
Reverse form (for PTR / rDNS)	118
synamisc — what the OS knows	118
DNS servers	118
Local IP addresses	119
Proxy settings	119
Wake-on-LAN	119
What is <i>not</i> here	119
Summary	120

Recipes	121
How to use this section	121
The recipes	121
Where the depth lives	121
Recipes — Web	123
Download a URL to a file	123
POST form data	123
Fetch JSON over HTTPS	124
Recipes — Email	126
Send a plain-text mail	126
Send a mail with an attachment	127
Fetch and list an inbox	128
Recipes — Network tools	129
Resolve a hostname	129
Ping a host	130
A minimal TCP client	130
A threaded echo server	131
Recipes — Serial	133
Open a port and do a request/response	133
Visual Synapse	135
What it adds over raw Synapse	135
How it uses Synapse	135
Status & caveats	135
Appendix — Synapse in Two Hours	136
1. Hello, socket — a minimal TCP client	136
Sending a request	137
2. One line of HTTP — the convenience layer	137
When you need the details — THTTPSend	138
Posting data	139
3. Your first server — a threaded echo/line server	140
The per-connection handler thread	140
The listener loop	141
4. Going encrypted — TLS by adding one word	142
STARTTLS — upgrading a live connection	142
5. Where to next	142

Introduction

The Synapse Cookbook — articles about network programming with the Ararat Synapse TCP/IP library for Object Pascal.

This cookbook began as a community wiki that later went offline; version 0.1 was reconstructed from the Internet Archive. It was always meant as a practical, example-first companion to the library — not a formal reference.

Brand new to this? Start with [Getting Started](#) — it assumes no networking knowledge and has you talking to the internet in a few minutes. From there, the **Appendix — Synapse in Two Hours** goes deeper (HTTP, a threaded server, TLS). Readers who want to understand *why* the library is shaped the way it is should read [The Architecture of Synapse](#) — it is the heart of this cookbook, and the reason the recipes are so short.

Status (2026 revival): this Markdown edition is being migrated, section-by-section, from the recovered 0.1 text (`releases/synapsecookbook-0.1.txt`). Code examples are to be re-verified against current FPC + Ararat Synapse.

About Synapse

[Ararat Synapse](#) is a lightweight, blocking (and optionally non-blocking) TCP/IP library for Object Pascal — Delphi, Kylix, and Free Pascal / Lazarus — by **Lukas Gebauer**. It provides socket classes and a range of client protocol implementations without visual components or external dependencies. This cookbook teaches how to use it; all credit for the library belongs to its author.

Getting Started

New to Synapse — or to network programming in general? Start here. In a few minutes you will have a program that talks to the internet. No prior networking knowledge assumed.

A 60-second mental model

Networking is simpler than it sounds:

- A **host** is a computer on a network, named by an address like `example.com` or `93.184.216.34`. Turning a name into an address is **DNS**.
- A **port** is a numbered door on that host. Web servers listen on port 80 (443 for HTTPS), mail on 25, and so on.
- A **socket** is your end of a connection to `host:port`. You open it, send bytes, receive bytes, close it. That's it.

A **client** opens connections (your browser); a **server** waits for them (the web server). Synapse does both. Everything else in this book is detail on top of those four words: host, port, socket, bytes.

What you need

1. A **Pascal compiler**. [Free Pascal](#) (FPC) with or without the [Lazarus](#) IDE, or Delphi. All are supported.
2. **Ararat Synapse** itself — it is a set of source units, nothing to compile or install as a library. Get it from github.com/geby/synapse (or the [homepage](#)) and unzip it somewhere, e.g. `~/synapse`.

That's the whole setup. Synapse has **no external dependencies** for the basics (see [Rolling its own crypto](#)); you only add an SSL unit later if you want TLS.

Hello, network — fetch a web page

The friendliest possible start. This downloads a page and prints it:

```
program hello_net;
uses httpsend;           // Synapse's HTTP client
var
  page: string;
begin
  if HttpGetText('http://example.com/', page) then
    WriteLn(page)
  else
```

```

    WriteLn('request failed');
end.

```

HttpGetText is a one-line convenience: give it a URL and a string, it fills the string with the response body and returns True/False. You just made an HTTP request. (For HTTPS, add one unit — see below.)

Hello, socket — the raw version

To see the socket underneath, connect to a host and read what it says. Here we open a plain TCP connection and read the greeting a server sends:

```

program hello_socket;
uses blksocket;           // Synapse's socket classes
var
    sock: TTCPBlockSocket;
begin
    sock := TTCPBlockSocket.Create;
    try
        sock.Connect('example.com', '80');           // host, port (as strings)
        if sock.LastError <> 0 then
            begin
                WriteLn('connect failed: ', sock.LastErrorDesc);
                Exit;
            end;
        sock.SendString('GET / HTTP/1.0' + CRLF + CRLF); // ask for the page
        WriteLn(sock.RecvString(3000));                // read one line, wait up to 3 s
    finally
        sock.Free;
    end;
end.

```

Two habits to notice, because they run through all of Synapse:

- **Ports are strings** ('80'), not integers — Synapse also accepts service names.
- **Check LastError after operations.** Synapse *reports* problems rather than raising exceptions, which keeps your code a straight line. LastError = 0 means success; LastErrorDesc is the human-readable reason. This is the whole error model — no try/except gymnastics required.

Compiling it

With FPC on the command line — point it at the Synapse source with -Fu:

```

fpc -Fu~/synapse hello_net.pas
./hello_net

```

With Lazarus — Project → Project Options → Compiler Options → Paths, and add ~/synapse to *Other unit files (-Fu)*. Then Run.

That's it — the same -Fu path idea, once per project.

Going encrypted (one line)

Want HTTPS instead of HTTP? Add a single SSL unit to `uses` and use an `https://` URL — nothing else changes:

```
uses httpsend, ssl_openssl;           // that extra unit is the entire TLS wiring
...
HttpGetText('https://example.com/', page);
```

That one line is Synapse's [SSL plugin seam](#) at work — TLS is a plugin you switch on by including it.

Where to go next

- [Synapse in Two Hours](#) — the fuller tutorial: HTTP GET/POST, a threaded echo *server*, TLS.
- [The Architecture of Synapse](#) — *why* it is shaped this way; short and worth it.
- [Recipes](#) — copy-paste solutions to common tasks (send mail, download a file, resolve a name, ping a host, talk to a serial port).

You now know the whole shape: open a socket to `host:port`, send and receive bytes, check `LastError`, close. Everything else is a protocol class doing that for you.

The Architecture of Synapse

Before the recipes, the design — because Synapse’s design is the reason the recipes are so short. Ararat Synapse is a masterclass in doing a great deal with very little: no visual components, no framework, no external runtime, and (for the essentials) no external libraries at all. Just Object Pascal, a clean set of classes, and a few genuinely clever seams.

It was written in an era when **async did not exist** as a language feature and preemptive **timeslicing was still being researched and proven**. Synapse’s answer was not to wait for the future — it was to build a *blocking* socket library with precise timeouts, designed from the start to be driven by threads. That decision aged remarkably well: the code is linear and readable (no callback spaghetti, no state machines), and concurrency is just “run another thread.”

Four ideas carry the whole library:

1. **One blocking socket, every transport.** TBlockSocket and its descendants put TCP, UDP, raw IP — and even **serial ports** — behind a single blocking, timeout-driven API. Learn one class, use them all. → [Blocking model & threads](#)
2. **A thin OS seam.** synsock isolates every platform difference (Winsock, Unix libc, Windows CE) behind one internal interface, so the socket classes above it are written **once** and compile everywhere. → [Cross-platform: synsock](#)
3. **TLS as a swappable plugin, bolted on mid-stream.** SSL/TLS is *not* baked into the socket. A single global metaclass (SSLImplementation) is overridden simply by which `ssl_*` unit you place in uses. Any block socket can be upgraded to TLS on a live connection (STARTTLS-style). This is the library’s most elegant hack. → [The SSL plugin seam](#)
4. **Batteries included, dependencies excluded.** MD5, MD4, SHA-1, HMAC, Base64, quoted-printable, DES/3DES — all **hand-rolled** in synacode/synacrypt, so authentication and encoding work with zero external libraries. → [Rolling its own crypto](#)

On top of those four seams sit the protocol clients — SMTP, POP3, IMAP, HTTP, FTP, NNTP, LDAP, SNMP, SNTTP, DNS, ping, telnet — each a thin, readable class that speaks its protocol over a TBlockSocket. That is the whole shape of the library: **thin, orthogonal layers, each doing one thing well**. Once you see the seams, the rest of this cookbook is just usage.

Why celebrate the architecture at all? Because it teaches transferable design: push platform differences down into a seam, express optional features (TLS) as pluggable metaclasses instead of flags, keep the I/O model simple (blocking + threads) and let the OS scheduler do the hard part. These lessons outlive Object Pascal.

The Blocking Model & Threads

Synapse is a **blocking** socket library. That is a deliberate design choice, not a limitation, and understanding it is the key to using Synapse well.

The era it came from

When Synapse was designed, `async/await` did not exist as a language construct, and reliable preemptive **timeslicing** of threads was still being researched and proven. Two schools competed for network I/O:

- **Non-blocking / select loops** — one thread juggling many sockets via a state machine. Powerful, but the code becomes a tangle of callbacks and saved state.
- **Blocking + threads** — each connection handled by straight-line code in its own thread; the OS scheduler interleaves them.

Synapse bet on the second, *provided* you had dependable timeouts and threads. That bet aged well: the code reads top-to-bottom like a description of the protocol, and scaling to many connections is “spawn more threads.”

What “blocking with timeouts” means

Every read/write takes a timeout in milliseconds. A call returns when the operation completes **or** the timeout elapses — it never hangs forever:

```
sock.Connect(host, port);           // blocks until connected or ConnectTimeout
line := sock.RecvString(5000);      // blocks up to 5 s for a CRLF-terminated line
sock.SendString('HELLO' + CRLF);    // blocks until sent
if sock.LastError <> 0 then ...      // every call reports status, no exceptions-as-flow
```

The timeout is what makes blocking *safe*: no runaway waits, and you can build keep-alive and cancellation on top of it. `CanRead(timeout)` / `CanWrite(timeout)` let you poll readiness without committing to a read — the ingredients of a select loop, if you want one, without leaving the blocking API.

Why this makes concurrency simple

Because each socket’s code is linear and self-contained, concurrency is just threading:

```
type
  TConnThread = class(TThread)
    FSock: TTCPPBlockSocket;
```

```
    procedure Execute; override;    // straight-line protocol code here
end;
```

One connection, one thread, one readable Execute. No shared state machine, no re-entrancy puzzles per socket. This is exactly the model that *pastella* (and Visual Synapse's servers) built on: a listener thread accepts, then hands each connection to its own handler thread, each speaking the protocol in plain sequential code.

The transferable lesson: a simple I/O model plus the OS scheduler often beats a clever single-threaded event loop — in *readability* always, and in throughput more often than people expect. Synapse chose boring-and-correct, and it is still in service two decades later.

Practical notes

- Give every socket a **thread of its own** for servers; never share one `TBlockSocket` across threads.
 - Choose timeouts per protocol step; they double as your liveness policy.
 - Check `LastError/LastErrorDesc` after operations — Synapse reports rather than throws, keeping the linear flow intact.
-

Cross-Platform: the synsock Seam

Synapse runs on Windows, Linux, other Unixes, and Windows CE, across Delphi, Kylix, and Free Pascal — and the socket classes are written **once**. The trick is a single thin unit that absorbs every platform difference: synsock.

The idea

Operating systems disagree about sockets: Windows has Winsock (ws2_32.dll, an explicit WSACleanup lifecycle, SOCKET handles), Unix has BSD sockets in libc (file descriptors, errno), Windows CE has its own quirks. If those differences leak upward, every socket class needs `{IFDEF}` clutter.

synsock is the **seam** that stops the leak. It exposes one internal socket API — the calls blksock needs — and implements it per platform behind conditional compilation:

```
unit synsock;
{$IFDEF WINCE}
  ...windows CE backend...
{$ELSE}
  {$I sslinux.inc}    // unix/linux backend, pulled in by include
{$ENDIF}
```

Above synsock, blksock.pas is platform-agnostic. It calls the seam's functions and never names a specific OS. Port to a new platform → implement the seam once, and every socket class and protocol client comes along for free.

Why a seam beats scattered IFDEFs

- **One place to be right.** Platform bugs live in synsock, not smeared across 50 units.
- **The layer above stays readable.** TTCPBlockSocket reads like sockets, not like a compatibility matrix.
- **New platforms are additive.** You extend the seam; you don't touch the library on top of it.

This is the same principle as a Hardware Abstraction Layer, or a driver interface: *define the narrow contract the upper layers need, then satisfy it per target*. Synapse applied it to the OS socket API long before “platform abstraction layer” was a buzzword — and it is why one Pascal codebase quietly serves every target at once.

Companion seams

synsock handles sockets; two sibling units finish the portability story:

- **synafpc** — smooths *compiler* differences (Delphi vs Kylix vs Free Pascal): types, RTL calls, and dialect quirks, so the same source compiles on each.
- **synachar / synaicnv** — charset conversion, so text protocols behave the same regardless of the host's locale.

Together: synsock abstracts the OS, synafpc abstracts the compiler, synachar abstracts the locale. Three narrow seams, and the rest of Synapse is written once for all of them.

The SSL Plugin Seam

This is Synapse's most elegant idea, and worth studying even if you never write Pascal: **TLS is not a feature of the socket — it is a plugin selected at link time and attached to a live connection.**

The problem it solves

A naive socket library bakes SSL in: `#ifdef` around OpenSSL, a `UseSSL`: Boolean flag, and the library now *depends* on one TLS implementation forever. Change your mind (OpenSSL → SChannel → a hardware library) and you rewrite the socket.

Synapse refuses that. The block socket knows *nothing* about any specific TLS library. It only knows an abstract contract.

How it works

In `blksock.pas`:

```
type
  TCustomSSL = class;           // abstract TLS contract
  TSSLClass = class of TCustomSSL; // a metaclass (class reference)

var
  SSLImplementation: TSSLClass = TSSLNone; // global default: "no TLS"
```

Three moving parts:

1. **TCustomSSL** — an abstract base class defining what a TLS provider must do (Connect, Accept, Shutdown, read/write, certificate handling...).
2. **TSSLNone** — a null-object implementation. With no TLS unit included, this is what you get: a socket that simply passes bytes through, unencrypted.
3. **SSLImplementation** — a single global **metaclass variable**. Every socket creates its TLS object from it:

```
FSSL := SSLImplementation.Create(self); // in the socket constructor
```

The trick: each concrete provider unit **sets that global in its own initialization**. `ssl_openssl.pas`, `ssl_schannel.pas`, `ssl_cryptlib.pas`, `ssl_streamsec.pas`, `ssl_libssh2.pas`, and more each do, in effect:

```
initialization
  SSLImplementation := TSSLOpenSSL; // (or TSSLChannel, TSSLCryptLib, ...)
```

So **you choose your TLS backend simply by which unit appears in uses**. No flag, no factory call, no configuration:

```
uses blksock, ssl_openssl;    // <-- this line is the entire choice
```

Include `ssl_openssl` and every socket transparently gains OpenSSL. Swap it for `ssl_schannel` and the *same socket code* now uses Windows' native TLS. Include nothing and you get `TSSLNone` — plaintext, zero TLS dependency. The socket classes never change.

TLS “midway” — upgrading a live socket

Because TLS is a layer attached to an already-open `TBlockSocket`, you can start a connection in plaintext and **upgrade it to TLS mid-stream** — exactly what protocols like SMTP STARTTLS, IMAP, FTP-over-TLS, and LDAP need:

```
sock.Connect(host, '25');      // plaintext
// ... negotiate STARTTLS in the protocol ...
sock.SSLDoConnect;            // upgrade this live socket to TLS
```

`SSLDoConnect` hands the existing connection to the plugin, which performs the handshake over it. The protocol clients (`smtpsend`, `imapsend`, ...) use exactly this to implement opportunistic encryption.

Why it is beautiful

- **Zero coupling.** The socket depends on an *interface* (`TCustomSSL`), never a library. OpenSSL is not a dependency of Synapse — it is a dependency of *your choice to include* `ssl_openssl`.
- **Link-time polymorphism, no cost.** The selection is a metaclass assignment at unit initialization — no runtime factory, no config parsing.
- **Extensible by outsiders.** Anyone can add a new TLS provider by writing one unit that descends `TCustomSSL` and sets `SSLImplementation`. The core never learns their name. (The ~20 `ssl_*` units in the tree are exactly this, accreted over 20 years.)
- **Graceful default.** No TLS unit → `TSSLNone` → it still compiles and runs, just unencrypted. Nothing to stub.

This is the null-object pattern and link-time strategy selection, done in a few lines of Object Pascal, years before those names were common vocabulary.

Rolling Its Own Crypto & Encoding

Open a mail client library today and it pulls in a crypto dependency for the message digests that authentication needs. Synapse pulls in **nothing** — the hashes, HMACs, ciphers, and transfer encodings are all hand-written in Object Pascal, in synacode and synacrypt. For the essentials, Synapse has *no external dependency at all*.

What's in the box

From synacode.pas — digests, MACs, and encodings, all from scratch:

```
function MD5(const Value: AnsiString): AnsiString;  
function MD4(const Value: AnsiString): AnsiString;  
function SHA1(const Value: AnsiString): AnsiString;  
function HMAC_MD5(Text, Key: AnsiString): AnsiString;  
function HMAC_SHA1(Text, Key: AnsiString): AnsiString;  
function MD5LongHash(const Value: AnsiString; Len: integer): AnsiString;  
function SHA1LongHash(const Value: AnsiString; Len: integer): AnsiString;
```

plus Base64, quoted-printable, UU/XX, and the other MIME transfer encodings the mail and HTTP clients need.

From synacrypt.pas — symmetric ciphers (DES, 3DES, and friends) implemented in pure Pascal, for the protocols and SASL mechanisms that require them.

Why roll your own here

In the era Synapse was built, you could not assume a crypto library was present and portable across Windows, Linux, Kylix, and CE. The choices were: depend on something that might not be there, or **implement the handful of algorithms you actually need**, portably, once. Synapse chose the second — and it is exactly why pop3send, smtpsend, imapsend, and the SASL/APOP/CRAM-MD5 authentication paths *just work* with nothing else installed.

Note the deliberate scope: these are the digests and ciphers protocols require for **authentication and encoding**, not a general-purpose crypto suite. For transport encryption (TLS), Synapse does the opposite and delegates to a best-in-class library through the [SSL plugin seam](#) — because reimplementing TLS would be reckless, while reimplementing MD5 is a weekend and removes a dependency forever.

The lesson (and the caveat)

- **The lesson:** know which wheels are worth reinventing. A 200-line MD5 that erases a fragile, non-portable dependency is a *good* trade. Synapse reinvents precisely the small, stable, spec-frozen algorithms — and delegates the large, evolving, security-critical one (TLS).
- **The caveat for today:** MD5, MD4, SHA-1, and single-DES are here because the *protocols* still specify them (APOP, CRAM-MD5, legacy auth), not because they are secure choices for new designs. Use them to speak the protocols that demand them; do not reach for them as your application's security primitives. Modern strength lives behind the TLS plugin.

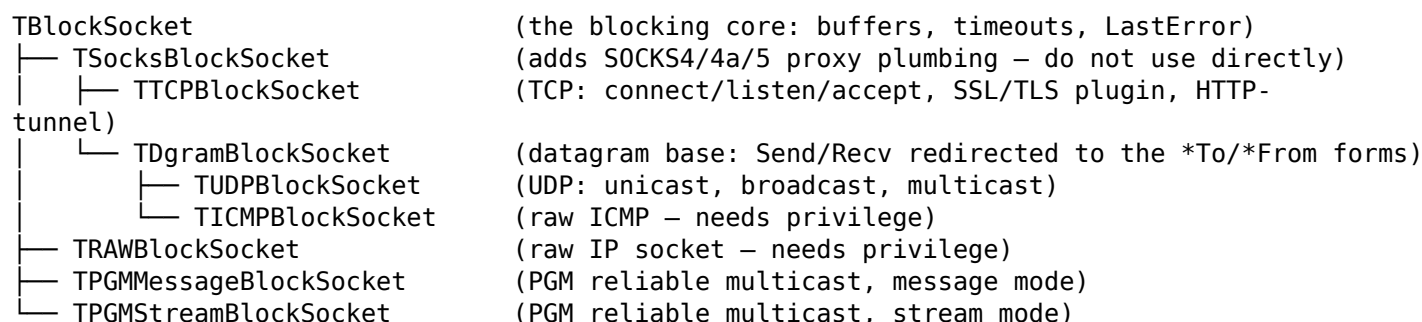
Read `synacode.pas` end to end at least once — it is a compact, readable tour of how these algorithms actually work, unobscured by macros or assembly. As teaching material, it is worth as much as the cookbook chapters that use it.

The Socket Class Family

Synapse’s socket layer is one file — `blksock.pas` — and one idea: **a single blocking base class, `TBlockSocket`, specialised by protocol**. Everything you send or receive, in any protocol, over TCP, UDP, ICMP, or a raw socket, goes through a descendant of that base. Learn the base once and every socket in the tree behaves the same way.

The hierarchy

The real tree, straight from `blksock.pas`:



Two things are worth reading off that diagram:

- **SOCKS lives in the *middle* of the tree, not at the top.** `TBlockSocket` itself knows nothing about proxies. `TSocksBlockSocket` inserts the proxy machinery, and both the TCP and the datagram families inherit from it — so proxy support comes “for free” to the sockets that can use it, while `TRAWBlockSocket` (which cannot) simply skips that layer by descending straight from `TBlockSocket`.
- **`TDgramBlockSocket` is a helper, not a socket you instantiate.** It exists only to redirect the connection-style `SendBuffer/RecvBuffer` onto the connectionless `SendBufferTo/RecvBufferFrom`, so datagram code can use the same high-level helpers (`SendString`, `RecvPacket`) as stream code. You create `TUDPSocket` or `TICMPSocket`; you never create `TDgramBlockSocket` or `TSocksBlockSocket` directly — their own source comments say as much (“Do not use this class directly”).

What unifies them

Every socket in the family shares one contract, defined on `TBlockSocket`:

A blocking API with timeouts. Reads and writes block until they complete or a millisecond timeout elapses; nothing hangs forever. See [The Blocking Model & Threads](#) for why that choice ages so well.

The `LastError` model — report, don't throw. After any operation you check `LastError` (an integer socket error code) and `LastErrorDesc` (a human string). Exceptions are opt-in: set `RaiseExcept := True` and `Synapse` raises `ESynapseError` on failure; leave it `False` (the default) and your protocol code stays linear.

```
sock.Connect('example.org', '80');
if sock.LastError <> 0 then
  Writeln('connect failed: ', sock.LastErrorDesc);
```

Two tiers of I/O on every socket. Low-level `SendBuffer/RecvBuffer` move raw bytes; high-level helpers built on an internal `LineBuffer` (`SendString`, `RecvString`, `RecvPacket`, `RecvTerminated`, `SendInteger`, `RecvBlock`, the stream methods...) do the framing for you. You can mix the two tiers freely on the same socket because the high-level side always drains `LineBuffer` first.

Shared knobs. Timeouts (`SetTimeout`, `ConnectionTimeout`), readiness polls (`CanRead`, `CanWrite`), byte counters (`RecvCounter`, `SendCounter`), bandwidth throttles (`MaxSendBandwidth`, `MaxRecvBandwidth`), a soft-abort `StopFlag`, and IPv4/IPv6 selection via the `Family` property — all live on the base and work identically for TCP, UDP, and the rest.

Choosing one

You want to...	Use
Speak a stream protocol (HTTP, SMTP, a custom line protocol)	<code>TTCPBlockSocket</code>
Wrap that stream in TLS/SSL or SSH	<code>TTCPBlockSocket</code> + an <code>ssl_*</code> plugin
Send/receive datagrams, broadcast, or multicast	<code>TUDPBlockSocket</code>
Send ICMP (ping-style) packets	<code>TICMPBlockSocket</code>
Craft raw IP packets yourself	<code>TRAWBlockSocket</code>
Reach any of the above through a SOCKS proxy	set <code>SocksIP</code> on the TCP/UDP socket you already have

The rest of this chapter walks the four you will actually reach for: [the base class](#), [TCP](#), [UDP](#), and [the SOCKS proxy layer](#) that quietly sits between them.

TBlockSocket — the base class

TBlockSocket is the root of the whole family. You almost never instantiate it directly — you create a TTCPBlockSocket or TUDPBlockSocket — but *every* method and property you use lives here or is overridden from here. This chapter is the reference you come back to.

The lifecycle

A Synapse socket has a simple, explicit life:

Create → Connect / Bind+Listen → Send/Recv ... → CloseSocket → Free

- **constructor Create** — makes the object and loads the socket library, but does *not* create an OS socket yet. There is also CreateAlternate(Stub: string) for loading a non-default socket provider.
- **Connect(IP, Port: string) / Bind(const IP, Port: string)** — the OS socket is created lazily on first use. Both take strings, numeric or symbolic: '192.168.0.1' or 'cosi.nekde.cz', '80' or 'http'.
- **CloseSocket** — closes the OS handle. It is virtual (TCP overrides it to shut down TLS first). The destructor calls it for you, so Free is enough, but closing explicitly when you are done is good manners.
- **AbortSocket** — the hard version: abandon any work and destroy the handle.

```
var sock: TTCPBlockSocket;  
begin  
  sock := TTCPBlockSocket.Create;  
  try  
    sock.Connect('example.org', '80');  
    if sock.LastError = 0 then  
      { ...talk... };  
  finally  
    sock.CloseSocket;  
    sock.Free;  
  end;  
end;
```

The LastError model

Synapse **reports** rather than throws. After any operation:

```
property LastError: Integer read FLastError;      // 0 = success  
property LastErrorDesc: string read FLastErrorDesc;
```

LastError is a standard socket error code (e.g. WSAETIMEDOUT on a timeout). LastErrorDesc is a human string for it, and the class function GetErrorDesc(ErrorCode) / method GetErrorDescEx turn any code into text. ResetLastError clears both back to the non-error state.

If you *prefer* exceptions, set RaiseExcept := True and Synapse raises ESynapseError whenever a call fails. The default is False — the linear, check-after-each-step style that keeps protocol code readable.

Sending — low level and high level

Low level, raw bytes in your buffer:

```
function SendBuffer(const Buffer: TMemory; Length: Integer): Integer; virtual;  
procedure SendByte(Data: Byte); virtual;
```

TMemory is Synapse's pointer-to-bytes type; you pass the address of your data and a length. SendBuffer returns the number of bytes actually sent.

High level, framing done for you:

```
procedure SendString(Data: AnsiString); virtual;    // sends bytes as-is – NO terminator added  
procedure SendInteger(Data: Integer); virtual;      // four bytes  
procedure SendBlock(const Data: AnsiString); virtual; // 4-byte length prefix, then data  
procedure SendStreamRaw(const Stream: TStream); virtual;  
procedure SendStream(const Stream: TStream); virtual; // length-framed, pairs with RecvStream
```

Note the honest caveat baked into SendString: **it adds no line terminator**. If your protocol is line-based, you append CRLF yourself:

```
sock.SendString('USER alice' + CRLF);
```

Receiving — low level and high level

Low level (you must know how much is coming, or check WaitingData first, or you risk a deadlock):

```
function RecvBuffer(Buffer: TMemory; Length: Integer): Integer; virtual;  
function RecvByte(Timeout: Integer): Byte; virtual;
```

High level, all take a millisecond Timeout and set LastError to WSAETIMEDOUT if it lapses:

```
function RecvBufferEx(Buffer: TMemory; Len, Timeout: Integer): Integer; virtual;  
function RecvBufferStr(Len, Timeout: Integer): AnsiString; virtual;  
function RecvString(Timeout: Integer): AnsiString; virtual;    // reads up to a CRLF, strips it  
function RecvTerminated(Timeout: Integer; const Terminator: AnsiString): AnsiString; virtual;  
function RecvPacket(Timeout: Integer): AnsiString; virtual;    // whatever is waiting, dynamic s  
function RecvInteger(Timeout: Integer): Integer; virtual;  
function RecvBlock(Timeout: Integer): AnsiString; virtual;    // reads a SendBlock frame  
procedure RecvStream(const Stream: TStream; Timeout: Integer); virtual;
```

The three you will reach for most:

- **RecvString** — a CRLF-terminated line, returned *without* the CRLF. The bread and butter of text protocols (SMTP, POP3, HTTP status lines). Under the hood it calls RecvTerminated(Timeout, CRLF).
- **RecvPacket** — “give me whatever is there right now.” Ideal when you do not know the length ahead of time — reading a chunk of a TCP stream, or a whole UDP datagram.

- **RecvBufferEx** — “give me exactly Len bytes, waiting up to Timeout.” For binary framing where the length is known.

The high-level functions share an internal LineBuffer (exposed as a property), which is why you can mix them: a RecvString that over-reads leaves the surplus in LineBuffer for the next call.

ConvertLineEnd := True relaxes RecvString to accept lone CR or lone LF as a line end too, not only CRLF — handy against sloppy peers.

Readiness and waiting

You do not usually need these — the Recv* timeouts already call them internally — but they are the ingredients of a hand-rolled select loop:

```
function CanRead(Timeout: Integer): Boolean; virtual; // data waiting, or (TCP) an incoming c
function CanReadEx(Timeout: Integer): Boolean; virtual; // also true if LineBuffer has data
function CanWrite(Timeout: Integer): Boolean; virtual; // send buffer has room
function WaitingData: Integer; virtual; // bytes ready to read (or datagram len)
```

Timeout = 0 means “test and return immediately”; -1 means “wait, possibly forever.” GroupCanRead polls a whole TSocketList at once — a real select() over many sockets, if you want one without leaving the blocking API.

Timeouts and other knobs

```
property ConnectionTimeout: Integer; // connect timeout in ms (0 = system default)
procedure SetTimeout(Timeout: Integer); // send + recv timeout (if the provider supports it)
procedure SetSendTimeout(Timeout: Integer);
procedure SetRecvTimeout(Timeout: Integer);
property MaxLineLength: Integer; // cap on RecvString/RecvTerminated – anti-DoS
property StopFlag: Boolean; // set True to soft-abort a long operation
property RecvCounter: int64; // bytes received this connection
property SendCounter: int64; // bytes sent this connection
property Family: TSocketFamily; // SF_Any (default), SF_IP4, SF_IP6
```

MaxLineLength deserves a mention: it bounds how much RecvString / RecvTerminated will buffer before erroring, so a hostile peer that never sends a terminator cannot make you allocate memory until you fall over. Default 0 means unlimited — set it on any server that reads lines from untrusted input.

A worked example — a tiny POP3-style dialogue

Line protocols map almost one-to-one onto RecvString/SendString:

```
uses blksock, synautil;
```

```
procedure FetchGreeting;
var
  sock: TTCPCBlockSocket;
  line: AnsiString;
begin
  sock := TTCPCBlockSocket.Create;
  try
```

```

sock.ConnectionTimeout := 5000;
sock.Connect('mail.example.org', '110');
if sock.LastError <> 0 then
begin
  Writeln('connect: ', sock.LastErrorDesc);
  Exit;
end;

line := sock.RecvString(15000);           // server greeting, e.g. '+OK POP3 ready'
Writeln('S: ', line);

sock.SendString('USER alice' + CRLF); // remember: we add CRLF ourselves
Writeln('S: ', sock.RecvString(15000));

sock.SendString('QUIT' + CRLF);
Writeln('S: ', sock.RecvString(15000));

Writeln(sock.RecvCounter, ' bytes received');
finally
  sock.CloseSocket;
  sock.Free;
end;
end;

```

Every line of that is base-class API — `TTCPBlockSocket` only adds the transport underneath. That is the point of `TBlockSocket`: the vocabulary is fixed once, and the protocol descendants and your own code both speak it unchanged.

TTCPBlockSocket — TCP

TTCPBlockSocket is the workhorse of the family: a blocking TCP socket that speaks stream protocols top-to-bottom, upgrades to TLS through the [SSL plugin seam](#), and can route itself through a SOCKS or HTTP proxy without your protocol code noticing. It descends TSocksBlockSocket, so it inherits everything in [the base class](#) plus the proxy plumbing.

Declared features, from the source:

IPv4, IPv6, SSL/TLS or SSH (depending on used plugin), SOCKS5 proxy (outgoing and limited incoming), SOCKS4/4a proxy (outgoing and limited incoming), TCP through HTTP proxy tunnel.

The client pattern

Connect, check LastError, talk, close. That is the whole shape:

```
uses blksock, synautil;

procedure HttpHead(const Host: string);
var
  sock: TTCPBlockSocket;
  line: AnsiString;
begin
  sock := TTCPBlockSocket.Create;
  try
    sock.ConnectionTimeout := 5000;
    sock.Connect(Host, '80');
    if sock.LastError <> 0 then
      begin
        Writeln('connect: ', sock.LastErrorDesc);
        Exit;
      end;

    sock.SendString('HEAD / HTTP/1.0' + CRLF +
      'Host: ' + Host + CRLF + CRLF);

    repeat
      line := sock.RecvString(10000);
      if sock.LastError = 0 then
        Writeln('S: ', line);
    until (sock.LastError <> 0) or (line = '');
  finally
    sock.CloseSocket;
```

```

    sock.Free;
end;
end;

```

Connect(IP, Port) creates the OS socket if needed and blocks until the connection is up or ConnectionTimeout expires. Ports may be numeric ('80') or symbolic ('http'). Set Family to SF_IP4 or SF_IP6 if you want to pin the address family; the default SF_Any picks based on how the host resolves.

The listen / accept pattern

A TCP server binds, listens, then accepts. Accept returns a raw OS socket handle (TSocket) for the new connection — you wrap it in a fresh TTCPCBlockSocket by assigning its Socket property:

```

procedure RunServer;
var
    listener, client: TTCPCBlockSocket;
begin
    listener := TTCPCBlockSocket.Create;
    try
        listener.CreateSocket;
        listener.SetLinger(True, 10000);
        listener.Bind('0.0.0.0', '8080');    // cAnyHost, any interface
        listener.Listen;
        if listener.LastError <> 0 then
            begin
                Writeln('bind/listen: ', listener.LastErrorDesc);
                Exit;
            end;
        while True do
            begin
                if listener.CanRead(1000) then    // 1 s poll so we can check for shutdown
                    begin
                        client := TTCPCBlockSocket.Create;
                        client.Socket := listener.Accept;    // adopt the accepted handle
                        HandleClient(client);    // in real code: hand to a thread (below)
                    end;
                end;
            finally
                listener.CloseSocket;
                listener.Free;
            end;
        end;
    end;

```

CanRead on the listener is how you tell an incoming connection is waiting without blocking. Accept forever — the same readiness poll from the base class.

The threaded server

The idiomatic Synapse server gives every connection its own thread of straight-line protocol code. This is exactly the model described in [The Blocking Model & Threads](#): the listener accepts, then hands each connection to a handler thread.

```

uses blksock, synautil, Classes;

type
  TClientThread = class(TThread)
  private
    FSock: TTCPBlockSocket;
  public
    constructor Create(AHandle: TSocket);
    procedure Execute; override;
  end;

constructor TClientThread.Create(AHandle: TSocket);
begin
  FSock := TTCPBlockSocket.Create;
  FSock.Socket := AHandle;           // adopt the handle Accept returned
  FreeOnTerminate := True;
  inherited Create(False);
end;

procedure TClientThread.Execute;
var
  line: AnsiString;
begin
  try
    line := FSock.RecvString(30000);
    while (FSock.LastError = 0) and (line <> '') do
    begin
      FSock.SendString('echo: ' + line + CRLF);
      line := FSock.RecvString(30000);
    end;
  finally
    FSock.CloseSocket;
    FSock.Free;
  end;
end;

```

The listener loop becomes: `TClientThread.Create(listener.Accept)` for each connection. One socket, one thread, one readable `Execute` — no shared state machine. **Never share a single `TTCPBlockSocket` across threads**; give each connection its own.

SSL/TLS upgrade

`TTCPBlockSocket` never encrypts by itself. TLS is a plugin attached at link time — include an `ssl_*` unit and the socket gains it transparently. The full story is in [The SSL Plugin Seam](#); the socket-side methods are:

```

constructor CreateWithSSL(SSLPlugin: TSSLClass); // pick a plugin per-socket
procedure SSLDoConnect;                        // upgrade a live client socket to TLS
function  SSLAcceptConnection: Boolean;        // server side: start TLS on an accepted soc
procedure SSLDoShutdown;                       // downgrade back to plain TCP
property  SSL: TCustomSSL read FSSL;           // the plugin instance (config: certs, keys,

```

Implicit TLS (HTTPS-style — encrypted from the first byte): include `ssl_openssl`, then call `SSLDoConnect` right after `Connect`:

```
uses blksock, ssl_openssl;    // <-- this line chooses OpenSSL
```

```
sock.Connect('example.org', '443');  
sock.SSLDoConnect;  
if sock.LastError <> 0 then  
    Writeln('TLS handshake failed: ', sock.LastErrorDesc);
```

Opportunistic TLS (STARTTLS-style — start plaintext, negotiate, then upgrade the *same* live connection):

```
sock.Connect('mail.example.org', '25');    // plaintext SMTP  
sock.SendString('STARTTLS' + CRLF);  
sock.RecvString(10000);                    // '220 ready to start TLS'  
sock.SSLDoConnect;                        // upgrade in place
```

Server side, configure certificates on the SSL object before you accept, then turn the accepted socket into a TLS session:

```
client.Socket := listener.Accept;  
client.SSL.CertificateFile := 'server.crt';  
client.SSL.PrivateKeyFile := 'server.key';  
if not client.SSLAcceptConnection then  
    Writeln('TLS accept failed: ', client.LastErrorDesc);
```

Other TCP-specific methods

- **GetRemoteSinIP / GetRemoteSinPort / GetLocalSinIP / GetLocalSinPort** — who you are talking to and from, once connected.
- **SetLinger(Enable, Linger)** (from the base) — control how long close waits to flush unsent data; stream-socket only.
- **OnAfterConnect** — an event fired right after a successful TCP connect, useful for logging or per-connection setup.
- **WaitingData / RecvPacket** are overridden to account for buffered TLS data, so they stay correct once you are inside a TLS session.

Honest caveats

- **SendString** adds **no** terminator — you append CRLF for line protocols (see the base class chapter). This is not TCP-specific but it bites here most.
 - Synapse is optimised for **synchronous** use. **NonBlockMode** exists but “not all functions work properly in it” — the source’s own warning. Prefer blocking + threads.
 - In **SOCKS** mode a server can accept only **one** connection, and **Accept** reuses the listening socket rather than making a new one — a limitation of **SOCKS BIND**, not of Synapse. For real servers, proxy the clients, not the listener.
-

TUDPBlockSocket — UDP

TUDPBlockSocket is the connectionless member of the family. There is no connect handshake, no stream, no “connection closed” — just datagrams in and datagrams out. It descends TDgramBlockSocket (which descends TSocksBlockSocket), and that middle class does one clever thing for you: it **redirects the ordinary SendBuffer/RecvBuffer onto the connectionless SendBufferTo/RecvBufferFrom**, so all the high-level helpers from [the base class](#) — SendString, RecvPacket, and friends — just work on datagrams.

Declared features, from the source:

IPv4, IPv6, unicasts, broadcasts, multicasts, SOCKS5 proxy (only unicasts, outgoing and incoming).

Do I need a thread? No.

Unlike TCP, UDP is fine on your main thread. Sending a datagram costs about as much as handing the packet to the network adapter, and for receiving you can poll with a 0 ms timeout and simply check whether anything arrived. Blocking UDP in the main thread is perfectly workable.

The two addressing styles

Synapse gives you two ways to name the far end:

1. **Connect(IP, Port)** — for UDP this does *not* handshake. It just fills in the remote address (RemoteSin), so subsequent SendString / SendBuffer go to that peer. Convenient when you talk to one fixed target.
2. **SendBufferTo + RecvBufferFrom** — explicit per-datagram addressing. RecvBufferFrom fills RemoteSin with the *sender's* address, so you can reply to whoever just wrote to you. This is the natural server style.

```
function SendBufferTo(const Buffer: TMemory; Length: Integer): Integer; override;  
function RecvBufferFrom(Buffer: TMemory; Length: Integer): Integer; override;
```

Worked example — a client that sends and waits for a reply

The UDP echo service (port 7) is the classic demo: send a datagram, read the one that comes back.

```
uses blksock;
```

```
procedure UdpEcho(const Host, Msg: string);
```

```

var
  sock: TUDPBlockSocket;
  reply: AnsiString;
begin
  sock := TUDPBlockSocket.Create;
  try
    sock.Connect(Host, '7');           // fixes RemoteSin; no handshake happens
    sock.SendString(Msg);             // redirected to SendBufferTo(RemoteSin)
    if sock.LastError <> 0 then
    begin
      Writeln('send: ', sock.LastErrorDesc);
      Exit;
    end;

    reply := sock.RecvPacket(2000); // whole datagram, up to 2 s
    if sock.LastError = 0 then
      Writeln('got back: ', reply)
    else
      Writeln('no reply: ', sock.LastErrorDesc);
  finally
    sock.CloseSocket;
    sock.Free;
  end;
end;

```

RecvPacket is the right receive call for UDP: one call returns exactly one whole datagram (its size is dynamic, so you never guess a buffer length).

Worked example — a UDP server

A server binds a port and loops, replying to each sender. Because RecvBufferFrom (and therefore RecvPacket) records the sender in RemoteSin, SendString back out goes straight to them.

```

uses blksock;

procedure RunUdpServer(const Port: string);
var
  sock: TUDPBlockSocket;
  data: AnsiString;
begin
  sock := TUDPBlockSocket.Create;
  try
    sock.Bind('0.0.0.0', Port);       // cAnyHost = all interfaces
    if sock.LastError <> 0 then
    begin
      Writeln('bind: ', sock.LastErrorDesc);
      Exit;
    end;

    while True do
    begin
      data := sock.RecvPacket(1000); // 1 s poll
      if sock.LastError = 0 then

```

```

begin
    Writeln('from ', sock.GetRemoteSinIP, ':', sock.GetRemoteSinPort,
           ' -> ', data);
    sock.SendString('ack: ' + data);    // replies to RemoteSin (the sender)
end;
// on WSAETIMEDOUT, loop again – lets you check a shutdown flag
end;
finally
    sock.CloseSocket;
    sock.Free;
end;
end;

```

If you ever need to send to a peer you have not received from, set the target explicitly with SetRemoteSin(IP, Port) (inherited) before SendString.

Broadcast

UDP can hit every host on the local network. Enable it, then send to the broadcast address (cBroadcast = '255.255.255.255'):

```

sock := TUDPBlockSocket.Create;
try
    sock.EnableBroadcast(True);           // must be on before broadcasting
    sock.Connect('255.255.255.255', '9999');
    sock.SendString('DISCOVER');
    // ...then RecvPacket in a loop to collect replies from responders...
finally
    sock.CloseSocket;
    sock.Free;
end;

```

EnableBroadcast binds the socket if it is not already bound. Note from the source: broadcast is **not** available in SOCKS5 mode, and **IPv6 has no broadcast at all** — use multicast there instead.

Multicast

For group communication (and the IPv6 replacement for broadcast):

```

procedure AddMulticast(MCastIP: string);           // join a group
procedure DropMulticast(MCastIP: string);           // leave it
procedure EnableMulticastLoop(Value: Boolean);     // hear your own sends, or not
property MulticastTTL: Integer;                     // router hops (1 = local network only)

sock := TUDPBlockSocket.Create;
try
    sock.Bind('0.0.0.0', '5000');
    sock.AddMulticast('239.1.2.3');                 // join the group
    sock.MulticastTTL := 1;                          // stay on the local segment
    sock.SendString('hello group');                  // reaches all joined members
finally
    sock.DropMulticast('239.1.2.3');
    sock.CloseSocket;

```

```
sock.Free;  
end;
```

MulticastTTL is a genuine footgun in disguise: leave it at 1 unless you mean to cross routers, because raising it sprays your traffic outward — and many routers drop multicast anyway, so a high TTL is often disappointment plus risk.

Honest caveats

- **UDP is lossy and unordered by design.** Synapse does not add reliability. A RecvPacket timeout on a UDP socket usually means the datagram (or its reply) was simply dropped — retry logic is your job.
 - A datagram larger than your buffer in the low-level RecvBufferFrom is **truncated**; RecvPacket avoids this by sizing dynamically.
 - SOCKS support here is **unicast only** — no broadcast or multicast through a proxy.
-

TSocksBlockSocket — the proxy layer

TSocksBlockSocket is the quiet middle of the hierarchy. You never instantiate it — its own source comment says “Do not use this class directly” — but it sits between [the base class](#) and both TTCPBlockSocket and the datagram sockets, and it carries the machinery that lets any of them reach the network **through a SOCKS proxy**. The elegant part: it does this without adding a single new call to your protocol code.

The idea: proxying is configuration, not a code path

A naive design would give you a “connect through proxy” method distinct from a “connect direct” one. Synapse refuses that. Instead, the proxy is a set of **properties** on the socket. Set them and the ordinary Connect / Bind / Listen you already call quietly route through the proxy; leave them empty and the same code connects directly.

```
property SocksIP: string;           // proxy address – EMPTY = SOCKS disabled; setting it ENABLES
property SocksPort: string;         // proxy port, default '1080'
property SocksUsername: string;     // set if the proxy needs auth
property SocksPassword: string;
property SocksTimeout: integer;     // proxy-conversation timeout, default one minute
property SocksResolver: Boolean;    // let the PROXY resolve hostnames (default True)
property SocksType: TSocksType;     // ST_Socks5 (default) or ST_Socks4
```

The trigger is SocksIP: assigning any non-empty value turns SOCKS mode on; clearing it turns it off. There is no separate “enable” switch — the source is explicit that “assigning any value to this property enables SOCKS mode.”

Using it — the whole change is three lines

Take the TCP client from [the TCP chapter](#) and route it through a SOCKS5 proxy. The protocol code is untouched; only setup changes:

```
uses blksock;

var
  sock: TTCPBlockSocket;
begin
  sock := TTCPBlockSocket.Create;
  try
    sock.SocksIP   := '10.0.0.1'; // <-- these three lines are the entire
    sock.SocksPort := '1080';     //      difference from a direct connection
```

```

sock.SocksType := ST_Socks5;

sock.Connect('example.org', '80'); // now tunnelled through the proxy
if sock.LastError <> 0 then
begin
    Writeln('connect via proxy: ', sock.LastErrorDesc);
    Exit;
end;
sock.SendString('HEAD / HTTP/1.0' + CRLF + CRLF);
Writeln(sock.RecvString(10000));
finally
    sock.CloseSocket;
    sock.Free;
end;
end;

```

Connect internally opens the connection to SocksIP:SocksPort, performs the SOCKS handshake (authenticating with SocksUsername/SocksPassword if set), asks the proxy to reach example.org:80, and only then returns. Your SendString/RecvString see a normal live socket.

With auth:

```

sock.SocksIP      := '10.0.0.1';
sock.SocksUsername := 'alice';
sock.SocksPassword := 's3cret';

```

SOCKS4, SOCKS4a, SOCKS5 — and who resolves the name

SocksType picks the protocol, and it interacts with SocksResolver:

- **ST_Socks5** (default) — full SOCKS5, with username/password auth and remote name resolution.
- **ST_Socks4** with SocksResolver = True — uses **SOCKS4a**, the extension that lets the proxy resolve the hostname.
- **ST_Socks4** with SocksResolver = False — pure SOCKS4, which cannot carry a hostname, so the name is resolved **locally** first.

SocksResolver (default True) is the privacy-relevant knob: leave it on and DNS lookups happen at the proxy, so the target hostname never leaks from your machine. Turn it off and your host resolves the name before connecting.

It works for UDP too

Because TDgramBlockSocket also descends TSocksBlockSocket, TUDPBlockSocket inherits the same properties. Set SocksIP on a UDP socket and unicast datagrams route through a SOCKS5 UDP association — with the honest limitation, noted in [the UDP chapter](#), that **only unicast** works through a proxy: no broadcast, no multicast.

Status and internals

A few members let you inspect what happened:

```

property UsingSocks: Boolean read FUsingSocks; // True once a proxy is actually in use
property SocksLastError: integer read FSocksLastError; // error code returned BY the proxy

```

The class also exposes `SocksOpen`, `SocksRequest`, and `SocksResponse`, but its comments flag all three as “needed only in special cases (it is called internally)” — you drive them by setting properties and calling `Connect`, not by calling these directly.

Not to be confused with the HTTP tunnel

`TTCPBlockSocket` has a *second*, independent way to traverse a proxy — an HTTP CONNECT tunnel, configured with `HTTPTunnelIP` / `HTTPTunnelPort` / `HTTPTunnelUser` / `HTTPTunnelPass`. That lives on `TTCPBlockSocket`, not here, and the source is emphatic: **you cannot combine the two**. SOCKS mode (`SocksIP` set) and HTTP-tunnel mode (`HTTPTunnelIP` set) are mutually exclusive — pick one proxy mechanism per socket.

Why it is nicely designed

- **Proxying is orthogonal.** It rides on the same `Connect/Bind/Listen` every socket already has, so *any* code built on `TTCPBlockSocket` or `TUDPBlockSocket` — including Synapse’s own HTTP, SMTP, and FTP clients — gains proxy support for free, just by having the properties set.
 - **Placed exactly once, inherited exactly where it belongs.** Putting SOCKS on a mid-tree class means TCP and UDP get it while `TRAWBlockSocket` (which cannot proxy) simply descends past it. The hierarchy encodes the capability.
 - **Configuration over API surface.** No new verbs to learn; the proxy is data you set, not a control path you branch on. The same lesson the [SSL plugin seam](#) teaches, applied to proxies instead of TLS.
-

Synapse Protocol Classes

Above the socket layer sits the part of Synapse most people actually reach for: the protocol clients. SMTP, POP3, IMAP, HTTP, FTP, NNTP, LDAP, SNMP, SNTP, DNS, ping, telnet — each is a small, readable class that speaks one protocol over a `TTCPSocket`. There is no framework here, no plugin registry, no visual component to drop on a form. There is a class, you set a few properties, you call a few methods in order, and you read the result. That is the whole story, and it is the same story for every protocol in the list.

One shape, repeated

Almost every client descends from a single tiny base class, `TSynaClient` (defined in `blksock`). It contributes nothing protocol-specific — just the connection knobs every client needs:

```
TSynaClient = class(TObject)
published
    property TargetHost: string; // server IP or name (default 'localhost')
    property TargetPort: string; // server port (client sets its default)
    property IPInterface: string; // outgoing local address
    property Timeout: integer; // socket timeout, ms
    property Username: string; // auth, if the protocol needs it
    property Password: string;
    property OAuth2Token: string;
end;
```

A concrete client — `TPOP3Send`, `TSMTPSend`, `TLDAPSend`, and the rest — then owns a `TTCPSocket` (exposed as the `Sock` property) and adds the verbs of its protocol. The constructor picks the standard port for you, so you usually only need to set `TargetHost`, maybe `Username/Password`, and go. The socket is created and freed with the client; you never manage it directly, though `Sock` is right there when you want to attach an `OnStatus` hook or tune a timeout.

The request/response rhythm

The clients are *blocking* by design (see [The blocking model](#)). That makes their internals a linear script rather than a state machine, and it makes the public API read the same way. The pattern, across protocols, is:

1. **Login** — connect the socket, do the protocol greeting/handshake, and (if you supplied credentials) authenticate. Returns `Boolean`.
2. **Issue verbs** — `Stat`, `Retr`, `MailFrom`, `Search`, `HTTPMethod`... each sends a command, blocks for the reply, and returns `True/False`.

3. **Read the result** — most clients expose `ResultCode / ResultString` for the last reply, plus a protocol-appropriate payload holder (`FullResult`, `Document`, `SearchResult`, ...).
4. **Logout** — send the protocol’s goodbye and close the socket.

Because each verb blocks until its reply lands, you never poll and you never register a callback. Concurrency, when you want it, is “run the client on another thread” — nothing in the class needs to change.

TLS drops in from the side

None of these clients contain any TLS code. Encryption is supplied by the [SSL plugin seam](#): you add one `ssl_*` unit to your uses clause and the global `SSLImplementation` metaclass is swapped for a real backend.

uses

```
smtpsend, ssl_openssl;    // that one extra unit is the entire TLS wiring
```

With a plugin present, two idioms cover almost everything:

- **Implicit TLS (tunnel from the first byte).** Set `FullSSL := True` before `Login`. The socket does its TLS handshake immediately on connect — this is the “SSL on a dedicated port” style (POP3S :995, SMTPS :465, IMAPS :993, HTTPS :443, LDAPS :636).
- **Explicit TLS (STARTTLS-style upgrade).** Set `AutoTLS := True` and `Login` will, if the server advertises the capability, upgrade the *live* plaintext connection to TLS mid-session (POP3 STLS, SMTP/LDAP STARTTLS). Each client that supports it also exposes a public `StartTLS` method if you want to drive the upgrade yourself.

HTTPS is even quieter: `THTTPSend` upgrades automatically when the URL scheme is `https://`, so a plugin in uses is all it takes.

The clients

Each lives in its own unit in the Synapse source (one unit per protocol). Ports are the defaults each client’s constructor sets.

Protocol	Unit	Class	Default port	Notes
SMTP / ESMTP	smtpsend	TSMTPSend	25	STARTTLS, CRAM-MD5/PLAIN/LOGIN auth · chapter
POP3	pop3send	TPOP3Send	110	APOP, STLS, XOAUTH2 · chapter
IMAP4	imapsend	TIMAPSend	143	full mailbox client · chapter
HTTP / HTTPS	httpsend	THTTPSend	80	convenience <code>HttpGetText</code> etc. · chapter
FTP	ftpsend	TFTPSend	21	active/passive data connections · chapter

Protocol	Unit	Class	Default port	Notes
NNTP	nntpsend	TNNTPSend	119	Usenet news · chapter
LDAP v2/v3	ldapsend	TLDAPSend	389	bind/search, SASL DIGEST-MD5 · chapter
SNMP	snmpsend	TSNMPSend	161	plus traps on 162 · chapter
SNTP	sntpsend	TSNTPSend	123	network time · chapter
DNS	dnssend	TDNSSend	53	resolver client · chapter
Ping	pingsend	TPINGSend	—	ICMP echo (raw socket) · chapter
Telnet	tlntsend	TTelnetSend	23	terminal session · chapter
Syslog	slogsend	TSyslogSend	514	RFC-3164 logging (UDP) · chapter
ClamAV	clamsend	TClamSend	3310	virus-scan daemon client

The chapters that follow work through the most-used clients in detail — POP3, LDAP, SMTP, HTTP, IMAP, FTP, and DNS — with the rest surveyed in [other protocols](#). Once you have read one, the others hold no surprises: learn the rhythm once, and it repeats all the way down the table.

TPOP3Send

POP3 is the simplest of the mail protocols — a numbered mailbox you connect to, count, download, and delete from — and TPOP3Send (in pop3send) is a faithful, compact mirror of it. If you have read the [protocol-class overview](#), everything here will feel familiar: a TSynaClient descendant wrapping a TTCPBlockSocket, a Login, a handful of verbs, a Logout.

The class at a glance

The credentials and connection properties come from the base TSynaClient (TargetHost, TargetPort, UserName, Password, Timeout). TPOP3Send adds the POP3 verbs and a few result holders:

```
TPOP3Send = class(TSynaClient)
    function Login: Boolean;
    function Logout: Boolean;
    function Stat: Boolean;           // -> StatCount, StatSize
    function List(Value: Integer): Boolean; // 0 = all messages
    function Retr(Value: Integer): Boolean; // download message N
    function RetrStream(Value: Integer; Stream: TStream): Boolean;
    function Dele(Value: Integer): Boolean; // mark message N deleted
    function Top(Value, Maxlines: Integer): Boolean;
    function Uidl(Value: Integer): Boolean; // unique-id listing
    function Reset: Boolean;              // RSET
    function NoOp: Boolean;
    function Capability: Boolean;        // CAPA
    function StartTLS: Boolean;          // STLS upgrade
    function CustomCommand(const Command: string; MultiLine: Boolean): Boolean;

    property ResultCode: Integer;        // 1 = OK, 0 = error
    property ResultString: string;       // the server's +OK / -ERR line
    property FullResult: TStringList;   // multiline payload (message, listing...)
    property StatCount: Integer;        // messages in mailbox, after Stat
    property StatSize: Integer;         // total bytes, after Stat
    property ListSize: Integer;         // size of one message, after List(N)
    property TimeStamp: string;         // APOP challenge, if the server sent one
    property AuthType: TPOP3AuthType;  // POP3AuthAll | POP3AuthLogin | POP3AuthAPOP
    property AutoTLS: Boolean;
    property FullSSL: Boolean;
    property Sock: TTCPBlockSocket;
end;
```

A couple of things worth noting from the source. ResultCode is *not* an HTTP- style code — POP3 only ever answers +OK or -ERR, so ResultCode is simply 1 for success and 0 for failure,

and the raw reply line is in `ResultString`. Every multiline reply (a retrieved message, a LIST of the whole mailbox) lands in `FullResult`, a `TStringList`, with POP3's byte-stuffed leading dots already un-stuffed for you.

Authentication: where the hand-rolled crypto earns its keep

Login does the whole handshake for you, and its auth logic is a nice illustration of why Synapse [rolls its own crypto](#). The default `AuthType` is `POP3AuthAll`, which means *autodetect*:

- If the server's greeting carries an **APOP timestamp** (a `<...>` token, stored in the `TimeStamp` property), Login tries **APOP**: it hashes `MD5(TimeStamp + Password)` and sends the digest, so the password never crosses the wire. That MD5 is the one in synacode — no external library.
- Otherwise it falls back to plain **USER/PASS**.
- If you set `OAuth2Token` on the client, Login uses **XOAUTH2** instead.

You can force one path by setting `AuthType` to `POP3AuthLogin` (USER/PASS only) or `POP3AuthAPOP` (APOP only).

Honest caveat. USER/PASS sends your password in the clear, and APOP — though it hides the password — relies on MD5 and is long deprecated. Neither is safe on an untrusted network *by itself*. For real security, put the session inside TLS (below); then even USER/PASS is protected by the transport.

TLS: implicit tunnel or STLS upgrade

As with every Synapse client, TLS is a uses clause away — add an `ssl_*` plugin and pick one of two styles:

```
uses pop3send, ssl_openssl;
```

- **POP3S (implicit).** Set `FullSSL := True` and point `TargetPort` at 995. The socket does its TLS handshake the moment it connects.
- **STLS (explicit upgrade).** Set `AutoTLS := True`. During Login, if the server's CAPA advertises STLS, the live plaintext connection is upgraded to TLS before authentication — Synapse re-runs `Capability` afterwards so the post-TLS capability set is the one you see.

Worked example: fetch and print the first message

```
program PopFetch;
```

```
uses
```

```
  SysUtils, Classes,  
  pop3send, ssl_openssl;  // ssl_openssl supplies TLS for STLS / POP3S
```

```
var
```

```
  Pop: TPOP3Send;  
  i: Integer;
```

```
begin
```

```
  Pop := TPOP3Send.Create;
```

```
  try
```

```
    Pop.TargetHost := 'mail.example.com';  
    Pop.UserName   := 'alice';
```

```

Pop.Password    := 's3cret';
Pop.AutoTLS     := True;           // upgrade with STLS if offered

if not Pop.Login then
begin
    Writeln('Login failed: ', Pop.ResultString);
    Halt(1);
end;

// How many messages, and how many bytes total?
if Pop.Stat then
    Writeln(Format('Mailbox: %d messages, %d bytes',
        [Pop.StatCount, Pop.StatSize]));

if Pop.StatCount > 0 then
begin
    // Retrieve message #1 in full; it lands in FullResult, line by line.
    if Pop.Retr(1) then
    begin
        Writeln('--- message 1 ---');
        for i := 0 to Pop.FullResult.Count - 1 do
            Writeln(Pop.FullResult[i]);
        end
    else
        Writeln('RETR failed: ', Pop.ResultString);

        // To delete it instead (marked now, removed at QUIT):
        // Pop.Dele(1);
    end;

    Pop.Logout;           // sends QUIT and closes the socket
finally
    Pop.Free;
end;
end.

```

The retrieved message in `FullResult` is the raw RFC-822 text — headers, a blank line, then the body, exactly as the server stored it. To turn that into decoded subject/from/body fields (and to walk MIME attachments), feed it to a `TMimeMess` and call `DecodeMessage`; the mail-message chapter covers that.

Peeking without downloading

Two verbs let you inspect a mailbox cheaply:

- **Top(N, Lines)** downloads message N’s headers plus the first Lines lines of its body into `FullResult` — ideal for a message-list preview without pulling megabytes.
- **Uidl(N)** returns a stable *unique id* for a message (or, with 0, for the whole mailbox in `FullResult`). Because POP3 message numbers are only stable within a single session, UIDs are how a “leave mail on server, download only what’s new” client remembers what it has already seen.

Deletion is deferred

Dele(N) does not delete immediately — it *marks* the message, and the server only actually removes marked messages when the session ends cleanly with Logout (QUIT). Call Reset to clear all delete marks in the current session, and remember that dropping the connection without Logout abandons the deletions.

Streaming large messages

For a big message you do not want the whole thing buffered in a TStringList. RetrStream(N, AStream) writes message N straight into any TStream (a file, a memory stream) as it arrives, bypassing FullResult entirely — the right tool for large attachments.

TLDAPSend — LDAP directory queries

LDAP is a different animal from the mail protocols. Where POP3 and SMTP trade lines of ASCII text, LDAP speaks **ASN.1 BER**, a binary encoding, over the wire. Synapse hides all of that: TLDAPSend (in `ldapsend`) builds and decodes the BER packets internally — leaning on the `asn1util` unit — and hands you plain Pascal objects. You bind, you search, you read result objects. The binary protocol never surfaces.

The class at a glance

Connection and credentials come from the `TSynaClient` base (`TargetHost`, `TargetPort`, `UserName`, `Password`). TLDAPSend adds the directory operations and, crucially, a structured result model:

```
TLDAPSend = class(TSynaClient)
  function Login: Boolean;           // connect (+ optional StartTLS) -- does NOT bind
  function Bind: Boolean;           // simple bind (plaintext password!)
  function BindSasl: Boolean;       // SASL DIGEST-MD5 bind (password never sent)
  function Logout: Boolean;
  function Search(obj: AnsiString; TypesOnly: Boolean; Filter: AnsiString;
    const Attributes: TStrings): Boolean;
  function Compare(obj, AttributeValue: AnsiString): Boolean;
  function Add(obj: AnsiString; const Value: TLDAPAttributeList): Boolean;
  function Modify(obj: AnsiString; Op: TLDAPModifyOp;
    const Value: TLDAPAttribute): Boolean;
  function ModifyDN(obj, newRDN, newSuperior: AnsiString;
    DeleteoldRDN: Boolean): Boolean;
  function Delete(obj: AnsiString): Boolean;
  function Extended(const Name, Value: AnsiString): Boolean;
  function StartTLS: Boolean;

  property Version: integer;        // default 3
  property ResultCode: Integer;     // 0 = Success (LDAP result code)
  property ResultString: AnsiString; // human-readable description
  property SearchScope: TLDAPSearchScope; // SS_BaseObject|SS_SingleLevel|SS_WholeSubtree
  property SearchSizeLimit: integer;
  property SearchTimeLimit: integer;
  property SearchResult: TLDAPResultList; // <-- results land here
  property Referrals: TStringList;
  property AutoTLS: Boolean;
  property FullSSL: Boolean;
```

```
property Sock: TTCPBlockSocket;  
end;
```

Note the LDAP convention on ResultCode: **0 means Success** here (it is the LDAP protocol result code — 0 Success, 49 Invalid credentials, 32 No such object, and so on), the opposite of POP3's 1 = OK. ResultString gives you the human-readable form for logging.

Login, then bind — they are two steps

This is the one place TLDAPSend differs from the other clients in a way that trips people up. Its Login **only opens the connection** (and runs StartTLS if AutoTLS is set) — it does *not* authenticate:

```
function TLDAPSend.Login: Boolean;  
begin  
    Result := False;  
    if not Connect then Exit;  
    Result := True;  
    if FAutoTLS then  
        Result := StartTLS;  
end;
```

Authentication is a separate Bind (or BindSasl). So the sequence is always Login → Bind/BindSasl → your operations → Logout. An empty UserName and Password at Bind time is a valid **anonymous bind** — common for public, read-only directory lookups.

- **Bind** sends a *simple bind*: the password travels in a BER octet-string, in the clear. Fine over TLS, unsafe otherwise.
- **BindSasl** does a **SASL DIGEST-MD5** exchange (built with synacode's HMAC/MD5, no external library) so the password is never transmitted. Prefer it when you cannot use TLS — though DIGEST-MD5 is itself dated, so the real answer for a hostile network is TLS underneath.

TLS

The usual seam applies — add an ssl_* plugin and choose implicit or explicit:

```
uses ldapsend, ssl_openssl;
```

- **LDAPS (implicit)** — set FullSSL := True, TargetPort := '636'.
- **STARTTLS (explicit)** — set AutoTLS := True so Login upgrades the plain :389 connection before you bind, or call StartTLS yourself.

The result model

A Search fills SearchResult, a TLDAPResultList. The shape mirrors LDAP itself:

- SearchResult — a list of TLDAPResult objects, one per directory entry found. Indexed with [i], counted with .Count.
- Each TLDAPResult has an **ObjectName** (the entry's DN) and an **Attributes** list (TLDAPAttributeList).
- Each TLDAPAttribute is a TStringList descendant with an **AttributeName** and one line per value (attributes can be multi-valued). Attributes.Get(name) is a shortcut to the first value of a named attribute.

Worked example: bind and search

Find every person whose surname is Smith under a base DN, and print their name and email:

```
program LdapSearch;

uses
  SysUtils, Classes,
  ldap send, ssl_openssl;  // ssl_openssl enables StartTLS / LDAPS

var
  Ldap: TLDAPSend;
  Attrs: TStringList;
  i, j: Integer;
  Entry: TLDAPResult;
  Attr: TLDAPAttribute;
begin
  Ldap := TLDAPSend.Create;
  Attrs := TStringList.Create;
  try
    Ldap.TargetHost := 'ldap.example.com';
    Ldap.UserName := 'cn=reader,dc=example,dc=com';
    Ldap.Password := 'readonly';
    Ldap.AutoTLS := True;           // upgrade to TLS before binding

    if not Ldap.Login then         // connect (+ StartTLS)
    begin
      Writeln('Connect failed');
      Halt(1);
    end;

    if not Ldap.Bind then          // authenticate; 0 = Success
    begin
      Writeln('Bind failed: ', Ldap.ResultString);
      Halt(1);
    end;

    // Only fetch the attributes we care about (empty list = all attributes).
    Attrs.Add('cn');
    Attrs.Add('mail');

    Ldap.SearchScope := SS_WholeSubtree;
    if Ldap.Search('dc=example,dc=com', // base DN
                  False,                // TypesOnly = False -> return values
                  '(sn=Smith)',         // filter
                  Attrs) then
    begin
      Writeln(Ldap.SearchResult.Count, ' entries:');
      for i := 0 to Ldap.SearchResult.Count - 1 do
      begin
        Entry := Ldap.SearchResult[i];
        Writeln('DN: ', Entry.ObjectName);
        for j := 0 to Entry.Attributes.Count - 1 do
        begin
```

```

        Attr := Entry.Attributes[j];
        // Attr is a TStringList of values under Attr.AttributeName
        Writeln(' ', Attr.AttributeName, ' = ', Attr.Text);
    end;
end;
end
else
    Writeln('Search failed: ', Ldap.ResultString);

    Ldap.Logout;
finally
    Attrs.Free;
    Ldap.Free;
end;
end.

```

The Search arguments map straight onto the LDAP operation: the **base DN** to search under, **TypesOnly** (pass False to get attribute values, True for just the attribute names), an **RFC-4515 filter string** ((sn=Smith), (&(objectClass=person)(mail=*)), ...) which TLDAPSend translates to BER for you, and the list of attributes to return (leave it empty to ask for all of them). Scope is set separately via SearchScope, and you can cap a runaway query with SearchSizeLimit / SearchTimeLimit.

Writing to the directory

The same object drives the write side, each method a one-to-one LDAP operation: Add (create an entry from a TLDAPAttributeList), Modify (add/delete/replace an attribute's values via TLDAPModifyOp = MO_Add/MO_Delete/MO_Replace), ModifyDN (rename or move an entry), and Delete. All obey the same ResultCode = 0 success convention, and all require a bind with sufficient privilege first.

Honest caveat. A simple Bind sends the password in cleartext BER; even BindSasl's DIGEST-MD5 is a legacy mechanism. Treat TLS (LDAPS or STARTTLS) as mandatory for anything but an anonymous read against a public directory.

TSMTPSend — sending mail

TSMTPSend (in `smtpsend`) is the send half of the mail story. It speaks SMTP and ESMTP, handles the EHLO/HELO negotiation, authenticates with the best mechanism the server offers, and — with an SSL plugin in uses — upgrades the session to TLS. As with every Synapse client it is a thin `TSynaClient` descendant over a `TTCPSocket`; you set a few properties and call the verbs in the order the protocol expects.

The class at a glance

Credentials and connection come from `TSynaClient` (`TargetHost`, `TargetPort`, `UserName`, `Password`). `TSMTPSend` adds the SMTP verbs and ESMTP state:

```
TSMTPSend = class(TSynaClient)
function Login: Boolean;           // connect, EHLO/HELO, STARTTLS, AUTH
function Logout: Boolean;         // QUIT + close
function MailFrom(const Value: string; Size: Integer): Boolean; // MAIL FROM
function MailTo(const Value: string): Boolean; // RCPT TO
function MailData(const Value: TStrings): Boolean; // DATA + body
function Reset: Boolean;         // RSET
function NoOp: Boolean;
function Verify(const Value: string): Boolean; // VRFY
function Etrn(const Value: string): Boolean;
function StartTLS: Boolean;      // STARTTLS upgrade
function FindCap(const Value: string): string; // query an ESMTP capability

property ResultCode: Integer;    // the numeric SMTP reply code (250, 550...)
property ResultString: string;   // the last reply line
property FullResult: TStringList; // the full (maybe multiline) reply
property ESMTPcap: TStringList;  // capabilities from EHLO
property ESMTP: Boolean;         // True if the server spoke ESMTP
property AuthDone: Boolean;      // True if AUTH succeeded
property ESMTPSize: Boolean;
property MaxSize: Integer;       // server's max message size, if advertised
property SystemName: string;     // name we send in HELO/EHLO
property AutoTLS: Boolean;
property FullSSL: Boolean;
property Sock: TTCPSocket;
end;
```

Unlike POP3's 1/0, SMTP's `ResultCode` is the real three-digit reply code, so you can reason about it directly (250 OK, 354 "send your data", 550 rejected, ...). Synapse also parses RFC-3463 enhanced status codes into `EnhCode1/2/3`, with `EnhCodeString` giving a human sentence for the last one.

What Login does for you

Login is a lot of protocol folded into one call. Reading the source, it:

1. Connects (doing an immediate TLS handshake first if FullSSL is set).
2. Sends **EHLO**; if the server rejects ESMTP it falls back to plain HELO.
3. Parses the advertised capabilities into ESMTPcap.
4. If AutoTLS is set and the server advertises STARTTLS, upgrades the live connection to TLS and re-runs EHLO.
5. If you supplied a Username/Password, authenticates — **preferring CRAM-MD5**, then PLAIN, then LOGIN, based on what the server's AUTH capability lists. All three digests/encodings come from synacode (HMAC-MD5, Base64) — no external crypto library. AuthDone tells you whether auth actually happened.

The name announced in HELO/EHLO defaults to the socket's local name; override it via SystemName if you need a specific identity.

STARTTLS via the plugin

TLS is the usual [SSL plugin seam](#) — one unit in uses, then choose implicit or explicit:

```
uses smtpsend, ssl_openssl;
```

- **Submission with STARTTLS (explicit).** Set AutoTLS := True, use the submission port TargetPort := '587' (or plain 25). Login upgrades the connection before it authenticates, so your credentials go out encrypted.
- **SMTPS (implicit).** Set FullSSL := True, TargetPort := '465'. The TLS handshake happens the instant the socket connects.

Honest caveat. AUTH LOGIN and AUTH PLAIN hand the server your password (Base64 is encoding, not encryption). Only send credentials over a TLS-upgraded session — which is exactly why Login performs STARTTLS *before* AUTH when AutoTLS is on.

The send rhythm

An SMTP transaction is a fixed sequence of verbs, and TSMTPSend maps one method to each:

```
Login -> MailFrom(sender) -> MailTo(rcpt) [-> MailTo(rcpt2) ...] -> MailData(lines) ->
```

MailFrom takes an optional size (pass the byte length of your message, or 0); if the server advertised the SIZE extension it will be sent along so an over-limit message is rejected up front. MailTo is called once per recipient. MailData takes a TStrings holding the **complete RFC-822 message** — headers, a blank line, then the body — and handles SMTP's dot-stuffing for you.

Worked example: send a plain message

```
program Smtplib;
```

```
uses
```

```
  SysUtils, Classes,  
  smtpsend, ssl_openssl;  // ssl_openssl enables STARTTLS / SMTPS
```

```
var
```

```

Smtp: TSMTPSend;
Msg: TStringList;
begin
  Smtp := TSMTPSend.Create;
  Msg := TStringList.Create;
  try
    // Build the full RFC-822 message: headers, blank line, body.
    Msg.Add('From: Alice <alice@example.com>');
    Msg.Add('To: Bob <bob@example.org>');
    Msg.Add('Subject: Hello from Synapse');
    Msg.Add('Date: ' + Rfc822DateTime(Now)); // Rfc822DateTime is in synutil
    Msg.Add(''); // <-- header/body separator
    Msg.Add('This message was sent with TSMTPSend.');
```

```

    Smtp.TargetHost := 'smtp.example.com';
    Smtp.TargetPort := '587'; // submission port
    Smtp.UserName := 'alice@example.com';
    Smtp.Password := 's3cret';
    Smtp.AutoTLS := True; // STARTTLS before AUTH

    if not Smtp.Login then
    begin
      Writeln('Login failed: ', Smtp.ResultString);
      Halt(1);
    end;

    if Smtp.MailFrom('alice@example.com', Length(Msg.Text)) and
       Smtp.MailTo('bob@example.org') and
       Smtp.MailData(Msg) then
      Writeln('Sent. Server said: ', Smtp.ResultString)
    else
      Writeln('Send failed: ', Smtp.ResultString);

    Smtp.Logout;
  finally
    Msg.Free;
    Smtp.Free;
  end;
end.
```

The envelope addresses in MailFrom/MailTo are the SMTP-level sender and recipients; the From:/To: *header* lines inside the message are separate (what the reader sees). They are usually the same, but need not be — that split is how mailing lists and Bcc: work.

Composing the body with mimemess

Hand-assembling headers is fine for a one-line notification, but the moment you want a subject with non-ASCII characters, an HTML alternative, or a file attachment, reach for **TMimeMess** (in mimemess). It builds a correct MIME message and serialises it to the TStrings that MailData wants:

```

uses smtpsend, mimemess, mimepart, ssl_openssl;

var
```

```

Mime: TMimeMess;
Body: TStringList;
begin
  Mime := TMimeMess.Create;
  Body := TStringList.Create;
  try
    Mime.Header.From := 'alice@example.com';
    Mime.Header.ToList.Add('bob@example.org');
    Mime.Header.Subject := 'Report attached';

    Body.Add('Hi Bob, see the attached report.');
    Mime.AddPartText(Body, nil);           // text/plain part
    Mime.AddPartBinaryFromFile('report.pdf', nil); // attachment

    Mime.EncodeMessage;                  // builds the full message into Mime.Lines

    // Mime.Lines is now the complete RFC-822 message -> hand it to MailData:
    // Smtplib.MailData(Mime.Lines);
  finally
    Body.Free;
    Mime.Free;
  end;
end;

```

After `EncodeMessage`, `Mime.Lines` is exactly the `TStrings` `MailData` expects, so the two compose cleanly: `TMimeMess` owns *what the message is*, `TSMTPSend` owns *getting it to the server*. See the mail-message chapter for the full `TMimeMess` treatment (parts, alternatives, inline images, charset handling).

The shortcut functions

If you just want to fire one message off and do not need the object, `smtplib` exposes three module-level helpers that create a `TSMTPSend`, run the whole `Login/MailFrom/MailTo/MailData/Logout` dance, and free it:

```

function SendTo(const MailFrom, MailTo, Subject, SMTPHost: string;
  const MailData: TStrings): Boolean;
function SendToEx(const MailFrom, MailTo, Subject, SMTPHost: string;
  const MailData: TStrings; const Username, Password: string): Boolean;
function SendToRaw(const MailFrom, MailTo, SMTPHost: string;
  const MailData: TStrings; const Username, Password: string): Boolean;

```

`SendTo` builds the `From/To/Subject/Date` headers for you and needs only the body; `SendToEx` adds authentication; `SendToRaw` takes an already-complete message (headers and all — the natural partner for `TMimeMess.Lines`). Append a `:port` to `SMTPHost` to override the port. They are the “send and forget” path; drop to the object whenever you need TLS toggles, capability inspection, or per-recipient control.

THTTPSend — talking HTTP

THTTPSend (in `httpsend`) is probably the most-used class in Synapse, and it comes at two levels. For the common cases there are one-line module functions — `HttpGetText`, `HttpPostURL`, and friends — that create the object, do the transfer, and free it. When you need control over headers, methods, cookies, or the raw response, you drop to the object itself. Both sit on the same `TSynaClient`-over-`TTCPSocket` foundation as the rest of the protocol clients, and both get HTTPS for free from the [SSL plugin seam](#).

The class at a glance

THTTPSend diverges a little from the mail clients: there is no Login/Logout pair, because HTTP is request-response. Instead the engine is a single method, `HTTPMethod`, and everything around it is properties you set before and read after:

```
THTTPSend = class(TSynaClient)
    function HTTPMethod(const Method, URL: string): Boolean;

    property Headers: TStringList;           // request headers in / response headers out
    property Cookies: TStringList;           // parsed Set-Cookie, reused on next request
    property Document: TMemoryStream;        // request body in / response body out
    property MimeType: string;                // Content-Type for the request body
    property Protocol: string;                // '1.0' (default) or '1.1'
    property KeepAlive: Boolean;
    property UserAgent: string;
    property ResultCode: Integer;             // HTTP status code: 200, 404, 500...
    property ResultString: string;            // the status reason phrase
    property DownloadSize: int64;
    property UploadSize: int64;
    property Sock: TTCPSocket;
    property ProxyHost, ProxyPort, ProxyUser, ProxyPass: string;
end;
```

The one object you touch most is **Document**, a `TMemoryStream`. It is bidirectional: you fill it with the request body before a POST/PUT, and after any call it holds the response body. `ResultCode` is the real HTTP status code, so you branch on it directly.

HTTPS is automatic

Here Synapse is at its quietest. `HTTPMethod` calls `ParseURL` on the URL, and if the scheme is `https`, it does the TLS handshake for you — no property to set:

```
if UpperCase(Prot) = 'HTTPS' then
...           // connect with SSL
```

So all HTTPS needs is an SSL plugin in uses and an https:// URL:

```
uses httpsend, ssl_openssl;
...
HttpGetText('https://example.com/', Response); // just works
```

No plugin, and an https:// URL will fail at the handshake — the seam is present but empty. That is the whole “HTTPS wiring.”

The convenience functions

For everyday work you rarely need the object. httpsend exposes these module-level helpers (each spins up a THTTPSend, runs one transfer, frees it):

```
function HttpGetText(const URL: string; const Response: TStrings): Boolean;
function HttpGetBinary(const URL: string; const Response: TStream): Boolean;
function HttpPostBinary(const URL: string; const Data: TStream): Boolean;
function HttpPostURL(const URL, URLData: string; const Data: TStream): Boolean;
function HttpPostFile(const URL, FieldName, FileName: string;
    const Data: TStream; const ResultData: TStrings): Boolean;
```

- **HttpGetText** — GET a URL, load the body into a TStrings. Perfect for an API that returns text/JSON.
- **HttpGetBinary** — GET into any TStream (a file, a memory stream) for images, downloads, binaries.
- **HttpPostURL** — POST form data. It sets Content-Type: application/x-www-form-urlencoded, sends URLData as the body, and returns the response in Data.
- **HttpPostBinary** — POST a raw octet-stream body.
- **HttpPostFile** — POST a multipart/form-data file upload, boundary and all.

Worked example: a quick GET

```
program HttpGet;

uses
  SysUtils, Classes,
  httpsend, ssl_openssl; // ssl_openssl -> https:// works

var
  Response: TStringList;
begin
  Response := TStringList.Create;
  try
    if HttpGetText('https://example.com/', Response) then
      WriteLn(Response.Text)
    else
      WriteLn('Request failed');
  finally
    Response.Free;
  end;
end.
```

Worked example: a form POST

```
program HttpPost;

uses
  SysUtils, Classes,
  httpsend, synacode, synautil;  // synacode: EncodeURLElement; synautil: ReadStrFromStream

var
  Reply: TMemoryStream;
  FormData: string;
begin
  Reply := TMemoryStream.Create;
  try
    // application/x-www-form-urlencoded: key=value&key=value, values escaped.
    FormData := 'name=' + EncodeURLElement('Alice') +
               '&city=' + EncodeURLElement('São Paulo');

    if HttpPostURL('http://httpbin.org/post', FormData, Reply) then
    begin
      Reply.Position := 0;
      WriteLn('Response:');
      WriteLn(ReadStrFromStream(Reply, Reply.Size));  // synautil helper
    end
    else
      WriteLn('POST failed');
  finally
    Reply.Free;
  end;
end.
```

Object-level usage: full control

When you need custom headers, a specific method, a status code, or the response headers, use THTTPSend directly. The pattern is: set Headers/Document/ MimeType, call HTTPMethod(verb, url), then read ResultCode, Headers, and Document.

```
program HttpJsonApi;

uses
  SysUtils, Classes,
  httpsend, ssl_openssl;

var
  Http: THTTPSend;
  Body: string;
begin
  Http := THTTPSend.Create;
  try
    Http.Protocol := '1.1';
    Http.MimeType := 'application/json';  // Content-Type of our body
    Http.Headers.Add('Authorization: Bearer ' + 'my-token');
    Http.Headers.Add('Accept: application/json');
```

```

// Put the request body into Document.
WriteStream(Http.Document, '{"name":"Alice"}'); // synautil helper

if Http.HTTPMethod('POST', 'https://api.example.com/users') then
begin
    WriteLn('Status: ', Http.ResultCode, ' ', Http.ResultString);

    // Response body is in Document; rewind and read it out.
    Http.Document.Position := 0;
    SetLength(Body, Http.Document.Size);
    if Http.Document.Size > 0 then
        Http.Document.ReadBuffer(Body[1], Http.Document.Size);
    WriteLn(Body);

    // Response headers are in Headers after the call.
    WriteLn('--- response headers ---');
    WriteLn(Http.Headers.Text);
end
else
    WriteLn('Transport failed (no reply)');
finally
    Http.Free;
end;
end.

```

A few things the source makes clear, worth knowing:

- **Headers is reused in both directions.** You add request headers to it before the call; after the call it holds the *response* headers (the request ones having been sent and cleared). Read them straight back out of the same list.
- **HTTPMethod returns the transport result, not the HTTP result.** It returns True when a reply was received at all — even a 404 or 500. Always inspect ResultCode to learn what the server actually said; a True return with ResultCode = 404 is a perfectly normal “not found.”
- **Document is a TMemoryStream, not text.** For a text/JSON response, read its bytes out (as above) or LoadFromStream it into a TStringList. Remember to set Position := 0 before reading.
- **MimeType** becomes the request’s Content-Type; **Protocol** selects HTTP/1.0 (default) or 1.1; **KeepAlive** (on by default) lets you reuse one THTTPSend across several requests to the same host, saving reconnects.
- **Cookies** set by the server are parsed into the Cookies list and sent back automatically on the next request from the same object — a small session for free.
- **Proxies** go through ProxyHost/ProxyPort/ProxyUser/ProxyPass; Basic auth to the target is handled if you embed user:pass@ in the URL.

When to use which

Reach for the **convenience function** whenever a single GET or POST with default headers will do — it is a one-liner and manages the object’s lifetime for you. Reach for the **object** the moment you need custom headers, a non-GET/POST verb (PUT, DELETE, PATCH — HTTPMethod takes any method string), the status code, the response headers, cookie continuity, or keep-alive across calls. Both are the same engine; the functions are just THTTPSend with the common case pre-wired.

TIMAPSend — the mailbox client

TIMAPSend (in `imapsend`) is the one client in this chapter that does more than push a message through a pipe: it drives a *stateful* mailbox on the server. IMAP keeps your mail on the server and lets you browse folders, fetch individual messages or just their headers, search, and set flags — all without downloading the whole mailbox. TIMAPSend is Synapse’s IMAP4rev1 implementation (RFC-2060, RFC-2595), and like every client here it is a thin `TSynaClient` descendant over a `TTCPSocket`: set a few properties, call the verbs in order, read the result.

The class at a glance

Connection and credentials come from `TSynaClient` (`TargetHost`, `TargetPort`, `UserName`, `Password`, `OAuth2Token`). TIMAPSend adds the IMAP verbs and the selected-folder state:

```
TIMAPSend = class(TSynaClient)
    function Login: Boolean;           // connect, greeting, STARTTLS, LOGIN
    function Logout: Boolean;          // LOGOUT + close
    function NoOp: Boolean;             // NOOP (keep-alive)
    function Capability: Boolean;      // CAPABILITY -> fills IMAPcap

    // folders
    function List(FromFolder: string; const FolderList: TStrings): Boolean;
    function ListSubscribed(FromFolder: string; const FolderList: TStrings): Boolean;
    function ListSearch(FromFolder, Search: string; const FolderList: TStrings): Boolean;
    function CreateFolder(FolderName: string): Boolean;
    function DeleteFolder(FolderName: string): Boolean;
    function RenameFolder(FolderName, NewFolderName: string): Boolean;
    function SubscribeFolder(FolderName: string): Boolean;
    function UnsubscribeFolder(FolderName: string): Boolean;
    function SelectFolder(FolderName: string): Boolean;    // SELECT (read/write)
    function SelectROFolder(FolderName: string): Boolean;  // EXAMINE (read-only)
    function CloseFolder: Boolean;
    function StatusFolder(FolderName, Value: string): integer;
    function ExpungeFolder: Boolean;
    function CheckFolder: Boolean;

    // messages
    function FetchMess(MessID: integer; const Mess: TStrings): Boolean;    // whole message
    function FetchHeader(MessID: integer; const Headers: TStrings): Boolean; // headers only
    function MessageSize(MessID: integer): integer;
    function AppendMess(ToFolder: string; const Mess: TStrings): Boolean;
    function DeleteMess(MessID: integer): Boolean;    // mark \Deleted
```

```

function CopyMess(MessID: integer; ToFolder: string): Boolean;
function SearchMess(Criteria: string; const FoundMess: TStrings): Boolean;
function GetUID(MessID: integer; var UID: Integer): Boolean;

// flags
function GetFlagsMess(MessID: integer; var Flags: string): Boolean;
function SetFlagsMess(MessID: integer; Flags: string): Boolean;    // replace
function AddFlagsMess(MessID: integer; Flags: string): Boolean;    // +FLAGS
function DelFlagsMess(MessID: integer; Flags: string): Boolean;    // -FLAGS

function StartTLS: Boolean;
function FindCap(const Value: string): string;

property ResultString: string;           // status line of the last operation
property FullResult: TStringList;       // every reply line of the last operation
property IMAPcap: TStringList;          // server capabilities
property AuthDone: Boolean;
property UID: Boolean;                  // interpret MessID as UID, not sequence no.
property SelectedFolder: string;
property SelectedCount: integer;        // messages in the selected folder
property SelectedRecent: integer;
property SelectedUIDvalidity: integer;
property AutoTLS: Boolean;
property FullSSL: Boolean;
property Sock: TTCPBlockSocket;
end;

```

Note what is *not* here. Unlike SMTP or HTTP there is no numeric ResultCode — IMAP replies are tagged text (OK, NO, BAD), and the source models that directly: the internal IMAPcommand returns the uppercase status word and every public verb boils it down to = 'OK'. So a method returning True means the server tagged the command OK; when it returns False, ResultString holds the raw tagged line the server sent, which is your diagnostic.

What Login does for you

Reading the source, Login folds the whole session opening into one call:

1. Connects (doing an immediate TLS handshake first if FullSSL is set).
2. Reads the server greeting. * PREAUTH means the connection is already authenticated (and sets AuthDone); * OK means proceed to login; anything else fails.
3. Runs CAPABILITY and requires the server to advertise IMAP4rev1 — Synapse will not talk to a server that does not claim the base protocol.
4. If AutoTLS is set and the server advertises STARTTLS, upgrades the live connection to TLS and re-reads the capabilities.
5. Authenticates. With an OAuth2Token set it sends AUTHENTICATE XOAUTH2; otherwise it sends a plain LOGIN "user" "pass" (user and password escaped for the quoting rules — see below). AuthDone reports whether auth succeeded.

FullSSL vs. STARTTLS

TLS is the usual [SSL plugin seam](#) — one unit in uses, then pick implicit or explicit, exactly as described in the [chapter intro](#):

```
uses imapsend, ssl_openssl;
```

- **IMAPS (implicit).** Set `FullSSL := True`, `TargetPort := '993'`. The TLS handshake happens the instant the socket connects.
- **STARTTLS (explicit).** Set `AutoTLS := True`, keep the plain port 143. Login upgrades the connection *before* it sends LOGIN, so your credentials travel encrypted. You can also drive the upgrade yourself with the public `StartTLS` method (it checks `FindCap('STARTTLS')` first and returns `False` if the server never advertised it).

Honest caveat. LOGIN sends your username and password in the clear (IMAP does no hashing of its own here). Only ever log in over a TLS session — which is exactly why Login performs STARTTLS *before* LOGIN when AutoTLS is on.

Sequence numbers vs. UIDs

Every message verb takes a `MessID: integer`, and the `UID` property decides how the server reads it. With `UID := False` (the default), `MessID` is the message's *sequence number* within the selected folder — 1 is the first message, `SelectedCount` is the last. Sequence numbers are stable only for the current selection: expunging a message renumbers everything after it. With `UID := True`, Synapse prefixes each command with `UID` and `MessID` is the message's permanent unique identifier — stable across sessions, paired with `SelectedUIDvalidity` to detect a folder that was deleted and recreated. Reach for UIDs the moment you need an identifier that survives past the current SELECT.

Selecting a folder is a mode switch

IMAP is stateful: most message verbs only work once a folder is *selected*.

- **SelectFolder** issues SELECT and enters read/write state — flags and deletions take effect. After it returns, `SelectedCount`, `SelectedRecent`, and `SelectedUIDvalidity` describe the folder, and `SelectedFolder` names it.
- **SelectROFolder** issues EXAMINE — the same, but read-only, so you can browse without marking anything \Seen.
- **CloseFolder** leaves selected state (and silently expunges \Deleted messages, per the protocol).

`List` fills a `TStrings` with folder *names* (it filters out \Noselect containers and un-escapes quoted names for you); pass an empty `FromFolder` to list everything. `ListSubscribed` does the same over `LSUB`, and `ListSearch` takes a wildcard pattern.

Deleting is two steps

`DeleteMess` does **not** remove a message — it sets the \Deleted flag (`STORE ... +FLAGS.SILENT (\Deleted)`). The message physically goes away only when you call `ExpungeFolder` (or `CloseFolder`, which expunges on its way out). That two-step is IMAP's design, not a Synapse quirk: it lets a client “undelete” by clearing the flag before the expunge. The flag verbs are the general form — `AddFlagsMess/DelFlagsMess` add or remove flags (\Seen, \Flagged, \Answered, ...), `SetFlagsMess` replaces the whole set, and `GetFlagsMess` reads it back.

Worked example: fetch a message from a folder

The core rhythm is Login → SelectFolder → Fetch → Logout:

```

program ImapFetch;

uses
  SysUtils, Classes,
  imapsend, ssl_openssl;  // ssl_openssl enables IMAPS / STARTTLS

var
  Imap: TIMAPSend;
  Msg: TStringList;
begin
  Imap := TIMAPSend.Create;
  Msg := TStringList.Create;
  try
    Imap.TargetHost := 'imap.example.com';
    Imap.TargetPort := '993';           // IMAPS
    Imap.FullSSL := True;              // implicit TLS from the first byte
    Imap.UserName := 'alice@example.com';
    Imap.Password := 's3cret';

    if not Imap.Login then
      begin
        Writeln('Login failed: ', Imap.ResultString);
        Halt(1);
      end;

    // Enter the mailbox. After this, SelectedCount is the message count.
    if not Imap.SelectFolder('INBOX') then
      begin
        Writeln('Cannot select INBOX: ', Imap.ResultString);
        Imap.Logout;
        Halt(1);
      end;
    Writeln('INBOX holds ', Imap.SelectedCount, ' message(s).');

    if Imap.SelectedCount > 0 then
      begin
        // Fetch the newest message (highest sequence number) in full.
        if Imap.FetchMess(Imap.SelectedCount, Msg) then
          begin
            Writeln('--- message ', Imap.SelectedCount, ' ---');
            Writeln(Msg.Text);           // full RFC-822 message: headers + body
          end
        else
          Writeln('Fetch failed: ', Imap.ResultString);
        end;

    Imap.Logout;
  finally
    Msg.Free;
    Imap.Free;
  end;
end.

```

FetchMess retrieves the whole message (FETCH n (RFC822)) into the TStrings as a complete

RFC-822 message — headers, blank line, body. When you only need the envelope (a message-list view, say), `FetchHeader` fetches just the headers (`RFC822.HEADER`), and `MessageSize` returns the byte count without transferring anything. To turn the fetched `Msg` into structured headers, alternatives, and attachments, hand it to `TMimeMess.DecodeMessage` — the mail-message chapter covers that; `IMAPSend` owns *getting the bytes*, `TMimeMess` owns *what they mean*.

Searching

`SearchMess` sends IMAP's `SEARCH` and returns the matching message numbers (or UIDs, if `UID` is set) as a `TStrings`. The criteria string is IMAP's own search language — powerful, SQL-ish, but not SQL:

```
var
  Hits: TStringList;
begin
  Hits := TStringList.Create;
  try
    Imap.SelectFolder('INBOX');
    // messages from a given sender that are still unseen
    if Imap.SearchMess('FROM "bob@example.org" UNSEEN', Hits) then
      Writeln(Hits.Count, ' match(es): ', Hits.CommaText);
  finally
    Hits.Free;
  end;
end;
```

Each entry in `Hits` is a message number you can pass straight to `FetchMess`, `GetFlagsMess`, `CopyMess`, or `DeleteMess`.

Escaping and the raw command

IMAP folder and mailbox names are quoted strings, and characters like `"` and `\` must be escaped (RFC-2683). Synapse handles this for you inside every verb via `EscapeSpecialCharacters`, and un-escapes returned names via `UnescapeSpecialCharacters`; both are public if you build commands by hand. And you can always drop to the raw protocol: `IMAPcommand(cmd)` sends any tagged command and returns the status word, leaving the reply in `FullResult` — the escape hatch for server-specific extensions Synapse does not wrap.

When to use which verb

`SelectFolder/SelectROFolder` gate everything — call one first. Then `FetchHeader` + `MessageSize` to *browse* cheaply, `FetchMess` to *read* in full, `SearchMess` to *find*, the flag verbs to *mark*, and `DeleteMess` + `ExpungeFolder` (or `CloseFolder`) to *remove*. `NoOp` keeps an idle connection from timing out. It is the same blocking, one-verb-at-a-time rhythm as the rest of the chapter — just with a folder's worth of state hanging off the client between calls.

TFTPSend — file transfer over two connections

TFTPSend (in `ftpsend`) is the classic FTP client, and it is the one client in this chapter with a twist the others do not have: FTP uses *two* sockets. The **control** connection carries commands and replies; a separate **data** connection carries each file or listing. TFTPSend manages both — it owns two `TTCPSock`s (`Sock` for control, `DSock` for data) — but the two-channel dance is mostly hidden behind ordinary-looking verbs. As always it descends from `TSynaClient`, so you set a few properties and call methods in order.

The class at a glance

Connection and credentials come from `TSynaClient`. Uniquely, the constructor pre-fills anonymous credentials (`UserName := 'anonymous', Password := 'anonymous@<localname>'`), so a public server needs nothing but a host. TFTPSend adds the FTP verbs plus the data-channel machinery:

```
TFTPSend = class(TSynaClient)
    function Login: Boolean;           // connect, greeting, AUTH TLS, logon sequence
    function Logout: Boolean;         // QUIT + close
    function FTPCommand(const Value: string): integer; // raw command -> reply code

    // transfers
    function RetrieveFile(const FileName: string; Restore: Boolean): Boolean; // RETR
    function StoreFile(const FileName: string; Restore: Boolean): Boolean;   // STOR
    function StoreUniqueFile: Boolean;                                       // STOU
    function AppendFile(const FileName: string): Boolean;                   // APPE
    function List(Directory: string; NameList: Boolean): Boolean;           // LIST/NLST/MLSD

    // filesystem
    function RenameFile(const OldName, NewName: string): Boolean;
    function DeleteFile(const FileName: string): Boolean;
    function FileSize(const FileName: string): int64; // SIZE
    function ChangeWorkingDir(const Directory: string): Boolean; // CWD
    function ChangeToParentDir: Boolean; // CDUP
    function ChangeToRootDir: Boolean;
    function DeleteDir(const Directory: string): Boolean; // RMD
    function CreateDir(const Directory: string): Boolean; // MKD
    function GetCurrentDir: String; // PWD
    function NoOp: Boolean;

    // low-level data channel (only for custom commands)
```

```

function DataRead(const DestStream: TStream): Boolean;
function DataWrite(const SourceStream: TStream): Boolean;
procedure Abort;
procedure TelnetAbort;

property ResultCode: Integer;           // numeric reply code of the last command
property ResultString: string;          // its main reply line
property FullResult: TStringList;      // every line of the reply
property Sock: TTCPBlockSocket;         // control connection
property DSock: TTCPBlockSocket;        // data connection
property DataStream: TMemoryStream;    // in-memory transfer buffer (default sink/source)
property DirectFile: Boolean;           // transfer straight to/from a disk file instead
property DirectFileName: string;
property FtpList: TFTPList;             // parsed directory listing after List(...)
property PassiveMode: Boolean;          // default True
property BinaryMode: Boolean;           // default True (TYPE I); False = ASCII
property CanResume: Boolean;            // set by Login if the server supports REST
property AutoTLS: Boolean;              // explicit FTPS (AUTH TLS)
property FullSSL: Boolean;              // implicit FTPS (TLS from connect)
property IsTLS: Boolean;                // control channel is encrypted
property IsDataTLS: Boolean;            // data channel is encrypted
property TLSONData: Boolean;            // default True: encrypt data too
property ForceDefaultPort: Boolean;
property ForceOldPort: Boolean;         // disable EPSV/EPRT (breaks IPv6)
property UseMLSDList: Boolean;
// plus firewall/proxy: FWHost, FWPort, FWUsername, FWPassword, FWMode, Account
end;

```

FTP reply codes are the real three-digit numbers, so `ResultCode` reads directly (200 OK, 226 transfer complete, 550 no such file). Internally almost every verb is `(FTPCommand(...) div 100) = 2` — “did the server answer in the 2xx family?” — which is why you can reason about `ResultCode` the same way the class does.

What Login does for you

Reading the source, `Login`:

1. Connects the control socket (immediate TLS handshake if `FullSSL` is set).
2. Reads the greeting, waiting past any 1xx preliminary lines for the real reply; a non-2xx greeting fails.
3. If `AutoTLS` is set, sends AUTH TLS and upgrades the control connection.
4. Runs the **logon sequence** — USER / PASS / ACCT — driven by `FWMode`. With no firewall (`FWHost` empty) it is the plain three-step login; the other modes thread the login through a firewall/proxy, and `FWMode := -1` runs a `CustomLogon` sequence you supply.
5. On a TLS session, negotiates data-channel protection (PBSZ 0, then PROT P if `TLSONData`, else PROT C).
6. Sets binary type (TYPE I), stream mode, and *probes* for resume support with REST — setting `CanResume` accordingly.

The data-connection model

Every transfer — a file or a directory listing — needs a fresh data connection, and there are two ways to open one. This is the single most important thing to understand about FTP, and

PassiveMode (default **True**) picks between them:

- **Passive (PassiveMode := True).** The client asks the server “what port should I connect to?” (PASV, or EPSV for IPv6), the server answers with an address, and the *client* opens the data connection. Because the client makes both outbound connections, passive mode sails through client-side firewalls and NAT — which is why it is the default and almost always the right choice.
- **Active (PassiveMode := False).** The client opens a listening socket and tells the server where to connect back (PORT, or EPRT for IPv6); the *server* dials in. This needs the client to be reachable from the server, so it often trips over NAT. ForceDefaultPort pins the listen to port 20; ForceOldPort disables the EPSV/EPRT extensions (which also disables IPv6).

You rarely touch any of this. The high-level verbs (RetrieveFile, StoreFile, List) open, use, and close the data connection internally via DataSocket / DataRead / DataWrite. Those three lower-level methods are exposed only for the rare case of implementing an unsupported command by hand.

DataStream vs. DirectFile

Where do the bytes go? Two modes, chosen by DirectFile:

- **DirectFile := False (default).** All transfers run through DataStream, an in-memory TMemoryStream. After RetrieveFile it holds the downloaded bytes (rewound to position 0); before StoreFile you fill it with what to upload. Simple, and fine for anything that fits in RAM.
- **DirectFile := True.** Transfers stream straight to or from a disk file named by DirectFileName — a TFileStream under the hood — so a multi-gigabyte download never has to fit in memory. This is the mode the convenience functions use.

FTPS: explicit and implicit

TLS is again the [SSL plugin seam](#) — add a plugin, then choose:

uses ftpsend, ssl_openssl;

- **Explicit FTPS (AUTH TLS).** Set AutoTLS := True on the standard port 21. Login upgrades the control connection after connecting, then negotiates data-channel encryption. This is the modern default for secure FTP.
- **Implicit FTPS.** Set FullSSL := True (historically port 990); TLS starts at connect time.
- **TLSONData** (default True) controls whether the *data* connection is encrypted too. IsTLS and IsDataTLS report what actually happened after Login.

Honest caveat. Plain FTP sends your password — and every byte of every file — in the clear. Prefer AutoTLS := True, and check IsTLS (and IsDataTLS if the payload is sensitive) before trusting the session.

Worked example: download a file

The rhythm is Login → (navigate) → RetrieveFile → Logout. Here streaming straight to disk with DirectFile:

program FtpDownload;

```

uses
    SysUtils, Classes,
    ftpsend, ssl_openssl;    // ssl_openssl enables AUTH TLS / FTPS

var
    Ftp: TFTPSTransfer;
begin
    Ftp := TFTPSTransfer.Create;
    try
        Ftp.TargetHost := 'ftp.example.com';
        // TargetPort defaults to 21; UserName/Password default to anonymous.
        Ftp.UserName := 'alice';
        Ftp.Password := 's3cret';
        Ftp.AutoTLS := True;                // explicit FTPS

        if not Ftp.Login then
            begin
                Writeln('Login failed (', Ftp.ResultCode, '): ', Ftp.ResultString);
                Halt(1);
            end;

        Ftp.ChangeWorkingDir('/pub/reports');

        // Stream the download straight to a local file.
        Ftp.DirectFileName := 'report.pdf';
        Ftp.DirectFile      := True;

        if Ftp.RetrieveFile('report.pdf', False) then    // Restore=False: fresh download
            Writeln('Downloaded. Server said: ', Ftp.ResultString)
        else
            Writeln('Download failed (', Ftp.ResultCode, '): ', Ftp.ResultString);

        Ftp.Logout;
    finally
        Ftp.Free;
    end;
end.

```

Set `Restore := True` and, if the server advertised resume support (`CanResume`), `RetrieveFile` picks up where an interrupted download left off — it seeks to the end of the existing local file and sends REST. If you would rather keep the bytes in memory, leave `DirectFile := False` and read them from `DataStream` after the call (remember it is already rewound to position 0).

Worked example: upload a file

`StoreFile` is the mirror image. In memory this time:

```

program FtpUpload;

uses
    SysUtils, Classes,
    ftpsend, ssl_openssl;

var

```

```

Ftp: TFTPSEND;
begin
  Ftp := TFTPSEND.Create;
  try
    Ftp.TargetHost := 'ftp.example.com';
    Ftp.UserName   := 'alice';
    Ftp.Password   := 's3cret';
    Ftp.AutoTLS    := True;

    if not Ftp.Login then
    begin
      Writeln('Login failed: ', Ftp.ResultString);
      Halt(1);
    end;

    // Fill DataStream with the payload (DirectFile stays False).
    Ftp.DataStream.LoadFromFile('local-notes.txt');

    if Ftp.StoreFile('notes.txt', False) then
      Writeln('Uploaded. Server said: ', Ftp.ResultString)
    else
      Writeln('Upload failed (', Ftp.ResultCode, '): ', Ftp.ResultString);

    Ftp.Logout;
  finally
    Ftp.Free;
  end;
end.

```

Related store verbs: AppendFile (APPE) appends to an existing remote file instead of overwriting, and StoreUniqueFile (STOU) lets the server assign a unique name. Set BinaryMode := False before a transfer to switch to ASCII mode (TYPE A, which translates line endings) — leave it at the default True for anything that is not plain text.

Listing a directory

List fetches a directory over the data connection and, for a full listing, also *parses* it:

```

var
  i: Integer;
  rec: TFTPListRec;
begin
  if Ftp.List('/pub', False) then           // NameList=False -> full parsed listing
    for i := 0 to Ftp.FtpList.Count - 1 do
    begin
      rec := Ftp.FtpList.Items[i];
      if rec.Directory then
        Writeln(Format('%-30s <dir>', [rec.FileName]))
      else
        Writeln(Format('%-30s %12d', [rec.FileName, rec.FileSize]));
      end;
    end;
  end;
end.

```

With NameList := False, List issues LIST (or MLSD if you set UseMLSDList := True) and parses

each line into `FtpList`, a `TFTPList` of `TFTPListRec` — each record exposing `FileName`, `FileSize`, `Directory`, `FileTime`, `Permission`, and the `OriginalLine`. Directory listing formats are a notorious swamp (Unix, Windows, Novell, VMS...), and Synapse ships a table of masks to cope; lines it cannot parse land in `FtpList.UnparsedLines`. With `NameList := True` it issues `NLST` and you get bare names in `DataStream`, no parsing. `MLSD` (RFC-3659) is the machine-readable modern format — prefer it when the server supports it, since it sidesteps the parsing guesswork entirely.

The convenience functions

For a one-shot transfer, `ftpsend` exposes three module-level helpers that create a `TFTPSend`, run the whole `Login/transfer/Logout`, and free it:

```
function FtpGetFile(const IP, Port, FileName, LocalFile, User, Pass: string): Boolean;
function FtpPutFile(const IP, Port, FileName, LocalFile, User, Pass: string): Boolean;
function FtpInterServerTransfer(
    const FromIP, FromPort, FromFile, FromUser, FromPass: string;
    const ToIP, ToPort, ToFile, ToUser, ToPass: string): Boolean;
```

`FtpGetFile` and `FtpPutFile` are the download/upload examples above in one line — they use `DirectFile` internally, so they stream to/from `LocalFile` on disk. Pass an empty `User` to log in anonymously. `FtpInterServerTransfer` wires two servers together with a `PASV/PORT` pair so bytes flow server-to-server without passing through your machine (the classic “FXP” transfer). Reach for the functions for the common case; drop to the object the moment you need `TLS` toggles, resume, directory navigation, or the parsed listing.

A note on `ftpsend` — a different protocol

Do not confuse `ftpsend` with **`ftpsend`** (class `TFTPSend`). Despite the near-identical name, `TFTP` — *Trivial* File Transfer Protocol (RFC-1350) — is a completely separate, minimal protocol that runs over **`UDP`**, with no login, no directory listing, and no security. `TFTPSend` is its client *and* server, with `SendFile/RecvFile` and a `Data` buffer, and it is used mostly for bootstrapping network devices, not general file transfer. If you mean ordinary `FTP`, you want `ftpsend` / `TFTPSend` — this chapter.

TDNSSend

DNS is the one protocol client in this chapter that does *not* speak a line-based request/response conversation over TCP. It is a single binary packet out, a single binary packet back, over UDP. TDNSSend (in dnssend) hides all of that wire encoding behind one method — DNSQuery — and hands you back the answers as plain strings, one resource record per line. There is no Login, no Logout, no session: you create the object, point it at a resolver, and ask.

The class at a glance

Like every client in this chapter, TDNSSend descends from TSynaClient, so the connection knobs (TargetHost, TargetPort, Timeout, IPInterface) come from the base. The constructor sets TargetPort to 53. What TDNSSend adds is one query method and a set of read-only result holders:

```
TDNSSend = class(TSynaClient)
    constructor Create;
    destructor Destroy; override;

    {: Query TargetHost for QType records matching Name. Answers land in Reply,
    one record per line; multi-field records are comma-delimited. }}
    function DNSQuery(Name: AnsiString; QType: Integer;
        const Reply: TStrings): Boolean;

    property Sock: TUDPBlockSocket; // the UDP socket (for OnStatus, tuning)
    property TCPSock: TTCPBlockSocket; // the TCP socket, used only when UseTCP
    property UseTCP: Boolean; // send over TCP instead of UDP
    property RCode: Integer; // DNS reply code: 0 ok, 3 name error, ...
    property Authoritative: Boolean; // answer came from an authoritative server
    property Truncated: Boolean; // reply was truncated (see "UDP vs TCP")
    property AnswerInfo: TStringList; // detailed answer-section records
    property NameserverInfo: TStringList; // detailed authority-section records
    property AdditionalInfo: TStringList; // detailed additional-section records
end;
```

The QType argument is one of the QTYPE_* constants the unit exports. The ones you will reach for:

Constant	Value	Record	Asks for
QTYPE_A	1	A	IPv4 address of a host
QTYPE_NS	2	NS	authoritative name servers for a zone
QTYPE_CNAME	5	CNAME	canonical name (alias target)
QTYPE_SOA	6	SOA	zone start-of-authority

Constant	Value	Record	Asks for
QTYPE_PTR	12	PTR	host name for an address (reverse lookup)
QTYPE_MX	15	MX	mail exchangers for a domain
QTYPE_TXT	16	TXT	free-form text (SPF, verification, ...)
QTYPE_AAAA	28	AAAA	IPv6 address of a host
QTYPE_SRV	33	SRV	service location
QTYPE_AXFR	252	—	full zone transfer (TCP only)
QTYPE_ALL	255	*	any/all records

The full list (QTYPE_MD, QTYPE_HINFO, QTYPE_KX, and the rest) is in the const block at the top of `dnssend.pas`.

How answers are shaped

DNSQuery returns True when it got a well-formed reply with RCode = 0, and fills Reply with the answer records. The encoding is deliberately flat: **one record per line**, and where a record has several fields they are joined with commas. An MX record with preference 10 for mail.example.com, for example, arrives as the single line:

```
10,mail.example.com
```

An SRV record comes back as priority,weight,port,target. A and AAAA records are just the address as text; PTR and CNAME are just the name. Numbers and IP addresses are always converted to their string form for you — you never decode raw bytes.

The three *Info string lists give you the *detailed* view of the reply’s three sections (answer, authority, additional). Each line there is a comma-delimited name,type,ttl,data tuple, so if you need the TTL or the record type number alongside the value, read AnswerInfo instead of Reply.

Reverse lookups are automatic

You do not build `.in-addr.arpa` names by hand. If the Name you pass to DNSQuery is a literal IP address, TDNSSend detects that and rewrites it into the correct reverse-DNS form for you — 1.2.3.4 becomes 4.3.2.1.in-addr.arpa, and an IPv6 literal becomes the matching `.ip6.arpa` name. So a PTR lookup is just “pass the address with QTYPE_PTR”; the class does the inversion.

UDP vs TCP — and the truncation caveat

By default TDNSSend sends the query as a **single UDP datagram** and reads a **single UDP packet** back. That is the right transport for ordinary lookups and what you almost always want.

Two things push you to TCP:

- **Zone transfers.** QTYPE_AXFR streams an entire zone and only works over TCP; DNSQuery has special multi-packet handling for it, but *only* when UseTCP is set.
- **Truncated replies.** A UDP answer that overflows is flagged by the server, and TDNSSend surfaces that in the read-only Truncated property.

Honest caveat — no automatic TCP fallback. Unlike a full stub resolver, TDNSSend does **not** silently retry a truncated UDP answer over TCP. It exposes Truncated and

leaves the decision to you: if you see it set, set `UseTCP := True` and call `DNSQuery` again to get the complete reply over the TCP socket. `UseTCP` simply switches which socket the query uses; there is no hybrid mode.

Worked example: resolve a host (A record)

```
program DnsA;

uses
  SysUtils, Classes,
  dnssend;

var
  Dns: TDNSSend;
  Reply: TStringList;
  i: Integer;
begin
  Dns := TDNSSend.Create;
  Reply := TStringList.Create;
  try
    Dns.TargetHost := '8.8.8.8';           // any resolver you trust

    if Dns.DNSQuery('www.example.com', QTYPE_A, Reply) then
      for i := 0 to Reply.Count - 1 do
        Writeln('A: ', Reply[i])         // each line is one IPv4 address
      else
        Writeln('query failed, RCode=', Dns.RCode);
    finally
      Reply.Free;
      Dns.Free;
    end;
  end.
```

`TargetHost` here is the *resolver* you send the question to (a recursive server such as your ISP's, 8.8.8.8, 1.1.1.1, ...), not the host you are asking about. That distinction is the whole mental model of the class: `TDNSSend` is a DNS *client* that talks to a *server*; it does not itself walk the delegation tree.

Worked example: look up mail exchangers (MX)

An MX query returns preference, hostname lines, and for real use you want them tried lowest-preference first. You can sort them yourself, but the unit already ships a convenience that does exactly that:

```
{: Query DNSHost for the MX records of Domain, and return the mail-server
   host names in Servers, already sorted by preference (best first) and with
   the preference numbers stripped. }
function GetMailServers(const DNSHost, Domain: AnsiString;
  const Servers: TStrings): Boolean;
```

Using it:

```
program MxLookup;
```

```

uses
    SysUtils, Classes,
    dnssend;

var
    Servers: TStringList;
    i: Integer;
begin
    Servers := TStringList.Create;
    try
        // DNSHost = resolver to ask; second arg = domain whose mail we want.
        if GetMailServers('8.8.8.8', 'example.com', Servers) then
            for i := 0 to Servers.Count - 1 do
                Writeln(i + 1, '. ', Servers[i]) // already in preference order
            else
                Writeln('no MX records found');
        finally
            Servers.Free;
        end;
    end.

```

If you want the raw preference numbers, skip the helper and call `DNSQuery(Domain, QTYPE_MX, Reply)` directly — each line is then `pref,host`, and you sort them yourself. `GetMailServers` is just that call plus a normalise-and-sort pass, and reading its (short) body in `dnssend.pas` is the best possible tutorial on driving `DNSQuery` for a multi-field record.

No TLS here

Note what is *absent* from this chapter: there is no `ssl_*` plugin, no `FullSSL`, no `AutoTLS`. Classic DNS over UDP/53 is unencrypted, and `TDNSSend` implements exactly that. DoT / DoH are different protocols and are not part of this class. If you need the transport authenticated or private, that is a layer above what `TDNSSend` provides.

The rest of the protocol clients

The chapters so far worked through the clients most programs reach for. Synapse ships several more, and every one of them follows the same shape you have already learned: a `TSynaClient` descendant, a socket exposed as `Sock`, a default port set by the constructor, and a small set of verbs. This chapter is a guided tour of six of them — NNTP, SNMP, SMTP, ping, telnet, and syslog — with enough of each to place it and a minimal example. When you need the details, the unit source is short and readable in every case.

Everything below is verified against the units in the Synapse source (one unit per protocol).

NNTP — Usenet news (nntpsend, TNNTPSend, port 119)

What it's for. Reading and posting Usenet articles: selecting a newsgroup, downloading articles by number or message-id, listing groups, and posting.

Key calls. Login / Logout bracket the session (the same connect-and-greet rhythm as the mail clients; Login also runs `AUTHINFO USER/PASS` if you set `UserName/Password`, and will `STARTTLS` when `AutoTLS` is set and the server advertises it). Then `SelectGroup(name)` switches group, `GetArticle(id)` / `GetBody(id)` / `GetHead(id)` download into the `Data` string list, and `PostArticle` posts the article you have placed in `Data`. Results come back as `ResultCode` / `ResultString`, with multi-line payloads in `Data`.

uses nntpsend;

var

 Nntp: TNNTPSend;

 i: Integer;

begin

 Nntp := TNNTPSend.Create;

try

 Nntp.TargetHost := 'news.example.com';

if Nntp.Login **then**

begin

if Nntp.SelectGroup('comp.lang.pascal.misc') **then**

 Writeln('group selected: ', Nntp.ResultString);

if Nntp.GetArticle('1') **then** *// article #1 -> Data*

for i := 0 **to** Nntp.Data.Count - 1 **do**

 Writeln(Nntp.Data[i]);

 Nntp.Logout;

end;

```

finally
    Nntp.Free;
end;
end.

```

TLS is available exactly as elsewhere: add an `ssl_*` unit and set `FullSSL` (implicit) or `AutoTLS` (`STARTTLS` upgrade). `StartTLS` is public if you want to drive the upgrade yourself.

SNMP — device monitoring (snmpsend, TSNMPSend, port 161)

What it's for. Reading and writing values (MIB objects, addressed by OID) on network devices — switches, printers, UPSes — using a *community string* as the access credential.

Key calls. The class `TSNMPSend` owns a `Query` and a `Reply` record (`TSNMPRec`, each carrying a `Community`, a `PDUType`, and a list of MIB entries), and `SendRequest` performs one round trip. But for the common case you never touch the class directly — the unit exports three convenience functions that build the record, send, and read the answer for you:

```

function SNMPGet(const OID, Community, SNMPHost: AnsiString;
    var Value: AnsiString): Boolean;
function SNMPGetNext(var OID: AnsiString; const Community, SNMPHost: AnsiString;
    var Value: AnsiString): Boolean;
function SNMPSet(const OID, Community, SNMPHost, Value: AnsiString;
    ValueType: Integer): Boolean;

```

Reading a single value is one line:

```

uses snmpsend, asnlutil;    // asnlutil for the ASN1_* value-type constants

var
    V: AnsiString;
begin
    // sysDescr.0 with the read community 'public'
    if SNMPGet('1.3.6.1.2.1.1.1.0', 'public', '192.168.1.1', V) then
        Writeln('sysDescr = ', V);

    // Writing needs the value's ASN.1 type, e.g. ASN1_OCTSTR for a string,
    // ASN1_INT for an integer:
    // SNMPSet('1.3.6.1.2.1.1.5.0', 'private', '192.168.1.1', 'newname', ASN1_OCTSTR);
end.

```

Honest caveat. These functions default to SNMP **v1/v2c**, whose community string is sent in clear text and is not real security. (`TSNMPRec` does carry v3 username/auth fields for the SNMPv3 path — the `Community` string is unused there — but that is well beyond a one-liner.) The `ValueType` argument to `SNMPSet` is one of the `ASN1_*` constants from `asnlutil` (`ASN1_INT` = \$02, `ASN1_OCTSTR` = \$04, `ASN1_OBJID` = \$06, ...).

The unit also offers `SNMPGetTable` / `SNMPGetTableElement` for walking MIB tables, and `SendTrap` / `RecvTrap` for the trap side (port 162).

Sntp — network time (sntp.send, TSntp.Send, port 123)

What it's for. Asking an NTP server for the current time. It is a single UDP exchange, so like DNS there is no login — you create, call, read.

Key call. GetSntp sends the request and, on success, fills NTptime (a TDateTime, in UTC) and the raw NTPReply record. GetNTP does the fuller four-timestamp exchange that also yields NTPOffset and NTPDelay; GetBroadcastNTP listens for a broadcast time packet. If you set SyncTime := True, a successful call will also set the local clock (subject to MaxSyncDiff) — which of course needs the privilege to do so.

```
uses sntpsend;

var
  Sntp: TSntpSend;
begin
  Sntp := TSntpSend.Create;
  try
    Sntp.TargetHost := 'pool.ntp.org';
    if Sntp.GetSntp then
      Writeln('server time (UTC): ', DateTimeToStr(Sntp.NTPTime))
    else
      Writeln('no reply');
  finally
    Sntp.Free;
  end;
end.
```

Ping — ICMP echo (pingsend, TPINGSend, no TCP port)

What it's for. Testing reachability and round-trip time with an ICMP echo (“ping”). Because ICMP is not TCP or UDP, this client uses a raw socket rather than a normal port.

Key call. Ping(host) sends one echo and returns True on a reply, filling PingTime (ms), ReplyFrom, and ReplyError. For a throwaway check the unit exports a convenience:

```
function PingHost(const Host: string): Integer; // ms, or -1 on failure/timeout
uses pingsend;

var
  Ms: Integer;
begin
  Ms := PingHost('example.com');
  if Ms >= 0 then
    Writeln('round trip: ', Ms, ' ms')
  else
    Writeln('no reply');
end.
```

Honest caveat — needs privileges. Raw ICMP sockets are privileged. On Linux this means running as root or granting the binary the cap_net_raw capability; on Windows it needs the analogous rights. Without them the raw socket cannot be created and the ping fails regardless of whether the host is up. (On Windows, TPINGSend

can fall back to the IP Helper IcmpSendEcho path internally, but the privilege point still stands.) The unit also exports TraceRouteHost for a traceroute-style probe.

Telnet — terminal sessions (tlntsend, TTelnetSend, port 23)

What it's for. Driving a line-oriented telnet server: connecting, waiting for prompts, sending commands, and reading the response. TTelnetSend handles telnet option negotiation (IAC sequences) transparently and accumulates everything it sees in SessionLog.

Key calls. Login connects (it is just a connect — telnet has no protocol login step; you authenticate by scripting the login prompt yourself). Then WaitFor(text) blocks until text appears in the stream, Send(text) sends input, and RecvString reads a CRLF-terminated line. Logout closes the socket.

```
uses tlntsend;

var
  Tel: TTelnetSend;
begin
  Tel := TTelnetSend.Create;
  try
    Tel.TargetHost := '192.168.1.1';
    if Tel.Login then
      begin
        Tel.WaitFor('login:');
        Tel.Send('admin' + #13#10);
        Tel.WaitFor('Password:');
        Tel.Send('secret' + #13#10);
        Tel.WaitFor('$');           // shell prompt
        Tel.Send('uptime' + #13#10);
        Writeln(Tel.RecvString);
        Tel.Logout;
      end;
    finally
      Tel.Free;
    end;
  end.
end.
```

Caveat. Telnet is plaintext — credentials and session both cross the wire unencrypted. TTelnetSend does expose an SSHLogin method that reuses the socket's SSL layer to negotiate an SSHv2 transport (via the libssh2 plugin), if you have that plugin in uses; plain Login gives you classic telnet.

Syslog — remote logging (slogsend, TSyslogSend, port 514, UDP)

What it's for. Shipping a log line to a syslog collector as a single UDP packet (RFC 3164, with RFC 5424 also supported by the message class).

Key call. TSyslogSend carries a SysLogMessage (a TSyslogMessage with Facility, Severity,

Tag/AppName, and LogMessage), and DoIt sends it. For a one-shot line the unit exports a convenience wrapper:

```
function ToSysLog(const SyslogServer: string; Facil: Byte;
  Sever: TSyslogSeverity; const Content: string): Boolean;
```

TSyslogSeverity is the RFC enum: Emergency, Alert, Critical, Error, Warning, Notice, Info, Debug. Facility is the numeric code (e.g. 1 = user, 16 = local0).

```
uses slogsend;
```

```
begin
```

```
  // facility 16 (local0), severity Info, one line to the collector
```

```
  ToSysLog('192.168.1.10', 16, Info, 'backup completed');
```

```
end.
```

Driving the class directly lets you set the Tag/AppName and other header fields before calling DoIt:

```
uses slogsend;
```

```
var
```

```
  Log: TSyslogSend;
```

```
begin
```

```
  Log := TSyslogSend.Create;
```

```
  try
```

```
    Log.TargetHost := '192.168.1.10';
```

```
    Log.SysLogMessage.Facility := 16;
```

```
    Log.SysLogMessage.Severity := Warning;
```

```
    Log.SysLogMessage.Tag := 'myapp';
```

```
    Log.SysLogMessage.LogMessage := 'disk 85% full';
```

```
    Log.DoIt;
```

```
  finally
```

```
    Log.Free;
```

```
  end;
```

```
end.
```

Note. Classic syslog over UDP/514 is fire-and-forget and unencrypted: DoIt returning True means the packet was sent, not that anything received it. That is inherent to the protocol, not a limitation of the client.

Where to go next

Every client here rewards a glance at its unit source — they are small, and each convenience function (GetMailServers, SNMPGet, PingHost, ToSysLog) is itself a worked example of driving the underlying class. Once the request / response rhythm from the [overview](#) is in your hands, none of these hold surprises; they differ only in the verbs of their protocol.

MIME & Mail Messages

TSMTPSend gets a message to the server, but it does not care what the message *is* — MailData takes a TStrings holding a complete RFC-822 message and sends it verbatim. Producing that TStrings correctly — headers with non-ASCII subjects, a text body, an HTML alternative, file attachments, inline images — is a separate job, and it belongs to a separate pair of units:

- **mimepart** — TMimePart, one node of a MIME tree. Knows how to encode and decode a single part (its headers, its transfer encoding, its body) and holds a list of child parts.
- **mimemess** — TMimeMess, the whole message: a TMessHeader (the mail headers) plus a root TMimePart (the body tree), with convenience methods that build the common part shapes for you.
- **mimeinln** — the inline (RFC-2047) header encoding, so a Subject: or a display name can carry accented characters. TMessHeader calls into it for you; you rarely call it directly.

Together they are the “what the message is” half of the story; TSMTPSend ([TSMTPSend — sending mail](#)) is the “getting it there” half. The seam between them is one property: TMimeMess.Lines, a TStringList, is exactly the TStrings that TSMTPSend.MailData expects.

A message is a tree of parts

The core idea in mimepart is that a MIME message is a **tree**. Each TMimePart is a node; a node is either a leaf (text, HTML, a binary attachment) or a multipart container whose children are more parts. A simple “body plus attachment” message is a three-node tree:

```
multipart/mixed      <- root container
├── text/plain        <- the body
└── application/pdf   <- the attachment
```

TMimePart (in mimepart) carries the properties of one node. The ones you touch most:

```
TMimePart = class(TObject)
```

```
published
```

```
property Primary: string;           // 'text', 'multipart', 'message', 'application'...
property Secondary: string;         // 'plain', 'mixed', 'html', 'pdf'...
property PrimaryCode: TMimePrimary; // MP_TEXT, MP_MULTIPART, MP_MESSAGE, MP_BINARY
property EncodingCode: TMimeEncoding; // ME_7BIT, ME_8BIT, ME_QUOTED_PRINTABLE, ME_BASE64, ME_...
property CharSetCode: TMimeChar;    // charset for text parts
property Disposition: string;       // 'inline' or 'attachment'
property ContentID: string;         // the cid: for inline images
property FileName: string;         // attachment file name
property Boundary: string;         // multipart delimiter (multipart parts only)

property Lines: TStringList;         // the raw, encoded part (headers + body)
property Headers: TStringList;      // this part's header lines
```

```

    property PartBody: TStringList;    // the still-encoded body
    property DecodedLines: TMemoryStream; // the decoded body bytes
end;

```

Two TStringList/stream properties are the crux of encode-vs-decode: **Lines** is the wire form (encoded), and **DecodedLines** is the raw bytes. Encoding goes DecodedLines -> Lines; decoding goes Lines -> DecodedLines. You put content into DecodedLines and call EncodePart; you read decoded content out of DecodedLines after calling DecodePart.

The four primary types are a closed set:

```

TMimePrimary = (MP_TEXT, MP_MULTIPART, MP_MESSAGE, MP_BINARY);
TMimeEncoding = (ME_7BIT, ME_8BIT, ME_QUOTED_PRINTABLE, ME_BASE64, ME_UU, ME_XX);

```

An unrecognised primary type decodes to MP_BINARY; an unrecognised transfer encoding decodes to ME_7BIT.

The message: TMimeMess

You rarely build the tree by hand. TMimeMess (in mimemess) owns the root part and a header object, and hands you AddPart... helpers that create a child node, set its type, load its content, and encode it in one call:

```

TMimeMess = class(TObject)
    function AddPart(const PartParent: TMimePart): TMimePart;
    function AddPartMultipart(const MultipartType: String; const PartParent: TMimePart): TMimePart;
    function AddPartText(const Value: TStringList; const PartParent: TMimePart): TMimePart;
    function AddPartTextEx(const Value: TStringList; const PartParent: TMimePart;
        PartCharset: TMimeChar; Raw: Boolean; PartEncoding: TMimeEncoding): TMimePart;
    function AddPartHTML(const Value: TStringList; const PartParent: TMimePart): TMimePart;
    function AddPartTextFromFile(const FileName: String; const PartParent: TMimePart): TMimePart;
    function AddPartHTMLFromFile(const FileName: String; const PartParent: TMimePart): TMimePart;
    function AddPartBinary(const Stream: TStream; const FileName: string; const PartParent: TMimePart): TMimePart;
    function AddPartBinaryFromFile(const FileName: string; const PartParent: TMimePart): TMimePart;
    function AddPartHTMLBinary(const Stream: TStream; const FileName, Cid: string; const PartParent: TMimePart): TMimePart;
    function AddPartHTMLBinaryFromFile(const FileName, Cid: string; const PartParent: TMimePart): TMimePart;
    function AddPartMess(const Value: TStringList; const PartParent: TMimePart): TMimePart;
    function AddPartMessFromFile(const FileName: string; const PartParent: TMimePart): TMimePart;

    procedure EncodeMessage; virtual;    // build MessagePart + Header into Lines
    procedure DecodeMessage; virtual;    // parse Lines into Header + MessagePart
published
    property MessagePart: TMimePart;    // the root of the part tree
    property Lines: TStringList;        // the raw RFC-822 message
    property Header: TMessHeader;       // From/To/Subject/...
end;

```

The PartParent argument is how you place a node in the tree. Pass nil and the part becomes the **root** (MessagePart itself); pass an existing multipart part and the new part is added as its child. So the pattern for a multipart message is: create a multipart/mixed root with AddPartMultipart('mixed', nil), then add each leaf with that root as parent.

What each helper produces (read from the source). AddPartText makes a text/plain, inline, quoted-printable part and picks an ideal charset for the content. AddPartHTML makes a text/html, inline, quoted-printable, UTF-8 part. AddPartBinary/...FromFile makes an attachment, base64 part and derives the MIME

type from the file extension. AddPartHTMLBinary makes an inline base64 part with a Content-ID (Cid) — the referent for a inside an HTML part. AddPartMess embeds a whole message/rfc822. Each helper calls EncodePart + EncodePartHeader before returning, so the part is fully encoded the moment you get it back.

The headers: TMessHeader

TMimeMess.Header is a TMessHeader holding the parsed mail headers as typed fields, not raw strings:

```
TMessHeader = class(TObject)
```

```
published
```

```
  property From: string;  
  property ToList: TStringList;           // one recipient per line  
  property CCList: TStringList;          // one CC per line  
  property Subject: string;  
  property Organization: string;  
  property ReplyTo: string;  
  property MessageID: string;  
  property Date: TDateTime;  
  property XMailer: string;  
  property Priority: TMessPriority;        // MP_unknown, MP_low, MP_normal, MP_high  
  property CharSetCode: TMimeChar;       // charset for encoding the headers  
  property CustomHeaders: TStringList;   // everything not parsed into a field above  
end;
```

Note the shape: From, Subject, ReplyTo are single strings, but ToList and CCList are TStringLists — you .Add one address per line rather than assigning a comma-joined string. On encode, TMessHeader.EncodeHeaders joins them with , and runs each through the inline encoder. It also fills in sensible defaults you did not set: a Date: of Now if you left it 0, a MIME-Version: 1.0, and an X-mailer: identifying Synapse. Anything the parser does not recognise on decode lands in CustomHeaders, readable with FindHeader / FindHeaderList.

Encodings: transfer encoding vs. inline header encoding

MIME has two distinct encoding problems, and Synapse keeps them in two places:

1. **Transfer encoding of part bodies** — making arbitrary bytes safe for a 7-bit mail path. This is EncodingCode on a part: ME_QUOTED_PRINTABLE for mostly-ASCII text (the default for AddPartText/AddPartHTML), ME_BASE64 for binary attachments (the default for AddPartBinary). The actual base64/quoted-printable routines live in synacode (see [Rolling its own crypto & encoding](#)) and are driven by TMimePart.EncodePart / DecodePart.
2. **Inline encoding of header text** — a Subject: or a display name cannot carry raw 8-bit bytes, so RFC-2047 wraps them as =?charset?B?...?= / =?charset?Q?...?= words. That is mimeinln:

```
function InlineDecode(const Value: string; CP: TMimeChar): string;  
function InlineEncode(const Value: string; CP, MimeP: TMimeChar): string;  
function InlineCode(const Value: string): string;  
function InlineCodeEx(const Value: string; FromCP: TMimeChar): string;  
function InlineEmail(const Value: string): string;  
function InlineEmailEx(const Value: string; FromCP: TMimeChar): string;  
function NeedInline(const Value: AnsiString): boolean;
```

TMessHeader calls InlineCodeEx (for Subject/Organization) and InlineEmailEx (for addresses, which must keep the <addr> un-encoded) on your behalf, so a Subject := 'Æblegrød' just works. You reach for these directly only when hand-building headers outside TMessHeader.

Where to go next

- [Building a message](#) — assemble a multipart message with a text body and a file attachment, then feed Lines to TSMTPSend.MailData.
 - [Parsing a message](#) — take a raw received message, decode it, walk the part tree, and pull out the text and the attachments.
-

Building a Message

The goal: a mail with a plain-text body **and** a file attachment, ready to hand to TSMTPSend. That is a two-leaf tree under a multipart/mixed root, and TMimeMess builds it in a handful of calls.

The shape you are building

```
multipart/mixed      <- AddPartMultipart('mixed', nil) -> root
├─ text/plain        <- AddPartText(Body, root)
└─ application/pdf   <- AddPartBinaryFromFile('report.pdf', root)
```

The rule from the source: to add **more than one** subpart you must have a multipart parent, and the first argument to every AddPart... helper is that parent. Pass nil for the root; pass the multipart part for each leaf.

Worked example

```
program BuildMail;

uses
  SysUtils, Classes,
  mimemess, mimepart,
  smtpsend, ssl_openssl;  // ssl_openssl enables STARTTLS / SMTPS

var
  Mime: TMimeMess;
  Body: TStringList;
  Root: TMimePart;
  SmtP: TSMTPSend;
begin
  Mime := TMimeMess.Create;
  Body := TStringList.Create;
  try
    // --- Headers -----
    // From / ReplyTo / Subject are single strings; ToList / CCList take one
    // address per line. Subject is inline-encoded automatically, so non-ASCII
    // is fine.
    Mime.Header.From := 'alice@example.com';
    Mime.Header.ToList.Add('bob@example.org');
    Mime.Header.CCList.Add('carol@example.org');
    Mime.Header.Subject := 'Quarterly report (Æblegrød included)';
    // Date, Message-ID, MIME-Version and X-mailer are filled in for you on
```

```

// encode if you leave them unset.

// --- Body tree -----
// A message with an attachment needs a multipart/mixed container as root.
Root := Mime.AddPartMultipart('mixed', nil);

// The text body: text/plain, inline, quoted-printable, ideal charset.
Body.Add('Hi Bob,');
Body.Add('');
Body.Add('The quarterly report is attached as a PDF.');
```

Body.Add('');

Body.Add('-- Alice');

Mime.AddPartText(Body, Root);

// The attachment: MIME type derived from the extension, base64, marked as
// an attachment with this file name.

Mime.AddPartBinaryFromFile('report.pdf', Root);

// --- Serialise -----

// EncodeMessage merges Header into the root part and composes the whole
// tree into Mime.Lines.

Mime.EncodeMessage;

// Mime.Lines is now a complete RFC-822 message. Send it.

Smtp := TSMTPSend.Create;

try

Smtp.TargetHost := 'smtp.example.com';

Smtp.TargetPort := '587';

Smtp.UserName := 'alice@example.com';

Smtp.Password := 's3cret';

Smtp.AutoTLS := **True**; // STARTTLS before AUTH

if not Smtp.Login **then**

raise Exception.Create('SMTP login failed: ' + Smtp.ResultString);

if Smtp.MailFrom('alice@example.com', Length(Mime.Lines.Text)) **and**

Smtp.MailTo('bob@example.org') **and**

Smtp.MailTo('carol@example.org') **and**

Smtp.MailData(Mime.Lines) **then** // <-- the seam: Lines -> MailData

Writeln('Sent: ', Smtp.ResultString)

else

Writeln('Send failed: ', Smtp.ResultString);

Smtp.Logout;

finally

Smtp.Free;

end;

finally

Body.Free;

Mime.Free;

end;

end.

What the calls actually did

- **AddPartMultipart('mixed', nil)** created the root part, set it to multipart/mixed, gave it a fresh Boundary (via GenerateBoundary), and encoded its header. Its return value is the parent you pass to the leaves.
- **AddPartText(Body, Root)** saved Body into the new part's DecodedLines, set text/plain, inline, quoted-printable, chose an ideal charset for the text, then called EncodePart + EncodePartHeader. The part comes back fully encoded.
- **AddPartBinaryFromFile('report.pdf', Root)** loaded the file into a memory stream, derived the MIME type from the extension (application/PDF from the built-in table; unknown extensions fall back to application/octet-string), set Disposition := 'attachment' and FileName, base64-encoded it, and built the header.
- **EncodeMessage** lifted the root part's Content-* headers up next to the From/To/Subject/... headers, composed every part into the wire form, and copied the result into Mime.Lines.

Variations

HTML with an inline image. Swap the text leaf for an HTML leaf and add the image as an inline (not attachment) part carrying a Content-ID:

```
Root := Mime.AddPartMultipart('related', nil);           // 'related' ties HTML + cid
Html := TStringList.Create;
Html.Add('<p>See the chart: </p>');
Mime.AddPartHTML(Html, Root);
Mime.AddPartHTMLBinaryFromFile('chart.png', 'chart1', Root); // Cid = 'chart1'
```

AddPartHTMLBinary... sets Disposition := 'inline' and ContentID := Cid; the HTML references it as cid:chart1. Use multipart/related (not mixed) so mail clients know the image belongs to the HTML body.

Content from files. AddPartTextFromFile and AddPartHTMLFromFile are the same as their in-memory versions but LoadFromFile the content for you.

Precise charset / encoding control. AddPartText picks the charset and forces quoted-printable. When you need to override either — a fixed charset, a Raw (no charset conversion) part, or base64 text — use AddPartTextEx:

```
Mime.AddPartTextEx(Body, Root, UTF_8, {Raw=}False, ME_BASE64);
```

Caveats

- **Build the tree before EncodeMessage.** EncodeMessage composes whatever is in MessagePart at call time; add all parts first, then encode once. Call it again only after changing the tree.
- **Envelope vs. header addresses are separate.** MailFrom/MailTo are the SMTP envelope; Header.From/Header.ToList are what the reader sees. They are usually the same, but Bcc works precisely by adding a MailTo with no matching header line — so do not expect a Header.CCList entry to reach a recipient unless you also MailTo them.
- **A single-part message needs no multipart.** If you only have a text body, skip AddPartMultipart and call AddPartText(Body, nil) to make the text the root directly. The multipart wrapper is only needed once you have two or more leaves.

Parsing a Message

Receiving is the mirror of building. You have the raw RFC-822 bytes (from TPOP3Send, TIMAPSend, an .eml file — anywhere), and you want the parsed headers, the readable text, and the attachments written to disk. TMimeMess decodes the message; walking the TMimePart tree pulls out the pieces.

The two-step decode

TMimeMess.DecodeMessage does two things, from the source:

1. Header.DecodeHeaders(Lines) — parses the mail headers into From, ToList, Subject, Date, ... (and everything else into CustomHeaders).
2. MessagePart.DecomposeParts — parses the body into the part tree.

Crucially, DecomposeParts **does not decode the part bodies** — it only splits the tree and reads each part's headers. Decoding a body (base64 → bytes, quoted-printable → text, charset conversion) is a separate DecodePart call per part, deliberately, so you don't pay to decode a 10 MB attachment you are going to skip. So the rhythm is: DecodeMessage once, then DecodePart on each leaf you actually want.

Walking the tree

Two ways to visit every node:

- **Manual recursion** with GetSubPartCount / GetSubPart(i) — full control, easy to read.
- **The WalkPart hook** — assign an OnWalkPart handler and Synapse calls it for the part and every descendant (it propagates the hook down the tree for you).

This example uses manual recursion.

Worked example

```
program ParseMail;
```

```
uses
```

```
  SysUtils, Classes,  
  mimemess, mimepart;
```

```
// Recursively visit every part; decode text into Text, save attachments.
```

```
procedure HandlePart(Part: TMimePart; const Text: TStrings);
```

```
var
```

```
  i: Integer;
```

```

begin
  case Part.PrimaryCode of
    MP_MULTIPART:
      // A container – recurse into its children, nothing to decode here.
      for i := 0 to Part.GetSubPartCount - 1 do
        HandlePart(Part.GetSubPart(i), Text);

    MP_MESSAGE:
      // An embedded message/rfc822 – it too has subparts; recurse.
      for i := 0 to Part.GetSubPartCount - 1 do
        HandlePart(Part.GetSubPart(i), Text);

    MP_TEXT:
      begin
        Part.DecodePart; // Lines -> DecodedLines (+ charset conversion)
        // Only fold plain text into the body; skip 'html' if you prefer.
        if LowerCase(Part.Secondary) = 'plain' then
          begin
            Part.DecodedLines.Position := 0;
            Text.LoadFromStream(Part.DecodedLines);
          end;
        end;

    MP_BINARY:
      begin
        // An attachment (or inline image). Decode and write it out.
        Part.DecodePart;
        if Part.FileName <> '' then
          begin
            Part.DecodedLines.Position := 0;
            Part.DecodedLines.SaveToFile('inbox_' + ExtractFileName(Part.FileName));
            Writeln('Saved attachment: ', Part.FileName,
              ' (', Part.DecodedLines.Size, ' bytes)');
          end;
        end;
      end;
  end;
end;

var
  Mime: TMimeMess;
  Text: TStringList;
begin
  Mime := TMimeMess.Create;
  Text := TStringList.Create;
  try
    // Load the raw message into Lines. (Here from a file; in practice this is
    // whatever POP3/IMAP handed you.)
    Mime.Lines.LoadFromFile('received.eml');

    // Step 1: parse headers + split the part tree (bodies NOT yet decoded).
    Mime.DecodeMessage;

    Writeln('From: ', Mime.Header.From);
    Writeln('Subject: ', Mime.Header.Subject);
  finally
    Mime.Free;
    Text.Free;
  end;
end;

```

```

Writeln('Date:      ', DateTimeToStr(Mime.Header.Date));
Writeln('To:        ', Mime.Header.ToList.CommaText);

// Step 2: walk the tree, decoding parts on demand.
HandlePart(Mime.MessagePart, Text);

Writeln('--- body ---');
Writeln(Text.Text);
finally
  Text.Free;
  Mime.Free;
end;
end.

```

Reading the results

- **Headers** are on Mime.Header as typed fields the moment DecodeMessage returns — From, Subject, ToList, Date, and so on. Anything the parser did not special-case (e.g. Received:, X-Spam-Score:) is in CustomHeaders; fetch it with Header.FindHeader('X-Spam-Score') or, for a header that repeats, Header.FindHeaderList('Received', List).
- **The body** is the MessagePart tree. For a simple message the root is the text part (PrimaryCode = MP_TEXT, GetSubPartCount = 0); for a multipart message the root is MP_MULTIPART and the leaves are its children. The recursion above handles both without a special case.
- **Distinguishing body from attachment** is by Disposition and FileName: an attachment has Disposition = 'attachment' and a non-empty FileName; an inline image has Disposition = 'inline' and a ContentID. The example keys off FileName being set, which catches both cases where there is a file to write.

Using the WalkPart hook instead

If you prefer a flat callback to explicit recursion, assign OnWalkPart on the root and call WalkPart — Synapse invokes your handler for the root and every descendant:

```

type
  TCollector = class
    procedure OnPart(const Sender: TMimePart);
  end;

procedure TCollector.OnPart(const Sender: TMimePart);
begin
  if (Sender.PrimaryCode = MP_BINARY) and (Sender.FileName <> '') then
  begin
    Sender.DecodePart;
    Sender.DecodedLines.Position := 0;
    Sender.DecodedLines.SaveToFile(ExtractFileName(Sender.FileName));
  end;
end;

// ...
Coll := TCollector.Create;
Mime.MessagePart.OnWalkPart := Coll.OnPart;
Mime.MessagePart.WalkPart; // fires OnPart for the root and all subparts

```

WalkPart only calls your hook if OnWalkPart is assigned, and it copies the hook onto each child before descending — so one assignment on the root covers the whole tree.

Caveats

- **DecomposeParts splits, DecodePart decodes.** Reading DecodedLines without first calling DecodePart on that part gives you nothing useful — the decoded stream is only populated by DecodePart. Always decode the leaf before you read its DecodedLines.
 - **Reset the stream position.** DecodedLines is a TMemoryStream; set Position := 0 before LoadFromStream/reading, or SaveToFile (which seeks itself) as shown.
 - **Malformed real-world mail.** Broken senders (the source singles out some Outlook versions) omit charset labels; DefaultCharset on the part governs how such text is decoded — adjust it before DecodePart if you see mojibake.
 - **Depth limiting.** MaxSubLevel on a part caps how deep DecomposeParts will recurse (default -1 = unlimited). Set it to guard against pathological deeply-nested messages.
 - **Binary receive path.** If the raw message came from THTTPSend (headers and an 8-bit body stream held separately, no transfer encoding applied), use DecodeMessageBinary(AHeader, AData) instead of DecodeMessage — it is the 8-bit-native equivalent.
-

TBlockSerial — the sockets model, applied to RS-232

Here is the quietly lovely thing about Synapse: when its authors needed a serial port library, they did not invent a new vocabulary. They took the *exact* mental model of TBlockSocket — blocking calls, millisecond timeouts, a LastError you check after each step, RecvString/RecvTerminated for line protocols — and pointed it at a UART instead of a TCP stream. The unit's own header says so:

This class provides numerous methods with same name and functionality as methods of the Ararat Synapse TCP/IP library.

If you have read the [TBlockSocket](#) chapter, you already know most of TBlockSerial. This chapter is about the parts that are genuinely different — opening a port instead of connecting to a host, configuring baud/bits/parity, and driving the modem control lines — and about how little else changes.

The unit is synaser. The one class you use is TBlockSerial.

The lifecycle

A serial port has the same explicit life as a socket, with Connect opening a device and CloseSocket closing it:

Create → Connect(device) → Config(...) → Send/Recv ... → CloseSocket → Free

- **constructor Create** — makes the object. It does *not* open a port yet.
- **procedure Connect(comport: string)** — opens the named device. The parameters are whatever the OS had configured for that port; call Config afterwards if you need specific settings.
- **procedure CloseSocket** — closes the handle. The destructor calls it, so Free is enough, but closing explicitly is good manners (and releases any lock file — see privileges below).

Note the name: it really is CloseSocket, not ClosePort. That is the design consistency showing through — the method names were kept identical to the socket library on purpose.

uses synaser;

```
var ser: TBlockSerial;  
begin  
  ser := TBlockSerial.Create;  
  try  
    ser.Connect('/dev/ttyUSB0');  
    if ser.LastError <> 0 then  
      begin
```

```

        WriteLn('open: ', ser.LastErrorDesc);
    Exit;
end;
ser.Config(9600, 8, 'N', SB1, False, False); // 9600 8N1, no flow control
{ ...talk to the device... }
finally
    ser.CloseSocket;
    ser.Free;
end;
end;

```

Naming the device — ‘COM1’ vs ‘/dev/ttyS0’

Connect takes a device *name*, and Synapse is bi-dialect about it. From the source documentation on Connect:

Comport can be used in Windows style (COM2), or in Linux style (/dev/ttyS1). When you use windows style in Linux, then it will be converted to Linux name. And vice versa!

So Connect('COM2') on Linux is translated to /dev/ttyS1 (the conversion is zero-based on the Unix side: COM1 → /dev/ttyS0, COM2 → /dev/ttyS1), and a /dev/ttyS1 on Windows maps back to a COM number. The translation lives in the GetComNr helper and only recognises the COMn and /dev/ttySn forms.

That last point matters in practice: **anything that is not a COMn or /dev/ttySn name is passed through unchanged.** USB-serial adapters (/dev/ttyUSB0), ACM/CDC modems (/dev/ttyACM0), and ARM on-chip UARTs (/dev/ttyAMA0) are *not* rewritten — you give the literal path and it is opened as-is. In modern life that is most of what you connect to, so the usual advice is simply: **on Unix, name the real device path; on Windows, use COMn.** The Device property tells you the actual OS name that ended up being opened.

To discover what is present, the unit exports a free function:

```
function GetSerialPortNames: string; // comma-separated list of port names
```

On Unix it globs /dev/ttyS*, /dev/ttyUSB*, /dev/ttyAM* and /dev/ttyACM*; on Windows it reads them from the registry’s SERIALCOMM map. Handy for populating a “choose a port” dropdown.

A privilege caveat (Unix)

Opening a serial device on Linux/Unix normally requires membership in the dialout group (or equivalent), because /dev/ttyS* and /dev/ttyUSB* are owned root:dialout. If Connect comes back with LastError set to a permission error, that is almost always the cause — add the user to dialout rather than running as root. Synapse can also create a lock file under /var/lock to cooperate with other programs using the port; this is governed by the LinuxLock property (default True), and the /var/lock directory must be writable for it. On Windows there is no such group; access is governed by whether another process already holds the port open.

Config — baud, bits, parity, stop bits, flow

One call sets every line parameter. You must be connected first:

```
procedure Config(baud, bits: integer; parity: char; stop: integer;
    softflow, hardflow: boolean); virtual;
```

- **baud** — bits per second, an *actual number* (not an enum): 9600, 19200, 115200, 460800, up to 4000000 on capable hardware. Internally Synapse maps it to the nearest supported Bxxx termios rate on Unix.
- **bits** — data bits per character, typically 8 (or 7, 6, 5).
- **parity** — a single character: 'N' none, 'O' odd, 'E' even, 'M' mark, 'S' space. (Case-insensitive.)
- **stop** — stop bits, using the named constants **SB1** (1 stop bit, value 0), **SB1andHalf** (1.5, value 1), or **SB2** (2, value 2).
- **softflow** — enable XON/XOFF software handshake.
- **hardflow** — enable RTS/CTS hardware handshake.

So the ubiquitous “9600 8N1, no flow control” is:

```
ser.Config(9600, 8, 'N', SB1, False, False);
```

and “115200 8N1 with hardware handshake” is:

```
ser.Config(115200, 8, 'N', SB1, False, True);
```

Config may be called again at any time to reconfigure a live port on the fly. Under the hood it fills a TDCB (Windows’ Device Control Block, *simulated* on Unix so the same code path serves both) and pushes it with SetCommState; the inverse GetCommState reads the current settings back. You rarely touch those two directly.

Sending — identical to the socket API

Every send method has the same name and shape as on TBlockSocket:

```
function SendBuffer(buffer: pointer; length: integer): integer; virtual;
procedure SendByte(data: byte); virtual;
procedure SendString(data: AnsiString); virtual;    // bytes as-is – NO terminator added
procedure SendInteger(Data: integer); virtual;    // four bytes
procedure SendBlock(const Data: AnsiString); virtual;    // 4-byte length prefix, then data
procedure SendStreamRaw(const Stream: TStream); virtual;
procedure SendStream(const Stream: TStream); virtual;
```

The same honest caveat applies as with the socket: **SendString appends no terminator**. The source is explicit — “No terminator is appended by this method. If you need to send a string with CR/LF terminator, you must append the CR/LF characters.” Because it adds nothing, it doubles as your binary-send path.

```
ser.SendString('AT' + CRLF);    // you supply the CRLF
ser.SendByte($02);             // a single control byte
```

CR, LF, and CRLF are declared right in synaser, so you do not need to pull them from elsewhere.

Receiving — the same three you already reach for

```
function RecvBuffer(buffer: pointer; length: integer): integer; virtual;
function RecvBufferEx(buffer: pointer; length, timeout: integer): integer; virtual;
function RecvBufferStr(Length, Timeout: Integer): AnsiString; virtual;
function RecvPacket(Timeout: Integer): AnsiString; virtual;
function RecvByte(timeout: integer): byte; virtual;
function RecvTerminated(Timeout: Integer; const Terminator: AnsiString): AnsiString; virtual;
```

```

function RecvString(Timeout: Integer): AnsiString; virtual;
function RecvInteger(Timeout: Integer): Integer; virtual;
function RecvBlock(Timeout: Integer): AnsiString; virtual;

```

Every one that can wait takes a **millisecond Timeout** and sets **LastError** to **ErrTimeout** if it lapses — exactly the blocking-with-timeouts contract from the sockets chapter, and exactly what makes serial code safe to write in a straight line. The three you will use most:

- **RecvString(Timeout)** — reads up to a CR/LF terminator and returns the line *without* it. This is the workhorse for line-oriented instruments and modems. It is **RecvTerminated(Timeout, CRLF)** under the hood, and it keeps its own internal buffer, so do not interleave it with raw **RecvBuffer** calls on the same port (the source warns this can lose data).
- **RecvTerminated(Timeout, Terminator)** — the same idea for devices that end each reply with something other than CR/LF: an ETX (#3), a '>' prompt, a '#', whatever your protocol uses. Returns the payload without the terminator.
- **RecvPacket(Timeout)** — “give me whatever bytes are waiting right now.” Ideal when the reply has no fixed framing and you just want to drain the port.

If a device uses lone-CR or lone-LF line ends rather than strict CR/LF, set **ConvertLineEnd := True** and **RecvString** will accept any of them. **MaxLineLength** caps how much **RecvString/RecvTerminated** will buffer before erroring (default 0 = unlimited) — worth setting when a misbehaving device might never send a terminator.

Readiness and buffers

Also mirrored from the socket class — you seldom need these because the **Recv*** timeouts already poll internally, but they are here:

```

function CanRead(Timeout: integer): boolean; virtual;           // data available to read
function CanWrite(Timeout: integer): boolean; virtual;         // room to write
function CanReadEx(Timeout: integer): boolean; virtual;        // also true if LineBuffer has data
function WaitingData: integer; virtual;                        // bytes ready to read now
function WaitingDataEx: integer; virtual;                      // ... including LineBuffer
function SendingData: integer; virtual;                        // bytes still queued to send
procedure Flush; virtual;                                       // wait until the output buffer drain
procedure Purge; virtual;                                       // discard both buffers immediately

```

Timeout = 0 on the **Can*** calls means “test and return now”; **-1** means “wait, possibly forever.”

Purge is the one to remember for serial work: after a timeout or a protocol error, **Purge** throws away any half-received garbage in both directions so your next exchange starts clean. **Flush** blocks until the transmit buffer has been handed to the hardware.

The LastError model — reports, not exceptions

Identical philosophy to the sockets. After any operation:

```

property LastError: integer read FLastError;           // 0 = success
property LastErrorDesc: string read FLastErrorDesc;    // human-readable

```

On success **LastError** is 0. On failure it is either an OS error code or one of synaser’s own constants, which are worth recognising:

Constant	Meaning
ErrTimeout (9997)	A read/write timed out
ErrPortNotOpen (9994)	Operation attempted with no open port
ErrAccessDenied (9990)	Could not open the device (permissions / in use)
ErrAlreadyInUse (9992)	Port is locked by another process
ErrAlreadyOwned (9991)	This process already owns the lock
ErrWrongParameter (9993)	Bad Config parameter
ErrFrame (9999)	Framing error on the line
ErrOvrerrun / ErrRxOvr (10000/10001)	Receiver overrun
ErrRxParity (10002)	Parity error

As with the sockets, if you prefer exceptions set `RaiseExcept := True` and a failing call raises `ESynSerError` (carrying `ErrorCode`); the default `False` keeps the linear check-after-each-step style.

Modem control lines — RTS, DTR, and the status inputs

This is the part that has no socket equivalent, and Synapse exposes the RS-232 control lines as plain properties. The two *outputs* you drive are **write-only**:

```
property RTS: Boolean write SetRTSF;    // Request To Send (output)
property DTR: Boolean write SetDTRF;    // Data Terminal Ready (output)
```

and the *inputs* you read are read-only:

```
property CTS: Boolean read GetCTS;      // Clear To Send
property DSR: Boolean read GetDSR;      // Data Set Ready
property Carrier: Boolean read GetCarrier; // DCD – carrier detect
property Ring: Boolean read GetRing;    // RI – ring indicator
```

By default, a successful `Connect` **raises DTR** (and RTS too, unless you asked for hardware handshake). If you need the lines left exactly as found — because you are bit-banging DTR/RTS yourself, e.g. to control a device's reset or power line — set **ManualSignals := True** *before* connecting. Then `Connect` and `Config` never touch RTS/DTR, and their values are entirely yours:

```
ser.ManualSignals := True;
ser.Connect('/dev/ttyUSB0');
ser.DTR := False;    // hold the target in reset
Sleep(50);
ser.DTR := True;     // release
```

Related odds and ends: `EnableRTSToggle(True)` puts the driver in half-duplex RTS-toggle mode for RS-485 transceivers; `SetBreak(Duration)` sends a line break for `Duration` ms; `ModemStatus` returns the raw status word the input properties decode. There is also a small AT-command helper layer (`ATCommand`, `ATConnect`, `ATResult`) for talking to Hayes modems, which is where this library was born.

A worked example — 9600 8N1, command and terminated reply

Putting it together: open a USB-serial device, configure it, send a command, and read one CR/LF-terminated line back — the whole exchange in linear, timeout-safe code that reads like the socket examples because it *is* the same shape.

```
uses synaser;

function QueryInstrument(const Device, Cmd: string): string;
var
  ser: TBlockSerial;
begin
  Result := '';
  ser := TBlockSerial.Create;
  try
    // Open the port. Unix: a literal /dev path; Windows: 'COM3'.
    ser.Connect(Device);
    if ser.LastError <> 0 then
      begin
        Writeln('open ', Device, ': ', ser.LastErrorDesc);
        Exit;
      end;

    // 9600 baud, 8 data bits, no parity, 1 stop bit, no flow control.
    ser.Config(9600, 8, 'N', SB1, False, False);
    if ser.LastError <> 0 then
      begin
        Writeln('config: ', ser.LastErrorDesc);
        Exit;
      end;

    // Some instruments need a moment after the line settles.
    ser.Purge; // start from a clean slate

    // Send a command. We supply the terminator ourselves, as always.
    ser.SendString(Cmd + CRLF);

    // Read one CR/LF-terminated reply, waiting up to 2 seconds.
    Result := ser.RecvString(2000);
    if ser.LastError = ErrTimeout then
      Writeln('no reply within 2 s')
    else if ser.LastError <> 0 then
      Writeln('read: ', ser.LastErrorDesc);
  finally
    ser.CloseSocket;
    ser.Free;
  end;
end;

// usage:
// Writeln(QueryInstrument('/dev/ttyUSB0', '*IDN?')); // Unix
// Writeln(QueryInstrument('COM3', '*IDN?')); // Windows
```

If your device does not frame replies with CR/LF, swap the one line:

```
Result := ser.RecvTerminated(2000, '>');    // read up to a '>' prompt
```

That is the entire story. The port-opening and line-configuration parts are new; everything after Config — SendString, RecvString/RecvTerminated, the Timeout argument, the LastError check, Purge — is the socket vocabulary you already speak, now driving a UART. Synapse chose one I/O model and applied it everywhere, and *that* consistency is the feature.

Encoding & Crypto

Synapse carries its own digests, MACs, ciphers, transfer encodings, and charset tables — no external dependency for any of them. The architecture chapter [Rolling Its Own Crypto & Encoding](#) explains *why* that choice was made and where it stops (TLS is delegated to a plugin, not reimplemented). This section is the practical companion: **which function to call, what it returns, and the honest caveats.**

The three units

- **synacode** — digests (MD5, MD4, SHA1), keyed MACs (HMAC_MD5, HMAC_SHA1), and the MIME transfer encodings: Base64, quoted-printable, URL triplet encoding, UU/XX/yEnc, and CRC-16/CRC-32. Everything here operates on `AnsiString` and returns `AnsiString` (raw bytes, not text — see below).
- **synacrypt** — symmetric block ciphers (DES, 3DES, AES) as a small class hierarchy rooted at `TSynaBlockCipher`, with ECB/CBC/CFB/OFB/CTR modes.
- **synachar** — charset detection and conversion between `TMimeChar` values (ISO-8859-*, the CP125x Windows codepages, UTF-8/16/32, and dozens more), optionally backed by `iconv` on non-Windows platforms via `synaicnv`.

You usually don't call these directly

The protocol clients call them for you. A few real examples from the upstream source:

- `TP0P3Send.AuthAP0P` computes `StrToHex(MD5(FTimeStamp + FPassword))` for you (`pop3send.pas`).
- `TSMTPSend` does CRAM-MD5 as `EncodeBase64(HMAC_MD5(challenge, FPassword))` and AUTH LOGIN as `EncodeBase64(FUsername)` (`smtpsend.pas`).
- `TMimePart` encodes and decodes bodies with `EncodeBase64` / `EncodeQuotedPrintable` based on the part's declared transfer encoding (`mimepart.pas`).
- `THTTPSend` runs `DecodeURL` over the credentials in a `user:pass@host` URL (`httpsend.pas`).
- `mimemess` / `mimeinln` reach into `synachar` to encode non-ASCII mail headers correctly.

So reach into these units directly when you are **speaking a protocol Synapse doesn't wrap**, decoding a blob by hand, or converting a charset the higher layers didn't handle for you. The rest of the time, let the client do it.

The recipes

- [Hashing & HMAC](#) — MD5/SHA1/HMAC_MD5/HMAC_SHA1, why the output is raw bytes, and how to hex it.

- [Transfer encodings](#) — Base64, quoted-printable, URL encoding, and the exact character sets each one escapes.
- [Charsets](#) — TMimeChar, GetCPFromID, CharsetConversion, and how non-ASCII mail/HTTP text survives a round trip.

One caveat up front

MD5, MD4, SHA-1, and single-DES live here because the *protocols* still specify them — APOP, CRAM-MD5, legacy SASL, and older cipher suites. They are **not** security primitives for new designs. Use them to speak the protocol that demands them; reach for the [SSL plugin seam](#) when you need real transport strength. This caveat is repeated on each recipe because it matters at each call site.

Hashing & HMAC

All the digest and MAC functions live in `synacode` and share one signature shape: `AnsiString` in, `AnsiString` out.

```
function MD5(const Value: AnsiString): AnsiString;
function MD4(const Value: AnsiString): AnsiString;
function SHA1(const Value: AnsiString): AnsiString;
function HMAC_MD5(Text, Key: AnsiString): AnsiString;
function HMAC_SHA1(Text, Key: AnsiString): AnsiString;
function MD5LongHash(const Value: AnsiString; Len: integer): AnsiString;
function SHA1LongHash(const Value: AnsiString; Len: integer): AnsiString;
```

(All verified in `synacode.pas` interface, lines 216–236.)

The output is raw bytes, not text

This is the one thing to internalize. `MD5('abc')` does **not** return the 32-character hex string you see in most tools — it returns the **16 raw digest bytes** packed into an `AnsiString`. `SHA1` returns 20 raw bytes, `HMAC_MD5` 16, `HMAC_SHA1` 20. Printing that `AnsiString` gives you binary garbage.

That is deliberate: for most protocol work you want the raw bytes so you can Base64-them or feed them onward. When you want the familiar hex, convert explicitly with `StrToHex` from `synautl`:

```
function StrToHex(const Value: Ansistring): string; // synautl.pas
```

It walks the bytes as `IntToHex(Byte(...), 2)` and lowercases the result, so you get the conventional lowercase hex digest.

```
uses
    synacode, synautl;

var
    raw, hex: AnsiString;
begin
    raw := MD5('The quick brown fox jumps over the lazy dog');
    hex := StrToHex(raw);
    // hex = '9e107d9d372bb6826bd81d3542a419d6'
    WriteLn(hex);
end;
```

The same pattern gives you a hex SHA-1:

```
WriteLn(StrToHex(SHA1('abc')));
// a9993e364706816aba3e25717850c26c9cd0d89d
```

HMAC — keyed message authentication

HMAC_MD5 and HMAC_SHA1 take the message as Text and the secret as Key (note: both are plain value parameters, in that order). They too return raw MAC bytes.

```
uses
    synacode, synautil;

var
    mac: AnsiString;
begin
    mac := HMAC_MD5('message-to-authenticate', 'shared-secret-key');
    WriteLn(StrToHex(mac));    // 32 hex chars
end;
```

Real protocol usage — you often don't call these yourself

The mail clients already do the right thing. Two examples straight from upstream:

APOP (pop3send.pas, TP0P3Send.AuthAPOP) — the server's timestamp is concatenated with the password, MD5'd, and hex-encoded:

```
s := StrToHex(MD5(FTimeStamp + FPassword));
Result := CustomCommand('APOP ' + FUserName + ' ' + s, False);
```

CRAM-MD5 (smtpsend.pas) — the Base64 challenge is decoded, HMAC-MD5'd with the password as key, and the raw MAC is re-encoded to Base64:

```
s := DecodeBase64(s);           // server challenge
s := HMAC_MD5(s, FPassword);    // raw 16-byte MAC
FSock.SendString(EncodeBase64(FUsername + ' ' + StrToHex(s)) + CRLF);
```

Notice the two different tail conversions: APOP wants **hex**, CRAM-MD5's MAC is carried as **hex inside a Base64 blob**. The raw-bytes return value is what lets each protocol format it its own way — which is exactly why the functions don't hex for you.

Long-hash variants

MD5LongHash and SHA1LongHash take a Len and produce a digest stream of that length by iterating the hash (a PRF-style key stretch used by a few protocols). They are niche; reach for them only when a spec names them.

The honest caveat

MD5, MD4, and SHA-1 are here for protocol compatibility, not security. APOP, CRAM-MD5, and legacy SASL still specify them, so Synapse implements them to speak those protocols. All three are broken for collision resistance (MD5 and SHA-1 demonstrably so), and HMAC-MD5 / HMAC-SHA1, while not collision-dependent, are not what you would pick for a new authentication scheme.

Do **not** use these as your application's password hashing, integrity, or signing primitives. For those, use a modern KDF/AEAD outside Synapse, and for transport strength use the [SSL plugin seam](#). Everything on this page exists to let old servers authenticate you, nothing more. See the architecture chapter [Rolling Its Own Crypto & Encoding](#) for the reasoning behind that scope.

Transfer Encodings

Base64, quoted-printable, and URL encoding all live in `synacode`, and all share the `AnsiString` → `AnsiString` shape. These are the encodings the mail and HTTP clients lean on constantly, and they are the ones you are most likely to call by hand when decoding a blob yourself.

Base64

```
function EncodeBase64(const Value: AnsiString): AnsiString;
function DecodeBase64(const Value: AnsiString): AnsiString;
```

(`synacode.pas`, interface lines 179–182.) Straightforward and byte-safe — encode any bytes, decode back to the exact same bytes:

```
uses
    synacode;

var
    enc, dec: AnsiString;
begin
    enc := EncodeBase64('Aladdin:open sesame');
    // enc = 'QWxhZGRpbjpvcGVuIHNlc2FtZQ=='
    dec := DecodeBase64(enc);
    // dec = 'Aladdin:open sesame'
end;
```

Internally `EncodeBase64` is `Encode3to4(Value, TableBase64)` and `DecodeBase64` is `Decode4to3Ex(Value, ReTableBase64)` — the standard RFC alphabet A–Z a–z 0–9 + / with = padding. `DecodeBase64` is tolerant of characters outside the alphabet (line breaks in wrapped Base64 are skipped), so you can feed it MIME-wrapped input directly.

There is also a URL-safe-ish variant pair, `EncodeBase64mod` / `DecodeBase64mod`, which uses + and , instead of + and / (the “modified” IMAP-style alphabet). Use it only when a spec calls for it.

This is the workhorse of SASL and MIME: `TSMTPSend` sends `EncodeBase64(FUsername)` for AUTH LOGIN, and `TMimePart` Base64-encodes binary attachment bodies.

Quoted-printable

```
function EncodeQuotedPrintable(const Value: AnsiString): AnsiString;
function DecodeQuotedPrintable(const Value: AnsiString): AnsiString;
function EncodeSafeQuotedPrintable(const Value: AnsiString): AnsiString;
```

(synacode.pas, interface lines 137, 149, 153.) Quoted-printable leaves ASCII text mostly readable and escapes only the bytes that need it, as =XX triplets:

```
uses
    synacode;

var
    s: AnsiString;
begin
    s := EncodeQuotedPrintable('Caf'#$E9' - 5'#$80);
    // non-ASCII bytes become =E9, =80, etc.; '=' itself becomes =3D
    WriteLn(s);
    WriteLn(DecodeQuotedPrintable(s)); // round-trips back to the bytes
end;
```

The two encoders differ in **which characters they escape**:

- EncodeQuotedPrintable escapes = plus all non-ASCII bytes (#0..#31, #127..#255). This is the normal body encoding.
- EncodeSafeQuotedPrintable escapes a wider SpecialChar set as well (() [] < > : ; , @ / ? \ " _ and =), producing text safe to drop into places where those punctuation characters are structural — e.g. RFC-2047 encoded-words in mail headers.

DecodeQuotedPrintable is just DecodeTriplet(Value, '=') — it reverses either encoder. TMimePart calls these for text parts declared quoted-printable.

URL encoding

```
function EncodeURL(const Value: AnsiString): AnsiString;
function EncodeURLElement(const Value: AnsiString): AnsiString;
function DecodeURL(const Value: AnsiString): AnsiString;
```

(synacode.pas, interface lines 162, 158, 140. Exact names confirmed — note EncodeURLElement, not EncodeURIComponent or similar.) Same triplet mechanism, %XX delimiter. The two encoders differ, and the difference matters:

- **EncodeURL** escapes only URLSpecialChar — control bytes, high bytes, and the characters that are unsafe *anywhere* in a URL (< > " % { } | \ ^ [] backtick, space, #\$7F..#\$FF). It **leaves the URL structure alone**: /, ?, :, @, &, =, +, ;, # all pass through. Use it to clean up a whole URL without breaking it.
- **EncodeURLElement** escapes URLSpecialChar **plus** URLFullSpecialChar (; / ? : @ = & # +) — i.e. the reserved delimiters too. Use it to encode a single component (a query-string value, a path segment) that must not be interpreted as structure.

```
uses
    synacode;

begin
    WriteLn(EncodeURL('http://h/a b?x=1'));
    // EncodeURL keeps ':/?=' → http://h/a%20b?x=1

    WriteLn(EncodeURLElement('a b?x=1'));
    // EncodeURLElement escapes the reserved chars too → a%20b%3F%3D1
end;
```

httpsend.pas documents that the value put into a request field “must be encoded by the EncodeURLElement function,” and it runs DecodeURL over the userinfo it parses out of a

user:pass@host URL. DecodeURL reverses both encoders (it just undoes every %XX).

No security caveat here — but a correctness one

These are encodings, not ciphers: they provide **zero** confidentiality or integrity. Base64 is not encryption. The only pitfall is **round-tripping**: pick the right encoder for the context (EncodeURL for a whole URL vs EncodeURLElement for one component; plain vs safe quoted-printable), or you will either over-escape and corrupt structure, or under-escape and break the message. For where these sit in the larger design, see [Rolling Its Own Crypto & Encoding](#).

Charsets

synachar is Synapse's charset layer: it knows a large set of encodings, can guess which one a byte string is, and can convert between any two of them. This is what keeps an accented Subject: line or a Latin-1 HTTP body from turning into mojibake. The mail units (mimemess, mimeinln, mimepart) call into it for you; you touch it directly when you are decoding non-ASCII text the higher layers handed you raw.

Charsets are a TMimeChar enum, not numeric IDs

The set of supported charsets is the enumerated type TMimeChar (synachar.pas, line 101). Each member is the charset identifier — there is no separate idNN constant table; you name charsets by these enum values:

type

```
TMimeChar = (ISO_8859_1, ISO_8859_2, ISO_8859_3, ISO_8859_4, ISO_8859_5,
  ISO_8859_6, ISO_8859_7, ISO_8859_8, ISO_8859_9, ISO_8859_10, ISO_8859_13,
  ISO_8859_14, ISO_8859_15, CP1250, CP1251, CP1252, CP1253, CP1254, CP1255,
  CP1256, CP1257, CP1258, KOI8_R, CP895, CP852, UCS_2, UCS_4, UTF_8, UTF_7,
  UTF_7mod, UCS_2LE, UCS_4LE,
  // the following are supported through iconv only:
  UTF_16, UTF_16LE, UTF_32, UTF_32LE, C99, JAVA, ISO_8859_16, KOI8_U, ...);
```

(The list continues with the Mac charsets, the CJK encodings — EUC_JP, SHIFT_JIS, GB2312, BIG5, EUC_KR — and more. See the full declaration in synachar.pas.)

Two important companion sets in the same unit:

- IconvOnlyChars — charsets that only work when iconv is present (via synaicnv; non-Windows). On a build without iconv these are unavailable.
- NoIconvChars — [CP895, UTF_7mod], the two Synapse handles with its own internal routines regardless of iconv.

Name ↔ enum: GetCPFromID / GetIDFromCP

Wire formats carry charset *names* ("utf-8", "iso-8859-1", "windows-1252"), so you convert between the string label and the enum:

```
function GetCPFromID(Value: AnsiString): TMimeChar; // "iso-8859-2" -> ISO_8859_2
function GetIDFromCP(Value: TMimeChar): AnsiString; // ISO_8859_2 -> "ISO-8859-2"
```

(synachar.pas, interface lines 213, 216.) GetCPFromID is generous: it uppercases the input and matches against a large alias table (so "LATIN1", "CP819", "ISO8859-1" all resolve to ISO_8859_1), with special-cased names for CP895/Kamenicky and modified UTF-7. Feed it the charset token straight out of a MIME Content-Type or HTTP charset= parameter.

```

uses
    synachar;

var
    cs: TMimeChar;
begin
    cs := GetCPFromID('windows-1252');    // -> CP1252
    WriteLn(GetIDFromCP(cs));             // canonical name back
end;

```

Converting: CharsetConversion

The core conversion is:

```

function CharsetConversion(const Value: AnsiString;
    CharFrom: TMimeChar; CharTo: TMimeChar): AnsiString;

```

(synachar.pas, interface line 191.) Give it the bytes, the charset they are *in*, and the charset you want them *in*:

```

uses
    synachar;

var
    latin1, utf8: AnsiString;
begin
    latin1 := 'Caf'#$E9;                                // 'Café' in ISO-8859-1
    utf8 := CharsetConversion(latin1, ISO_8859_1, UTF_8);
    // utf8 now holds the UTF-8 bytes 'Caf' + C3 A9
end;

```

If you have a charset *name* rather than an enum, resolve it first:

```

from := GetCPFromID(headerCharsetName);
utf8 := CharsetConversion(rawBody, from, UTF_8);

```

The richer variants

```

function CharsetConversionEx(const Value: AnsiString; CharFrom: TMimeChar;
    CharTo: TMimeChar; const TransformTable: array of Word): AnsiString;

```

```

function CharsetConversionTrans(Value: AnsiString; CharFrom: TMimeChar;
    CharTo: TMimeChar; const TransformTable: array of Word;
    Translit: Boolean): AnsiString;

```

(interface lines 196, 202.) CharsetConversionEx applies an extra character replacement table during conversion — pass Replace_None for no substitution, or Replace_Czech (both declared in synachar) to strip Czech diacritics. CharsetConversionTrans adds a Translit flag: when False, unconvertible characters are *not* transliterated to approximations. Plain CharsetConversion is CharsetConversionEx with Replace_None and transliteration on.

Detection helpers

```

function NeedCharsetConversion(const Value: AnsiString): Boolean;
function GetCurCP: TMimeChar;

```

```

function GetCurOEMCP: TMimeChar;
function IdealCharsetCoding(const Value: AnsiString; CharFrom: TMimeChar;
    CharTo: TMimeSetChar): TMimeChar;

```

(interface lines 219, 206, 210, 223.)

- NeedCharsetConversion returns True only when the string has non-7-bit-ASCII bytes — a cheap gate so you skip conversion for pure-ASCII text.
- GetCurCP / GetCurOEMCP report the operating system's current charset (and the OEM/DOS-box charset on Windows), useful as a CharFrom when reading platform-local text.
- IdealCharsetCoding picks, from a candidate TMimeSetChar, the target charset that encodes Value with the fewest unconvertible characters — this is how the mail encoder chooses the smallest adequate charset for an outgoing header.

uses

synachar;

begin

```

if NeedCharsetConversion(body) then
    body := CharsetConversion(body, GetCPFromID(declaredCharset), UTF_8);
    // pure-ASCII bodies pass through untouched

```

end;

iconv: the reach beyond the built-in tables

Many of the TMimeChar members (everything in IconvOnlyChars — the CJK encodings, UTF-16/32, the Mac charsets) are handled by delegating to the system iconv through synaicnv. On a platform without iconv, or with the global DisableIconv flag set, those charsets are unavailable and conversions involving them will fail; the built-in ISO-8859-*, CP125x, KOI8, UTF-7/8, and UCS conversions always work. If you support arbitrary inbound charsets, test on your target platform with iconv actually present.

No security dimension — just correctness

Charset conversion is neither encryption nor sanitization: it changes byte *representation*, not meaning, and provides no safety guarantees. The failure mode is silent corruption — pass the wrong CharFrom, or run the bytes through conversion twice, and you get mojibake or lost characters. Always convert from the charset the data actually declares (from the MIME/HTTP header via GetCPFromID), convert once, and prefer UTF_8 as your internal target. For where this fits the overall design, see [Rolling Its Own Crypto & Encoding](#).

The Utility Units

Under every Synapse protocol class sits a thin layer of plain functions — no objects, no state, just helpers that the protocol code (and you) call directly. They are the parts of Synapse you end up reaching for even when you never open a socket: split a header line, format an RFC-822 date, validate an IP address, encode an ASN.1 packet, ask the OS which DNS server it uses.

This chapter covers the four units that make up that layer:

Unit	What it is for
<code>synautil.pas</code>	The general toolbox: string splitting, date/time, hex/binary, headers
<code>asn1util.pas</code>	BER/DER encode & decode — the bytes under LDAP and SNMP
<code>synaip.pas</code>	IP-address parsing and formatting (IPv4 and IPv6)
<code>synamisc.pas</code>	Host/system info: DNS servers, local IPs, proxy settings, Wake-on-LAN

How they fit together

None of these units depend on a socket. `synautil` depends on nothing but the RTL; `synaip` uses `synautil`; `asn1util` is self-contained; `synamisc` reaches into the OS. That independence is the point — the protocol classes are built *on top* of these functions, so anything a protocol can do to a string or an address, you can do too, in isolation, without instantiating anything.

- **synautil is the workhorse.** Almost every other unit in Synapse uses it. When you parse a MIME header, decode a `Date:` field, or Fetch tokens off a server response, you are calling `synautil`. It is the one to learn first.
- **asn1util is a translation layer.** ASN.1 BER is how LDAP and SNMP put structured data on the wire. The unit is small — a handful of `ASNEnc*` / `ASNItem` functions — because BER itself is a small, recursive grammar. You rarely call it directly (the LDAP and SNMP classes do), but understanding it demystifies both protocols.
- **synaip is pure conversion.** No I/O — it turns `'192.168.0.1'` into an integer and back, validates addresses, expands short-form IPv6. It never resolves a name (that is DNS, and lives elsewhere); `IsIP` explicitly rejects symbolic names.
- **synamisc asks the operating system questions.** Its answers (the system's DNS servers, every local IP, the configured proxy) come from Windows registry / IP Helper calls or from `/etc/resolv.conf`-style probing on POSIX, so the same function name returns platform-specific truth.

A note on strings

Everything here works with `AnsiString` — byte strings, not UTF-16. Synapse treats the wire as bytes and these helpers do too: `StrToHex`, `ASNObject`, `IPToID`, and the `Fetch` family all move

raw octets. That is deliberate and it is why the same code compiles on FPC and every Delphi from 7 up.

The rest of the chapter: [the synautil toolbox](#), [ASN.1 with asnlutil](#), and [IP and host helpers](#).

The synutil Toolbox

synutil.pas is the single most-used non-socket unit in Synapse. It is a flat collection of functions — no classes — that the protocol code leans on for string surgery, date formatting, and byte packing. This page walks the ones you will actually reach for. Every signature below is copied from the unit's interface section.

Add it to your uses and call directly:

```
uses synutil;
```

Splitting strings

The two most-called functions in the whole unit split a string on a delimiter:

```
{:Returns a portion of the "Value" string located to the left of the "Delimiter" string. If a delimiter is not found, results is original string.}
```

```
function SeparateLeft(const Value, Delimiter: string): string;
```

```
{:Returns the portion of the "Value" string located to the right of the "Delimiter" string. If a delimiter is not found, results is original string.}
```

```
function SeparateRight(const Value, Delimiter: string): string;
```

Both search for the *first* occurrence of Delimiter. If it is absent, SeparateLeft returns the whole string and SeparateRight also returns the whole string — so pair them carefully.

```
host := SeparateLeft('example.org:443', ':'); // 'example.org'
port := SeparateRight('example.org:443', ':'); // '443'
```

```
// No delimiter present:
```

```
SeparateLeft('plainhost', ':'); // 'plainhost'
SeparateRight('plainhost', ':'); // 'plainhost'
```

GetBetween — respects nesting

```
{:Get string between PairBegin and PairEnd. This function respect nesting.}
```

```
function GetBetween(const PairBegin, PairEnd, Value: string): string;
```

Unlike a naive left/right split, GetBetween matches balanced pairs:

```
GetBetween('(', ')', 'Hi! (hello(yes!))'); // 'hello(yes!)'
```

The Fetch family — consume tokens left to right

Fetch is destructive: it removes the returned token *and* the delimiter from Value (a var parameter), so you can loop until the string is empty.

```
{:Fetch string from left of Value string.}
function Fetch(var Value: string; const Delimiter: string): string;

{:Like fetch, but working with binary strings, not with text.}
function FetchBin(var Value: string; const Delimiter: string): string;

{:Fetch string from left of Value string. This function ignore delimiters inside quotations.}
function FetchEx(var Value: string; const Delimiter, Quotation: string): string;
```

Fetch trims surrounding spaces from the token; FetchBin does not (it is for binary data where every byte matters); FetchEx skips delimiters that fall inside a quoted run.

```
var
    s, tok: string;
begin
    s := 'alice, bob, carol';
    while s <> '' do
        begin
            tok := Fetch(s, ',');    // 'alice', then 'bob', then 'carol'
            Writeln(tok);
        end;
    end;
```

Trimming — spaces only

Standard Trim also strips control characters. Synapse's variants remove *only* spaces, which matters when a protocol field may legitimately end in a tab or a control byte:

```
{:Like TrimLeft, but remove only spaces, not control characters!}
function TrimSPLeft(const S: string): string;
{:Like TrimRight, but remove only spaces, not control characters!}
function TrimSPRight(const S: string): string;
{:Like Trim, but remove only spaces, not control characters!}
function TrimSP(const S: string): string;
```

Dates and times

Synapse speaks the date formats the protocols require. The two you will use most are the RFC-822 formatter (for mail and HTTP headers) and the general decoder.

```
{:Returns current time in format defined in RFC-822. ... (Example 'Fri, 15 Oct 1999 21:14:56 +0200')}
function Rfc822DateTime(t: TDateTime): string;

{:Same as Rfc822DateTime, but GMT timezone is used.}
function Rfc822DateTimeGMT(t: TDateTime): string;
```

```
{:Returns date and time in format defined in RFC-3339 "yyyy-mm-ddThh:nn:ss.zzz"}
function Rfc3339DateTime(t: TDateTime): string;
```

Formatting a header value:

```
Writeln('Date: ', Rfc822DateTime(Now));
// Date: Sat, 11 Jul 2026 14:03:22 +0200
```

Going the other way, `DecodeRfcDateTime` is the forgiving parser — it accepts RFC-822/1123, RFC-850, and C `asctime()` shapes, and applies the timezone correction so the result is always a comparable `TDateTime`:

```
{:Decode various string representations of date and time to TDateTime type.
This function do all timezone corrections too!}
function DecodeRfcDateTime(Value: string): TDateTime;

dt := DecodeRfcDateTime('Sun, 06 Nov 1994 08:49:37 GMT');
dt := DecodeRfcDateTime('Sunday, 06-Nov-94 08:49:37 GMT');
dt := DecodeRfcDateTime('Sun Nov 6 08:49:37 1994'); // all three parse
```

Supporting pieces you can call on their own:

```
{:Return your timezone bias from UTC time in minutes.}
function TimeZoneBias: integer;
{:Return your timezone bias ... in string representation like "+0200".}
function TimeZone: string;
{:Decode three-letter month name to month number (0 if no match).}
function GetMonthNumber(Value: String): integer;
{:Decode time from string with ':' separator ("hh:mm" or "hh:mm:ss").}
function GetTimeFromStr(Value: string): TDateTime;
{:Decode TimeZone string (CEST, GMT, +0200, ...) to offset.}
function DecodeTimeZone(Value: string; var Zone: integer): Boolean;
```

Header lists

When you have a `TStringList` of raw `Name: value` header lines, these convert in place to and from the `Name=value` form that `TStringList.Values` understands:

```
{:Convert lines in stringlist from 'name: value' form to 'name=value' form.}
procedure HeadersToList(const Value: TStrings);
{:Convert lines in stringlist from 'name=value' form to 'name: value' form.}
procedure ListToHeaders(const Value: TStrings);
```

Related, for pulling a single parameter out of a header value:

```
{:Returns parameter value from string in format:
parameter1="value1"; parameter2=value2}
function GetParameter(const Value, Parameter: string): string;

charset := GetParameter('text/html; charset="utf-8"', 'charset'); // 'utf-8'
```

Email address extraction

```
{:Returns only the e-mail portion of an address from the full address format.}
function GetEmailAddr(const Value: string): string;
{:Returns only the description part from a full address format.}
function GetEmailDesc(Value: string): string;
```

```
GetEmailAddr('"someone" <nobody@somewhere.com>'); // 'nobody@somewhere.com'
GetEmailDesc('"someone" <nobody@somewhere.com>'); // 'someone'
```

Hex, binary, and byte packing

```
{:Returns hexadecimal digits for the bytes in "Value".}
function StrToHex(const Value: AnsiString): string;
{:Returns a string of binary "Digits" representing "Value".}
function IntToBin(Value: Integer; Digits: Byte): string;
{:Returns an integer equivalent of the binary string in "Value".}
function BinToInt(const Value: string): Integer;
```

The Code*/Decode* pair packs integers into big-endian byte strings — handy for binary protocols:

```
{:Return two characters representing the value in byte format. (High-endian)}
function CodeInt(Value: Word): AnsiString;
{:Decodes two characters at "Index" of "Value" to a Word.}
function DecodeInt(const Value: AnsiString; Index: Integer): Word;
{:Return four characters representing the value in byte format. (High-endian)}
function CodeLongInt(Value: LongInt): AnsiString;
{:Decodes four characters at "Index" of "Value" to a LongInt.}
function DecodeLongInt(const Value: AnsiString; Index: Integer): LongInt;

StrToHex(CodeInt(258)); // '0102' (two big-endian bytes)
DecodeInt(CodeInt(258), 1); // 258 (Index is 1-based)
```

Line-terminator and quoting helpers

```
{:Return position of string terminator in string. Possible terminators are:
CRLF, LFCR, CR, LF. The one found is returned in Terminator.}
function PosCRLF(const Value: AnsiString; var Terminator: AnsiString): integer;

{:Remove quotation from Value string. If not quoted, returns it unchanged.}
function UnquoteStr(const Value: string; Quote: Char): string;
{:Quote Value string. If Value contains Quote chars, they are doubled.}
function QuoteStr(const Value: string; Quote: Char): string;

{:Replaces all "Search" found within "Value" with "Replace".}
function ReplaceString(Value, Search, Replace: AnsiString): AnsiString;
{:Like POS, but from the right side of Value.}
function RPos(const Sub, Value: String): Integer;
{:Return count of Chr in Value string.}
function CountOfChar(const Value: string; Chr: char): integer;
```

Note on CRLF. The CR, LF, and CRLF constants themselves are not declared in synautil — they live in blksock.pas (CRLF = CR + LF). synautil gives you PosCRLF to *find* any of the four line terminators in a buffer, but for the literal constant you uses blksock.

URL parsing

```
{:Parses a URL to its various components.}
function ParseURL(URL: string; var Prot, User, Pass, Host, Port, Path,
    Para: string): string;

var Prot, User, Pass, Host, Port, Path, Para: string;
begin
    ParseURL('http://user:pw@example.org:8080/path?q=1',
        Prot, User, Pass, Host, Port, Path, Para);
    // Prot='http' User='user' Pass='pw' Host='example.org'
    // Port='8080' Path='/path' Para='q=1'
end;
```

Streams and temp files

```
{:Read a string of "len" bytes from a stream.}
function ReadStrFromStream(const Stream: TStream; len: integer): AnsiString;
{:Write a string to a stream.}
procedure WriteStrToStream(const Stream: TStream; Value: AnsiString);
{:Return a unique temp-file name in "Dir" with the given "prefix".}
function GetTempFile(const Dir, prefix: String): String;
```

IP validation lives next door

synautil does *not* define IsIP / IsIP6 — those are in [synaip](#), which synautil does not even use. If you want to validate an address, reach for that unit.

When to use what

You want to...	Call
Split host:port	SeparateLeft / SeparateRight
Loop tokens off a list, consuming as you go	Fetch
Pull text between balanced brackets	GetBetween
Format a header date	Rfc822DateTime
Parse any RFC date string	DecodeRfcDateTime
Read charset= out of a Content-Type	GetParameter
Turn Name: value lines into Values[]	HeadersToList
Pack an integer big-endian for a binary wire	CodeInt / CodeLongInt
Hex-dump a byte string	StrToHex
Break a URL into pieces	ParseURL

ASN.1 BER with asnlutil

LDAP and SNMP do not send text. They send **ASN.1 BER** — a compact, recursive, type-length-value binary encoding. `asnlutil.pas` is Synapse's implementation of that encoding: a small set of functions that turn Pascal values into BER bytes and back. The LDAP (`ldapsend.pas`) and SNMP (`snmpsend.pas`) classes are built directly on it; you rarely call it yourself, but knowing it is what turns those protocols from magic into plumbing.

uses `asnlutil`;

The TLV idea

Every ASN.1 element is three parts laid end to end:

+-----+-----+-----+-----+			
Type	Length	Contents	
1 byte	1+ bytes	Length bytes	
+-----+-----+-----+-----+			

- **Type** — one byte naming the kind of value (integer, string, sequence...).
- **Length** — how many content bytes follow. Short lengths (< 128) fit in one byte; longer ones use a multi-byte form flagged by the high bit.
- **Contents** — the value, whose interpretation depends on Type. For the *constructed* types (like SEQUENCE) the contents are themselves more TLV elements — that recursion is the whole grammar.

The type constants

`asnlutil` defines the BER type tags it understands:

```
ASN1_BOOL      = $01;
ASN1_INT       = $02;
ASN1_OCTSTR    = $04;    // octet string
ASN1_NULL      = $05;
ASN1_OBJID     = $06;    // object identifier (an OID)
ASN1_ENUM      = $0a;
ASN1_SEQ       = $30;    // sequence (constructed)
ASN1_SETOF     = $31;    // set-of (constructed)
ASN1_IPADDR    = $40;    // SNMP application types below
ASN1_COUNTER   = $41;
ASN1_GAUGE     = $42;
ASN1_TIMETICKS = $43;
ASN1_OPAQUE    = $44;
ASN1_COUNTER64 = $46;
```

The \$40-range tags are SNMP's application-specific types; the rest are universal ASN.1.

Encoding

The workhorse is `ASNObject` — it wraps any content in its `Type` and `Length`, producing a finished element:

```
{:Encodes ASN.1 object to binary form.}  
function ASNObject(const Data: AnsiString; ASNTYPE: Integer): AnsiString;
```

Its whole body is the TLV definition made literal:

```
Result := AnsiChar(ASNTYPE) + ASNEncLen(Length(Data)) + Data;
```

You build the *content* with the typed encoders, then wrap it:

```
{:Encodes a signed integer to ASN.1 binary}  
function ASNEncInt(Value: Int64): AnsiString;  
{:Encodes unsigned integer into ASN.1 binary}  
function ASNEncUInt(Value: Integer): AnsiString;  
{:Encodes the length of an ASN.1 element to binary.}  
function ASNEncLen(Len: Integer): AnsiString;  
{:Encodes an OID item to binary form.}  
function ASNEncOIDItem(Value: Int64): AnsiString;
```

Encoding an integer element:

```
var  
    elem: AnsiString;  
begin  
    elem := ASNObject(ASNEncInt(42), ASN1_INT);  
    // bytes: $02 $01 $2A (type=INT, length=1, value=42)  
    WriteLn(StrToHex(elem)); // '02012a' (StrToHex from synautil)  
end;
```

Because SEQUENCE content is just concatenated elements, you nest by wrapping a concatenation:

```
var  
    body, seq: AnsiString;  
begin  
    body := ASNObject(ASNEncInt(1), ASN1_INT)  
        + ASNObject('hello', ASN1_OCTSTR);  
    seq := ASNObject(body, ASN1_SEQ);  
    // seq is a SEQUENCE containing an INTEGER and an OCTET STRING  
end;
```

Decoding

Decoding walks the buffer with a **var Start cursor** — each call advances it past the element it read, so you decode a sequence by calling repeatedly:

```
{:Beginning with the "Start" position, decode the ASN.1 item of the next element  
in "Buffer". Type of item is stored in "ValueType."}  
function ASNItem(var Start: Integer; const Buffer: AnsiString;  
    var ValueType: Integer): AnsiString;
```

```
{:Decodes length of next element in "Buffer" from the "Start" position.}
```

```
function ASNDecLen(var Start: Integer; const Buffer: AnsiString): Integer;
{:Decodes an OID item of the next element from the "Start" position.}
function ASNDecOIDItem(var Start: Integer; const Buffer: AnsiString): Int64;
```

ASNItem returns the decoded value as a string and reports its BER type through ValueType. Start is **1-based** (Pascal string indexing):

```
var
    pos, vt: Integer;
    value: AnsiString;
begin
    pos := 1;
    value := ASNItem(pos, elem, vt);
    // for the '02012a' buffer above:
    //   vt   = ASN1_INT ($02)
    //   value = '42'   (integer types come back as their decimal string)
    //   pos   = 4      (advanced past the element)
end;
```

For a SEQUENCE, first decode the outer element (which for constructed types returns the raw inner bytes), then loop ASNItem over those with a fresh cursor.

OIDs and MIBs

SNMP object identifiers are dotted-number MIB strings like 1.3.6.1.2.1.1.1.0. asnlutil converts between the string and its BER binary form:

```
{:Encodes an MIB OID string to binary form.}
function MibToId(Mib: String): AnsiString;
{:Decodes MIB OID from binary form to string form.}
function IdToMib(const Id: AnsiString): String;

var
    oid: AnsiString;
begin
    oid := MibToId('1.3.6.1.2.1.1.1.0'); // BER-encoded OID bytes
    Writeln(IdToMib(oid));               // '1.3.6.1.2.1.1.1.0' (round-trip)
end;
```

To place an OID on the wire, wrap it as an ASN1_OBJID element:

```
oidElem := ASNObject(MibToId('1.3.6.1.2.1.1.1.0'), ASN1_OBJID);
```

Debugging

When a decode goes wrong, dump the buffer in human-readable form:

```
{:Convert ASN.1 BER encoded buffer to human readable form for debugging.}
function ASNdump(const Value: AnsiString): AnsiString;
```

ASNdump walks the TLV structure and prints each element's type and value — the fastest way to see what an LDAP or SNMP packet actually contains.

Summary

You want to...	Call
Wrap content as a typed element	ASNObject(data, ASN1_*)
Encode a signed / unsigned integer	ASNEncInt / ASNEncUInt
Encode an OID from a MIB string	MibToId
Read the next element from a buffer	ASNItem(Start, buf, vt)
Turn an OID back into dotted form	IdToMib
Inspect a packet while debugging	ASNdump

You will most often *see* these functions inside `ldapsend.pas` and `snmpsend.pas` rather than call them — but every LDAP filter and every SNMP varbind on the wire is exactly this: `ASNObject` calls nested inside an `ASN1_SEQ`, unwound on the far end by `ASNItem`.

IP Addresses and Host Info

Two small units cover the address end of Synapse. `synaip.pas` is pure conversion — strings to addresses and back, no I/O. `synamisc.pas` asks the operating system what it knows: your DNS servers, your local IPs, your proxy. Between them they answer “is this a valid address?”, “what does it look like in binary?”, and “what is my machine configured to use?”.

synaip — parse and format addresses

uses synaip;

This unit does no networking. It never resolves a name — `IsIP` explicitly rejects symbolic hostnames — it only transforms address *text*.

Validation

```
{:Returns TRUE, if "Value" is a valid IPv4 address. Cannot be a symbolic Name!}
function IsIP(const Value: string): Boolean;
{:Returns TRUE, if "Value" is a valid IPv6 address. Cannot be a symbolic Name!}
function IsIP6(const Value: string): Boolean;

IsIP('192.168.0.1');    // True
IsIP('example.org');    // False - a name, not an address
IsIP6('::1');           // True
IsIP6('192.168.0.1');   // False - that is IPv4
```

A common pattern is to decide whether you already have an address or still need to resolve a name:

```
if IsIP(target) or IsIP6(target) then
    // use it directly
else
    // hand it to DNS
```

IPv4 conversion

```
{:Convert IPv4 address from their string form to binary.}
function StrToIp(value: string): integer;
{:Convert IPv4 address from binary to string form.}
function IpToStr(value: integer): string;

n := StrToIp('192.168.0.1');    // packed into an integer
s := IpToStr(n);                // '192.168.0.1' (round-trip)
```

IPv6 conversion

IPv6 addresses are 16 bytes, so they go through a byte-array type, `TIp6Bytes = array [0..15] of Byte`:

```
{:Convert IPv6 address from their string form to binary byte array.}
function StrToIp6(value: string): TIp6Bytes;
{:Convert IPv6 address from binary byte array to string form.}
function Ip6ToStr(value: TIp6Bytes): string;
{:Expand short form of IPv6 address to long form.}
function ExpandIP6(Value: AnsiString): AnsiString;
```

`ExpandIP6` turns the compressed `::` form into the full eight-group form — useful when you need to compare two addresses textually:

```
ExpandIP6('2001:db8::1');
// '2001:0DB8:0000:0000:0000:0000:0000:0001'
```

Reverse form (for PTR / rDNS)

```
{:Returns a string with the "Host" ip address converted to binary form.}
function IPToID(Host: string): AnsiString;
{:Convert IPv4 address to reverse form.}
function ReverseIP(Value: AnsiString): AnsiString;
{:Convert IPv6 address to reverse form.}
function ReverseIP6(Value: AnsiString): AnsiString;
```

`ReverseIP` produces the reversed-octet order that reverse-DNS lookups use:

```
ReverseIP('192.168.0.1'); // '1.0.168.192'
```

`IPToID` returns the raw binary (byte-string) form of an address — the shape DNS records and some protocol fields carry it in.

synamisc — what the OS knows

```
uses synamisc;
```

Where `synaip` is pure math, `synamisc` reaches into the operating system. The same function name returns platform-specific truth: on Windows it reads the registry and IP Helper API; on POSIX it probes the system's resolver configuration.

DNS servers

```
{:Autodetect current DNS servers used by the system. If more than one DNS server
is defined, then the result is comma-delimited.}
function GetDNS: string;

Writeln('System DNS: ', GetDNS);
// e.g. '192.168.0.1,8.8.8.8'
```

This is the function to call when you want to do your own DNS query but need to know which server to ask — Synapse's own resolver uses it for exactly that.

Local IP addresses

```
{:Return all known IP addresses on the local system. Addresses are  
comma-delimited.}  
function GetLocalIPs: string;  
{:Return all known IP addresses of the required type on the local system.}  
function GetLocalIPsFamily(value: TSocketFamily): string;  
  
GetLocalIPs returns every address the host has; GetLocalIPsFamily filters by family  
(TSocketFamily, e.g. IPv4-only or IPv6-only):  
  
Writeln('This host: ', GetLocalIPs);  
// e.g. '127.0.0.1,192.168.0.42,::1'
```

Proxy settings

Proxy discovery is Windows-oriented (it reads Internet Explorer / WinHTTP config) and returns a record:

```
TProxySetting = record  
  Host: string;  
  Port: string;  
  Bypass: string;  
  ResultCode: integer;  
  Autodetected: boolean;  
end;  
  
{:Read Internet Explorer 5.0+ proxy setting for given protocol. Windows only!}  
function GetIEProxy(protocol: string): TProxySetting;  
  
{IFDEF MSWINDOWS}  
{:Autodetect system proxy setting for a specified URL. Windows only!}  
function GetProxyForURL(const AURL: WideString): TProxySetting;  
{ENDIF}
```

GetProxyForURL is compiled only on Windows (**{IFDEF MSWINDOWS}**); on other platforms it does not exist, so guard your calls with the same define.

Wake-on-LAN

A genuinely miscellaneous member — send the magic packet that powers on a sleeping machine:

```
{:Turn on a network computer that supports Wake-on-LAN. You need the MAC  
address; you can also give a target IP (broadcast is used if omitted).}  
procedure WakeOnLan(MAC, IP: string);  
  
WakeOnLan('00:11:22:33:44:55', ''); // broadcast on the local network
```

What is *not* here

Neither unit resolves a hostname to an address — there is no GetIPByName. Name resolution is DNS, and in Synapse that lives in the socket layer (synsock / blksock) and the DNS client (dnssend.pas), not in these helpers. synaip validates and converts addresses you already have; synamisc tells you which DNS server *would* answer such a query. The lookup itself is a socket's job.

Summary

You want to...	Call	Unit
Check a string is a literal IPv4 / IPv6	IsIP / IsIP6	synaip
Pack / unpack an IPv4 address	StrToIp / IpToStr	synaip
Pack / unpack an IPv6 address	StrToIp6 / Ip6ToStr	synaip
Expand :: shorthand	ExpandIP6	synaip
Build a reverse-DNS name	ReverseIP / ReverseIP6	synaip
Find the system's DNS servers	GetDNS	synamisc
List this host's own IPs	GetLocalIPs	synamisc
Read the configured proxy (Windows)	GetIEProxy / GetProxyForURL	synamisc
Power on a machine by MAC	WakeOnLan	synamisc

Recipes

This is the section you copy from. Each recipe is a **single common task** — download a file, send a mail, resolve a hostname — answered with a minimal but complete pascal snippet you can drop into a program, compile, and adjust. No narrative build-up; just problem, code, and the one gotcha that bites people.

How to use this section

- **Each recipe is self-contained.** The snippet names every unit it needs in uses and checks status the Synapse way (LastError / a Boolean result / ResultCode). Paste it, change the host and data, compile.
- **HTTPS/TLS is a uses line.** Wherever a recipe talks to an https:// URL or a TLS mail server, add `ssl_openssl` (or `ssl_openssl3`) to uses — that one line registers the SSL plugin globally and nothing else in the code changes. See [the SSL plugin seam](#).
- **Snippets are trimmed for clarity.** Error handling is shown once per pattern, not belt-and-braces on every line. For the *why* behind a class, and its full property surface, follow the cross-link to its chapter.
- **The Synapse rhythm.** Create the object, call one method, check status, read the results off properties, free the object. Every recipe is a variation on that.

The recipes

Web — HTTP with THTTPSend and the httpsend one-liners. - Download a URL to a file - POST form data - Fetch JSON over HTTPS

Email — sending with TSMTPSend/TMimeMess, reading with TPOP3Send. - Send a plain-text mail - Send a mail with a file attachment - Fetch and list an inbox

Network tools — DNS, ICMP, and raw sockets. - Resolve a hostname (TDNSSend) - Ping a host (TPINGSend / PingHost) - A minimal TCP client and a threaded echo server

Serial — talking to a serial port with TBlockSerial. - Open a port and do a request/response

Where the depth lives

These recipes are the fast path. When you need the full API — headers, cookies, keep-alive, streaming receives, the blocking model — the chapters have it:

- [HTTP — THTTPSend](#)
- [SMTP — TSMTPSend](#)
- [POP3 — TPOP3Send](#)
- [MIME messages](#)

- [Socket classes](#)
 - [Serial ports](#)
 - [Synapse in Two Hours](#) — the guided on-ramp
-

Recipes — Web

HTTP tasks with `httpsend`. The module-level helpers (`HttpGetText`, `HttpGetBinary`, `HttpPostURL`) cover the common cases in one line; drop to `THTTPSend` when you need headers or the status code. Full API in [THTTPSend — talking HTTP](#).

Download a URL to a file

Problem: fetch a remote resource (image, archive, anything binary) and save it to disk.

```
program Download;

{$mode objfpc}{$H+}

uses
  Classes,
  httpsend, ssl_openssl;  // ssl_openssl -> https:// works

var
  FileStream: TFileStream;
begin
  FileStream := TFileStream.Create('synapse.zip', fmCreate);
  try
    if HttpGetBinary('https://example.com/synapse.zip', FileStream) then
      WriteLn('Saved ', FileStream.Size, ' bytes')
    else
      WriteLn('Download failed');
  finally
    FileStream.Free;
  end;
end.
```

`HttpGetBinary(const URL: string; const Response: TStream): Boolean` streams the body straight into any `TStream`, so it never buffers the whole file in memory. **Gotcha:** a `True` return only means a reply arrived — for a broken link the server may hand you a 404 page as the body. If that matters, use the object and check `ResultCode` (next recipes).

POST form data

Problem: submit an HTML form (`application/x-www-form-urlencoded`).

```
program PostForm;
```

```

{$mode objfpc}{$H+}

uses
  Classes,
  httpsend, synacode, synautil;    // synacode: EncodeURLElement; synautil: ReadStrFromStream

var
  Reply: TMemoryStream;
  FormData: string;
begin
  Reply := TMemoryStream.Create;
  try
    // key=value&key=value, with every value percent-escaped.
    FormData := 'user=' + EncodeURLElement('Alice') +
      '&city=' + EncodeURLElement('São Paulo');

    if HttpPostURL('http://httpbin.org/post', FormData, Reply) then
    begin
      Reply.Position := 0;
      WriteLn(ReadStrFromStream(Reply, Reply.Size));
    end
    else
      WriteLn('POST failed');
  finally
    Reply.Free;
  end;
end.

```

HttpPostURL(const URL, URLData: string; const Data: TStream): Boolean sets the Content-Type to application/x-www-form-urlencoded, sends URLData as the body, and returns the response in Data. **Gotcha:** always run each value through EncodeURLElement (from synacode) — an unescaped & or space in a field corrupts the form.

Fetch JSON over HTTPS

Problem: call a JSON API with custom headers and read both the status code and the body.

program FetchJson;

```

{$mode objfpc}{$H+}

uses
  Classes,
  httpsend, ssl_openssl;

var
  Http: THTTPSend;
  Body: TStringList;
begin
  Http := THTTPSend.Create;
  Body := TStringList.Create;
  try
    Http.Headers.Add('Accept: application/json');
    Http.Headers.Add('Authorization: Bearer my-token');
  end;
end.

```

```

if Http.HTTPMethod('GET', 'https://api.example.com/status') then
begin
  WriteLn('Status: ', Http.ResultCode, ' ', Http.ResultString);
  Http.Document.Position := 0;           // rewind before reading
  Body.LoadFromStream(Http.Document);    // Document is a TMemoryStream
  WriteLn(Body.Text);
end
else
  WriteLn('Transport failed: ', Http.Sock.LastErrorDesc);
finally
  Body.Free;
  Http.Free;
end;
end.

```

HTTPMethod returns True for *any* reply, including 404/500 — branch on Http.ResultCode for the real outcome. **Gotcha:** Document is a TMemoryStream, not text; set Position := 0 before LoadFromStream, or you'll read zero bytes. To POST JSON, write the body into Document, set Http.MimeType := 'application/json', and call HTTPMethod('POST', URL).

Recipes — Email

Sending with `smtpsend`, composing MIME with `mimemess`, reading with `pop3send`. Depth in [SMTP — TSMTPSend](#), [POP3 — TPOP3Send](#), and [MIME messages](#).

Send a plain-text mail

Problem: fire off one text email without touching the SMTP object.

```
program SendMail;

{$mode objfpc}{$H+}

uses
  Classes, smtpsend;

var
  Msg: TStringList;
begin
  Msg := TStringList.Create;
  try
    Msg.Add('Hello from Synapse.');
```

```
    Msg.Add('This is the body.');
```

```
    // SendTo(MailFrom, MailTo, Subject, SMTPHost, MailData): Boolean
    if SendTo('me@example.com', 'you@example.com',
              'Test message', 'mail.example.com', Msg) then
      WriteLn('Sent')
    else
      WriteLn('Send failed');
```

```
  finally
    Msg.Free;
  end;
end.
```

`SendTo` builds the headers, connects, and delivers in one call. **Gotcha:** it takes **no** username/password — it's for open/relay-by-IP servers. For authenticated SMTP use `SendToEx(..., Username, Password)`, or the `TSMTPSend` object (`FullSSL := True` for SMTPS on port 465, or `ssl_openssl` in uses + `STARTTLS` for 587).

Send a mail with an attachment

Problem: build a multipart message with a text body and a file attachment, then hand it to TSMTPSend.

```
program SendAttachment;

{$mode objfpc}{$H+}

uses
  Classes, mimemess, mimepart, smtpsend;

var
  Mime: TMimeMess;
  Body: TStringList;
  SMTP: TSMTPSend;
begin
  Mime := TMimeMess.Create;
  Body := TStringList.Create;
  try
    Body.Add('Report attached.');

    Mime.Header.From := 'me@example.com';
    Mime.Header.ToList.Add('you@example.com');
    Mime.Header.Subject := 'Monthly report';

    Mime.AddPartText(Body, nil); // the text body
    Mime.AddPartBinaryFromFile('report.pdf', nil); // the attachment
    Mime.EncodeMessage; // -> Mime.Lines

    SMTP := TSMTPSend.Create;
    try
      SMTP.TargetHost := 'mail.example.com';
      SMTP.Username := 'me@example.com';
      SMTP.Password := 'secret';
      if SMTP.Login and
        SMTP.MailFrom('me@example.com', Length(Mime.Lines.Text)) and
        SMTP.MailTo('you@example.com') and
        SMTP.MailData(Mime.Lines) then
        WriteLn('Sent')
      else
        WriteLn('SMTP error: ', SMTP.ResultString);
      SMTP.Logout;
    finally
      SMTP.Free;
    end;
  finally
    Body.Free;
    Mime.Free;
  end;
end.
```

Gotcha: you must call Mime.EncodeMessage **before** reading Mime.Lines — that's the step that renders the parts into the encoded message. The recipient list on the envelope (MailTo) is separate from the To: header (Header.ToList); set both. For an authenticated/TLS server,

add `ssl_openssl` to uses and set `SMTP.FullSSL := True` (or `STARTTLS`) as the SMTP chapter shows.

Fetch and list an inbox

Problem: log into POP3, report how many messages are waiting, and show the first line of each.

```
program ListInbox;

{$mode objfpc}{$H+}

uses
  Classes, pop3send;

var
  POP3: TPOP3Send;
  i: Integer;
begin
  POP3 := TPOP3Send.Create;
  try
    POP3.TargetHost := 'mail.example.com';
    POP3.Username := 'me@example.com';
    POP3.Password := 'secret';

    if not POP3.Login then
    begin
      WriteLn('Login failed: ', POP3.ResultString);
      Exit;
    end;

    POP3.Stat; // fills StatCount / StatSize
    WriteLn(POP3.StatCount, ' messages, ', POP3.StatSize, ' bytes');

    for i := 1 to POP3.StatCount do
      if POP3.Retr(i) then // full message -> FullResult
        WriteLn(i, ': ', POP3.FullResult[0]);

    POP3.Logout;
  finally
    POP3.Free;
  end;
end.
```

Stat populates StatCount and StatSize; messages are numbered 1..StatCount and their text lands in the FullResult string list after Retr. **Gotcha:** Retr downloads the *whole* message — for just the headers use `Top(Index, 0)`, which is far cheaper on a big mailbox. For POP3S add `ssl_openssl` to uses and set `POP3.FullSSL := True` (port 995).

Recipes — Network tools

DNS lookups, ICMP pings, and raw TCP. The socket layer under all of it is the [socket classes](#); the client/server walkthrough is [Synapse in Two Hours](#).

Resolve a hostname

Problem: ask a DNS server for the A records of a name.

```
program Resolve;

{$mode objfpc}{$H+}

uses
  Classes, dnssend;

var
  DNS: TDNSSend;
  Answers: TStringList;
  i: Integer;
begin
  DNS := TDNSSend.Create;
  Answers := TStringList.Create;
  try
    DNS.TargetHost := '8.8.8.8';           // the DNS server to query

    // QTYPE_A = IPv4 address records (QTYPE_AAAA for IPv6, QTYPE_MX for mail).
    if DNS.DNSQuery('example.com', QTYPE_A, Answers) then
      for i := 0 to Answers.Count - 1 do
        WriteLn(Answers[i])
      else
        WriteLn('Query failed, RCode=', DNS.RCode);
    finally
      Answers.Free;
      DNS.Free;
    end;
  end.
```

DNSQuery(Name, QType, Reply) sends over UDP and fills Reply with one value per line.

Gotcha: you must set TargetHost to an actual resolver — Synapse does not read your system's /etc/resolv.conf. For mail servers specifically, GetMailServers(const DNSHost, Domain: AnsiString; const Servers: TStrings) queries MX and returns the hosts already sorted by preference.

Ping a host

Problem: measure round-trip time to a host with ICMP.

```
program Ping;

{$mode objfpc}{$H+}

uses
    pingsend;

var
    Ms: Integer;
begin
    Ms := PingHost('example.com');    // round-trip in ms, or -1 on failure
    if Ms >= 0 then
        WriteLn('Reply in ', Ms, ' ms')
    else
        WriteLn('No reply');
    end.
end.
```

PingHost wraps TPINGSend, returning the round-trip time in milliseconds or -1 if there's no reply. **Gotcha:** ICMP needs a raw socket, so on Unix this must run as root (or with CAP_NET_RAW). Use the TPINGSend object directly when you want to set Timeout/TTL or read ReplyFrom and PingTime.

A minimal TCP client

Problem: connect, send a line, read the reply — the bare socket pattern.

```
program TcpClient;

{$mode objfpc}{$H+}

uses
    blksock;

var
    Sock: TTCPBlockSocket;
begin
    Sock := TTCPBlockSocket.Create;
    try
        Sock.Connect('example.com', '80');
        if Sock.LastError <> 0 then
            begin
                WriteLn('Connect failed: ', Sock.LastErrorDesc);
                Exit;
            end;
        Sock.SendString('GET / HTTP/1.0' + CRLF + CRLF);    // CRLF is exported by blksock
        WriteLn(Sock.RecvString(5000));                      // first response line
    finally
        Sock.Free;    // destructor closes the socket
    end;
end.
```

Every call sets `LastError` (0 = success) rather than raising — check it after `Connect`. `RecvString(Timeout)` returns one CRLF-terminated line without the line ending, and sets `LastError` to a timeout code when the peer goes quiet.

A threaded echo server

Problem: accept many clients at once, one thread per connection.

```
program EchoServer;

{$mode objfpc}{$H+}

uses
  {$IFDEF UNIX} cthreads, {$ENDIF}
  Classes, blksock, synsock;  // synsock exports TSocket

type
  TEchoThread = class(TThread)
  private
    FSock: TTCPBlockSocket;
  public
    constructor Create(ClientSocket: TSocket);
    procedure Execute; override;
  end;

constructor TEchoThread.Create(ClientSocket: TSocket);
begin
  FSock := TTCPBlockSocket.Create;
  FSock.Socket := ClientSocket;  // adopt the handle Accept returned
  FreeOnTerminate := True;
  inherited Create(False);      // start immediately
end;

procedure TEchoThread.Execute;
var
  Line: string;
begin
  try
    repeat
      Line := FSock.RecvString(60000);
      if FSock.LastError <> 0 then Break;
      FSock.SendString('you said: ' + Line + CRLF);
    until Terminated;
  finally
    FSock.Free;
  end;
end;

var
  Listener: TTCPBlockSocket;
  Client: TSocket;
begin
  Listener := TTCPBlockSocket.Create;
```

```

try
  Listener.CreateSocket;
  Listener.SetLinger(True, 10000);
  Listener.Bind('0.0.0.0', '9000');
  if Listener.LastError <> 0 then
  begin
    WriteLn('Bind failed: ', Listener.LastErrorDesc);
    Exit;
  end;
  Listener.Listen;
  WriteLn('Listening on 9000...');
  repeat
    if Listener.CanRead(1000) then           // stay interruptible
    begin
      Client := Listener.Accept;
      if Listener.LastError = 0 then
        TEchoThread.Create(Client);          // fire and forget
      end;
    until False;
  finally
    Listener.Free;
  end;
end.

```

The listener only accepts and delegates; each connection runs plain sequential code in its own thread. **Gotcha:** never share one TBlockSocket across threads — give every connection its own, as above. The full walkthrough (including the TLS one-liner) is in [Synapse in Two Hours §3](#).

Recipes — Serial

TBlockSerial (in synaser) is the same blocking-with-timeouts model as the block sockets, aimed at a serial port instead of a network peer. Full API and platform notes in [the serial-ports chapter](#).

Open a port and do a request/response

Problem: open a serial port, send a command, and read the device's reply.

```
program SerialQuery;

{$mode objfpc}{$H+}

uses
  synaser;

var
  Ser: TBlockSerial;
  Reply: string;
begin
  Ser := TBlockSerial.Create;
  try
    // COM3 on Windows, /dev/ttyUSB0 on Linux – names are auto-translated.
    Ser.Connect('/dev/ttyUSB0');
    if Ser.LastError <> 0 then
      begin
        WriteLn('Open failed: ', Ser.LastErrorDesc);
        Exit;
      end;

    // baud, data bits, parity, stop bits, software flow, hardware flow.
    Ser.Config(9600, 8, 'N', SB1, False, False);

    Ser.SendString('STATUS?' + CRLF);
    Reply := Ser.RecvString(2000); // one CRLF-terminated line, 2 s timeout
    if Ser.LastError = 0 then
      WriteLn('Device said: ', Reply)
    else
      WriteLn('No reply: ', Ser.LastErrorDesc);
  finally
    Ser.Free; // destructor closes the port
  end;
end;
```

end.

Connect opens the port (check `LastError`), `Config(baud, bits, parity, stop, softflow, hardflow)` sets the line parameters, and `SendString/RecvString` mirror the socket API exactly — `RecvString(Timeout)` returns one CRLF-terminated line without the terminator. **Gotcha:** many devices are slow to wake, so allow a generous receive timeout and, for a fixed terminator other than CRLF, use `RecvTerminated(Timeout, Terminator)`; for raw byte counts use `RecvPacket/RecvBufferEx`. For a modem, `ATCommand('AT')` sends a command and returns the response in one call.

Visual Synapse

Where Synapse gives you socket classes and protocol *clients*, **Visual Synapse** is a layer built on top of it that adds two things Synapse deliberately leaves out: **RAD components** and **protocol servers**. It is a separate project — this chapter explains how it relates to the library the rest of this cookbook covers.

Live project: <https://github.com/yoctobyte/visualsynapse>

What it adds over raw Synapse

1. **RAD components.** Thin, drop-on-form wrappers around Synapse's client protocols (HTTP, UDP, DNS, ICMP, TCP, SMTP), with Object-Inspector-editable properties and event callbacks. Commands run on a background thread and a callback fires per result — the [blocking-plus-threads model](#) packaged as visual components.
2. **Servers.** Synapse ships clients; Visual Synapse adds hand-written protocol *server* implementations — HTTP, FTP, SMTP, telnet, and a SQL server component. Each is a listener that accepts connections and hands every one to its own handler thread, speaking the protocol in straight-line code over a `TTCPSocket` — exactly the pattern the architecture section describes.

It also historically carried **TPastella**, an experimental content-addressed peer-to-peer gossip layer built on the same socket foundation.

How it uses Synapse

Visual Synapse depends on Synapse; it does not replace it. The components and servers are built on `TBlockSocket` and friends, and they inherit Synapse's seams for free — including the [SSL plugin](#): a Visual Synapse server gains TLS the same way any Synapse socket does, by which `ssl_*` unit is in uses.

Status & caveats

Visual Synapse is a 2004–2008 project, revived for preservation. Treat its servers as historical: the 0.60 HTTP server has a known, unpatched directory-traversal, and the code predates modern transport hardening. See the project's own `SECURITY.md` before running anything. For learning the underlying library, the rest of this cookbook — grounded directly in Synapse — is the better guide; this chapter exists to place Visual Synapse in the map.

Appendix — Synapse in Two Hours

Welcome. This is the on-ramp. In one sitting you will write a TCP client, fetch a web page in a single line, run a threaded server, and switch the whole thing to TLS by adding one word to a uses clause. Everything here is real code against the real Ararat Synapse API — no pseudocode, no hand-waving. If you know a little Object Pascal and can compile with FPC or Delphi, you have everything you need.

Read the [architecture chapter](#) first if you want the *why*; this chapter is the *how*. The one idea to carry in with you: Synapse is **blocking with timeouts**. Every call either finishes its job or returns when your timeout elapses — it never hangs forever, and it reports status through `LastError` rather than throwing. That single decision is what keeps every example below short and linear. See [The blocking model & threads](#) for the full story.

1. Hello, socket — a minimal TCP client

The workhorse class is `TTCPBlockSocket`, declared in `blksock.pas`. It gives you TCP behind five methods you will use constantly:

- `Connect(IP, Port: string)` — open the connection (both arguments are strings; `Port` can be a number like '13' or a service name).
- `SendString(Data: AnsiString)` — write bytes.
- `RecvString(Timeout: Integer): AnsiString` — read one CRLF-terminated line, returning it *without* the line ending.
- `LastError: Integer` — 0 means success; anything else is the socket error code, with `LastErrorDesc` giving a human-readable string.
- `CloseSocket` — release the connection (the destructor calls it for you too).

Let's talk to a **daytime** service (TCP port 13), one of the oldest and simplest text protocols on the internet: connect, and the server sends back the current date and time as a line of text, then closes. No request needed.

```
program HelloSocket;

{$mode objfpc}{$H+}

uses
  blksock;

var
  Sock: TTCPBlockSocket;
  Line: string;
begin
```

```

Sock := TTCPBlockSocket.Create;
try
  // Connect blocks until the connection is up or ConnectionTimeout elapses.
  Sock.Connect('time.nist.gov', '13');
  if Sock.LastError <> 0 then
    begin
      WriteLn('Connect failed: ', Sock.LastErrorDesc);
      Exit;
    end;

  // Read lines until the server closes and the read times out.
  repeat
    Line := Sock.RecvString(5000); // wait up to 5 seconds for a line
    if Sock.LastError = 0 then
      WriteLn(Line);
  until Sock.LastError <> 0;

finally
  Sock.Free; // destructor closes the socket
end;
end.

```

Notice the **discipline**: after *every* operation we check `LastError` before trusting the result. That is the Synapse rhythm. `Connect` doesn't raise on failure — it sets `LastError` and returns, so you decide what to do. `RecvString` returns an empty string on timeout *and* sets `LastError` to `WSAETIMEDOUT`, which is exactly how we know the server has finished and it's time to stop reading.

That is a complete, correct network client. Twenty lines. This is what “blocking with timeouts” buys you: code that reads top-to-bottom like a description of the conversation.

Sending a request

The daytime server talks first. Most protocols want you to speak first. The pattern is identical — `SendString`, then `RecvString`:

```

Sock.Connect('example.com', '80');
if Sock.LastError = 0 then
  begin
    Sock.SendString('GET / HTTP/1.0' + CRLF + CRLF); // CRLF is declared in blksock
    Line := Sock.RecvString(5000);
    WriteLn(Line); // -> HTTP/1.1 200 OK
  end;

```

`CRLF` is a constant Synapse exports from `blksock` (`CR + LF`), so you never have to remember `#13#10`. You *could* speak the whole of HTTP this way, line by line. You shouldn't — because Synapse already did it for you. That's section 2.

2. One line of HTTP — the convenience layer

`httpsend.pas` sits on top of `TTCPBlockSocket` and speaks HTTP so you don't have to. For the common case there is a free function, `HttpGetText`, that does connect, request, read the response, and disconnect — all of it:

```

program OneLineHTTP;

{$mode objfpc}{$H+}

uses
  Classes, httpsend;

var
  Page: TStringList;
begin
  Page := TStringList.Create;
  try
    if HttpGetText('http://example.com/', Page) then
      WriteLn(Page.Text)
    else
      WriteLn('Request failed');
  finally
    Page.Free;
  end;
end.

```

HttpGetText(const URL: string; const Response: TStrings): Boolean returns True on a successful fetch and fills your TStrings with the response body. One call. Compare that to hand-rolling the request, parsing status lines, and handling headers — this is the payoff for the protocol classes.

When you need the details — THTTPSend

HttpGetText is a convenience wrapper around the real workhorse, THTTPSend. Reach for the class when you need headers, status codes, or anything other than a plain GET. The pattern is: create it, call HTTPMethod(Method, URL), then read the results off its properties:

- Document: TMemoryStream — the response body.
- Headers: TStringList — request headers going out, response headers coming back.
- ResultCode: Integer / ResultString: string — the HTTP status.
- MimeType: string — content type for the body you send.

```

program HttpGetClass;

{$mode objfpc}{$H+}

uses
  Classes, httpsend;

var
  HTTP: THTTPSend;
  Body: TStringList;
begin
  HTTP := THTTPSend.Create;
  Body := TStringList.Create;
  try
    if HTTP.HTTPMethod('GET', 'http://example.com/') then
      begin
        WriteLn('Status: ', HTTP.ResultCode, ' ', HTTP.ResultString);
        Body.LoadFromStream(HTTP.Document); // Document is a TMemoryStream
      end;
  end;
end.

```

```

        WriteLn(Body.Text);
    end
    else
        WriteLn('Failed: ', HTTP.Sock.LastErrorDesc);
    finally
        Body.Free;
        HTTP.Free;
    end;
end.

```

Note HTTP.Sock — THTTPSend exposes the underlying TTCPBlockSocket it drives, so the same LastError/LastErrorDesc discipline is right there when a request fails at the socket level.

Posting data

For a POST you load the request body into Document, set MimeType, and call HTTPMethod('POST' , URL):

```

program HttpPost;

{$mode objfpc}{$H+}

uses
    Classes, httpsend;

var
    HTTP: THTTPSend;
    Payload: string;
begin
    HTTP := THTTPSend.Create;
    try
        Payload := '{"name":"synapse","stars":9001}';
        HTTP.Document.Write(Pointer(Payload)^, Length(Payload)); // fill the request body
        HTTP.MimeType := 'application/json';

        if HTTP.HTTPMethod('POST', 'http://example.com/api') then
            WriteLn('Server said: ', HTTP.ResultCode, ' ', HTTP.ResultString)
        else
            WriteLn('POST failed: ', HTTP.Sock.LastErrorDesc);
        finally
            HTTP.Free;
        end;
    end.

```

Document is a TMemoryStream, so anything you can write to a stream — a string, a file, binary — becomes your request body. If you just want to submit a URL-encoded form, httpsend also exports the one-liner HttpPostURL(const URL, URLData: string; const Data: TStream): Boolean.

That's HTTP. The protocol classes (SMTP in smtpsend.pas, POP3, IMAP, FTP, and the rest) all follow the same shape: a thin class over a block socket, a handful of methods, status reported the Synapse way.

3. Your first server — a threaded echo/line server

A client is one conversation. A **server** is many at once, and here is where the blocking model earns its keep. The recipe:

1. Bind(IP, Port) — claim the address.
2. Listen — start accepting.
3. Accept — block until a client arrives; it returns a raw TSocket handle.
4. **Hand that handle to its own thread** and go straight back to Accept.

That fourth step is the whole trick. The listener's only job is to accept and delegate. Each accepted connection gets a TThread running plain, sequential protocol code — no shared state machine, no re-entrancy puzzles. One connection, one thread, one readable Execute.

The per-connection handler thread

type

```
TEchoThread = class(TThread)
private
    FSock: TTCPBlockSocket;
public
    constructor Create(ClientSocket: TSocket);
    procedure Execute; override;
end;
```

constructor TEchoThread.Create(ClientSocket: TSocket);

begin

```
    FSock := TTCPBlockSocket.Create;
    FSock.Socket := ClientSocket; // adopt the handle Accept handed us
    FreeOnTerminate := True;    // clean itself up when done
    inherited Create(False);    // start running immediately
```

end;

procedure TEchoThread.Execute;

var

Line: **string**;

begin

try

```
    FSock.SendString('Welcome. Type something; QUIT to leave.' + CRLF);
```

repeat

```
    Line := FSock.RecvString(60000); // up to 60 s of idle before we drop them
```

```
    if FSock.LastError <> 0 then
```

```
        Break;
```

```
        // timeout or disconnect
```

```
    if Line = 'QUIT' then
```

```
        Break;
```

```
    FSock.SendString('you said: ' + Line + CRLF);
```

```
    until Terminated;
```

finally

```
    FSock.Free; // closes the connection
```

end;

end;

Straight-line code. It reads exactly like the protocol it speaks, because a blocking socket lets it. Assigning to the Socket property is how you wrap the raw handle that Accept returns in a full TTCPBlockSocket.

The listener loop

```
program EchoServer;

{$mode objfpc}{$H+}

uses
  {$IFDEF UNIX} cthreads, {$ENDIF}    // FPC needs this unit for threads on Unix
  Classes, blksock, synsock;          // synsock exports TSocket

  { ... TEchoThread from above ... }

var
  Listener: TTCPSocket;
  ClientHandle: TSocket;
begin
  Listener := TTCPSocket.Create;
  try
    Listener.CreateSocket;
    Listener.SetLinger(True, 10000);
    Listener.Bind('0.0.0.0', '9000');    // all interfaces, port 9000
    if Listener.LastError <> 0 then
    begin
      WriteLn('Bind failed: ', Listener.LastErrorDesc);
      Exit;
    end;
    Listener.Listen;
    WriteLn('Listening on port 9000...');

    repeat
      // CanRead lets the accept loop stay responsive and interruptible.
      if Listener.CanRead(1000) then
      begin
        ClientHandle := Listener.Accept;
        if Listener.LastError = 0 then
          TEchoThread.Create(ClientHandle);    // fire and forget; thread owns it now
        end;
      until False;    // real code: break on a shutdown flag
    finally
      Listener.Free;
    end;
  end.
```

`CanRead(1000)` polls the listening socket for a pending connection with a one-second timeout, so the loop wakes up regularly instead of blocking forever in `Accept` — that's your hook for a clean shutdown flag. When a client is waiting, `Accept` returns instantly with its handle, we spin up a `TEchoThread`, and we're back at the top of the loop ready for the next one. The listener never does any real work; the threads do. That is how one small program serves hundreds of simultaneous connections.

Two rules for servers. Give every connection its own thread, and never share one `TBlockSocket` across threads. Stay inside those two lines and Synapse's concurrency stays as simple as the code above.

Connect to it with `telnet localhost 9000`, or point the client from section 1 at it. This is a real, working server.

4. Going encrypted — TLS by adding one word

Here is the most elegant trick in the library. You do **not** rewrite your client to speak HTTPS. TLS in Synapse is a *plugin*, selected purely by which unit you put in your uses clause. Add one:

```
uses
  Classes, httpsend,
  ssl_openssl;      // <-- this line turns on TLS
```

Just placing `ssl_openssl` (or `ssl_openssl3` for OpenSSL 3.x) in `uses` registers an SSL implementation globally. Now the very same `THTTPEnd` code from section 2 handles `https://` URLs — the block socket underneath negotiates TLS on your behalf when the scheme calls for it:

```
if HTTP.HTTPMethod('GET', 'https://example.com/') then // now works, unchanged
  WriteLn('Secure status: ', HTTP.ResultCode);
```

Nothing else in your program changes. That's the payoff of TLS-as-a-swappable-plugin: the socket classes were written once with a seam for encryption, and the `ssl_*` unit you link decides what fills it.

STARTTLS — upgrading a live connection

Some protocols (SMTP, IMAP, FTP) start in plaintext and upgrade the *existing* connection to TLS mid-stream. Synapse supports this directly: once you're connected, call `SSLDoConnect` on the block socket to perform the TLS handshake in place.

```
Sock.Connect('mail.example.com', '587');
Sock.SendString('STARTTLS' + CRLF); // tell the server we want to go secure
Sock.RecvString(5000);              // read its go-ahead
Sock.SSLDoConnect;                  // upgrade this live socket to TLS
if Sock.SSL.LastError <> 0 then
  WriteLn('TLS handshake failed: ', Sock.SSL.LastErrorDesc);
```

`SSLDoConnect` (declared on `TTCPSocket` in `blksock.pas`) runs the handshake over the connection you already have. From that point on your `SendString` and `RecvString` calls are encrypted, with no other change to your code. TLS-level status lives on the socket's SSL object (`Sock.SSL.LastError` / `Sock.SSL.LastErrorDesc`).

For the full picture of how the plugin seam works — the global `SSLImplementation` metaclass and why a `uses` line is all it takes — read [The SSL plugin seam](#).

5. Where to next

Two hours in, you can open connections, fetch and post over HTTP, run a threaded server, and encrypt any of it. That is the whole shape of Synapse — everything else is more protocols in the same mold.

- **The socket classes** — `TBlockSocket` and its family (TCP, UDP, raw, even serial), timeouts, `CanRead/CanWrite`, the receive functions beyond `RecvString` (`RecvTerminated`, `RecvPacket`). Start with [The blocking model & threads](#), then the socket-classes section for the full API surface.

- **The protocol classes** — SMTP (`smtpsend.pas`; try the `SendTo` free function for a fire-and-forget email), POP3, IMAP, FTP, NNTP, LDAP, DNS and more. Each is a thin readable class over a block socket, and each follows the exact rhythm you learned here: create, call a method, check status the Synapse way.
- **The architecture chapters** — [start here](#) for the four seams that make all of this so small: the blocking model, the synsock OS seam, the SSL plugin, and Synapse's own hand-rolled crypto.

The library rewards curiosity. Because every layer is thin and orthogonal, reading the source of any one class is a short trip — and now you know the vocabulary to follow it.
